

Exhibit 94

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

EXHIBIT 94

Domestic Industry Claim Chart for Claims of U.S. Patent No. 10,025,731 (the “’731 Patent”) Based on Practicing ’731 Products

As shown in the chart below, the Practicing ’731 Products, which include Netlist DDR5 VLP RDIMMs, Netlist Lightning DDR5 RDIMMs, Netlist DDR5 LP MRDIMMs, [REDACTED], and SK hynix RDIMMs and MRDIMMs, practice at least claim 1 of the ’731 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Practicing ’731 Products do not practice the ’731 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Practicing ’731 Products, such differences are insubstantial and the Practicing ’731 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Practicing ’731 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Practicing ’731 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

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Claim 1	Evidence of Use
[1.a] [i] A memory module [ii] operable to communicate data with a system memory controller via a memory bus [iii] in response to address and control signals from the system memory controller, comprising:	The Practicing '731 Products include [i] “a memory module.”  VLP RDIMM Ex. CC-33 (Netlist DDR5 DIMMs).  SERVER DDR5 RDIMM Ex. CC-34 (SK hynix Products).



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RDIMM

VIEW MORE



DDR5
Generation

16-256GB
Module Density

4800-6400Mbps and higher
Speed

DDR4
Generation

16-256GB
Module Density

2666-3200Mbps
Speed

Application

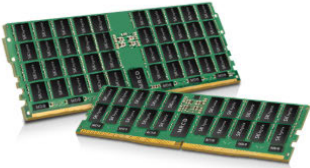
Server

Networking

- DIMM with a register (Register Clock Driver, RCD) for command/address signals between the DRAM and system memory controller
- Supports x4, x8 DRAM and x4 3DS stacks

MRDIMM

VIEW MORE



DDR5
Generation

32-128GB
Module Density

up to 8800Mbps
Speed

Application

Server

- DIMM with a register (Register Clock Driver, Data Buffer) for enabling simultaneous operation of two ranks, which increases bandwidth.
- Supports x4, x8 DRAM

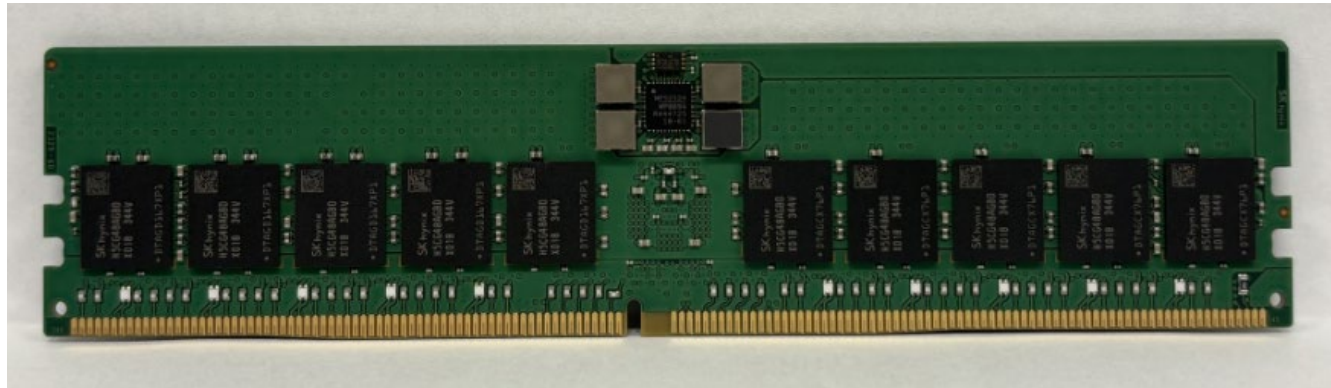
Ex. CC-35 (SK hynix Module).



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Ex. CC-36 at IMG_5114 (showing DDR5 RDIMM Product).

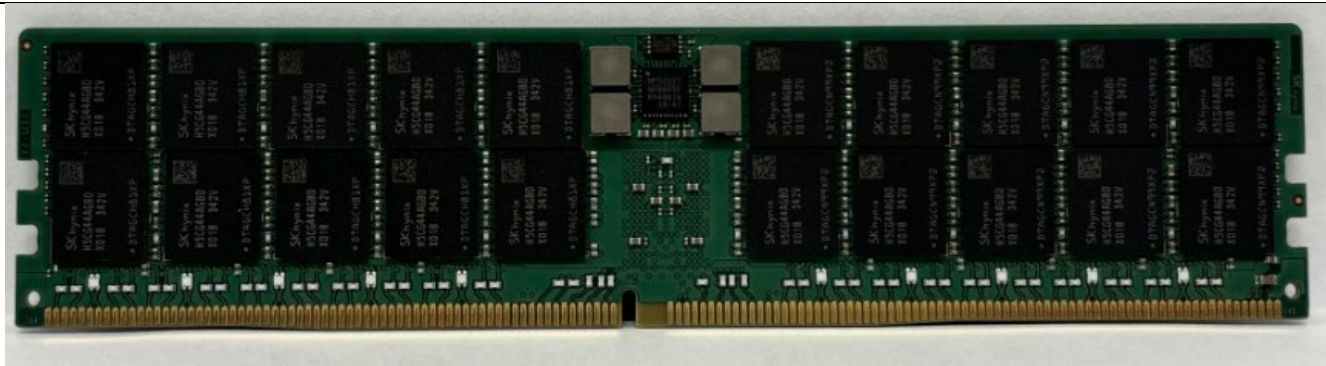


Ex. CC-36 at IMG_5117 (showing DDR5 RDIMM Product).



Ex. CC-36 at IMG_5112 (showing DDR5 RDIMM Overclock Product).

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Ex. CC-36 at IMG_5109 (showing DDR5 RDIMM Overclock Product).



Ex. CC-36 at IMG_5126 (showing Netlist DDR5 MRDIMMs).

[REDACTED]

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Ex. CC-36 at IMG_5122 (showing Netlist DDR5 MRDIMMs).

“DIMM” in RDIMM and MRDIMM stands for dual inline *memory module*. For example, Netlist’s DDR5 DIMMs web page is in the “Products / Memory Modules” section, advertising that “DDR5 DIMMs are the *physical modules that contain DDR5 memory chips*” and listing RDIMMs, VLP RDIMMs, and MRDIMMs as examples of the memory modules offered by Netlist. Ex. CC-33 (Netlist DDR5 DIMMs). Additionally, Netlist’s Lightning DDR5 DIMM web page advertises “*memory Lightning Modules*,” specifically listing DDR5 RDIMM Overclock Products and DDR5 RDIMM Ultra Low Latency RDIMM Products. Ex. CC-37 (Netlist Lightning DDR5 DIMM).

As another example, [REDACTED]

[REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 1.

As yet another example, [REDACTED]

[REDACTED] Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1.

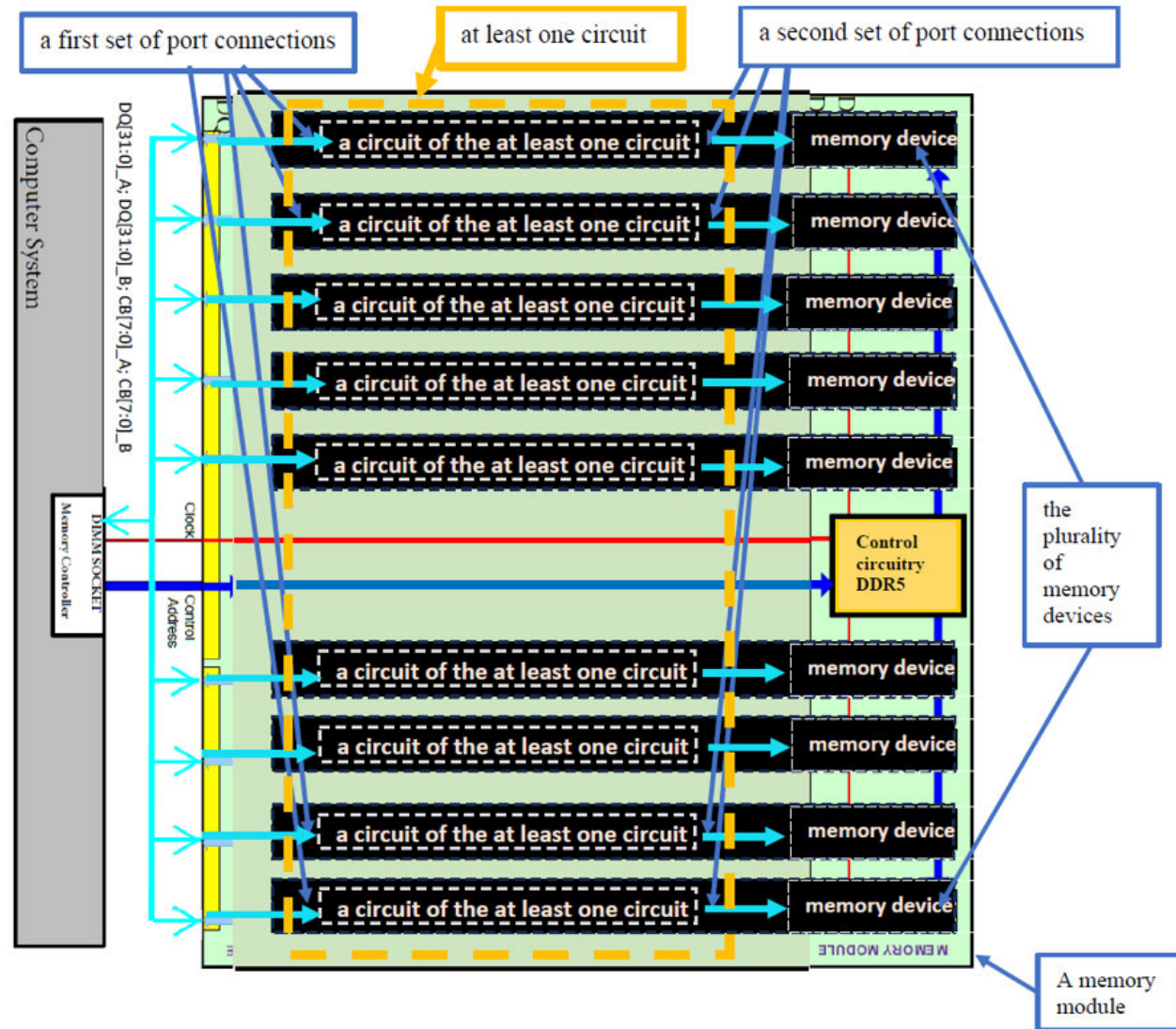
The Practicing ’731 Products are each [ii] “operable to communicate data” (e.g., read or write data) “with a system memory controller” (e.g., a memory controller in a host computer’s CPU) “via a



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	memory bus” (e.g., a pathway between a CPU and DRAMs for data communication). Illustrated below is a representative depiction of the Practicing ’731 Products’ memory module. The memory module includes a memory bus, depicted via at least the red, blue, and/or cyan lines between the computer system and components of the illustrated memory module.
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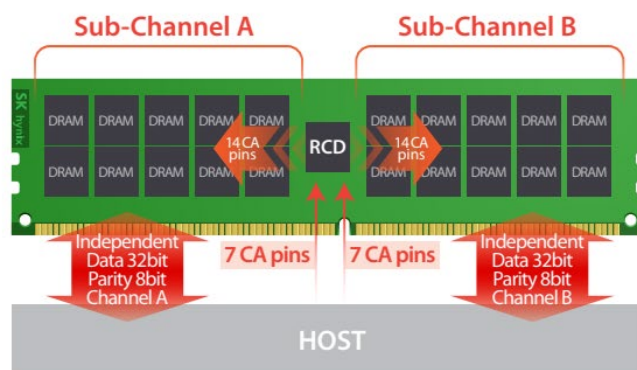
The DIMMs communicate data with a system memory controller via a memory bus. For example:

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“DDR5 DIMMs are the physical modules that contain *DDR5 memory chips and are used to expand the memory capacity of a computer system*. DDR5 DIMMs have larger burst lengths than DDR4, with a burst length of 16 vs. 8, which allows for a greater amount of data to be transferred *between the memory and the CPU*.” Ex. CC-33 (Netlist DDR5 DIMMs).

Netlist’s memory modules, such as Netlist’s DDR5 RDIMMs, communicate data with a host via the independent data 32 bit parity 8 bit Channel A and Channel B connections, as shown below:

Features



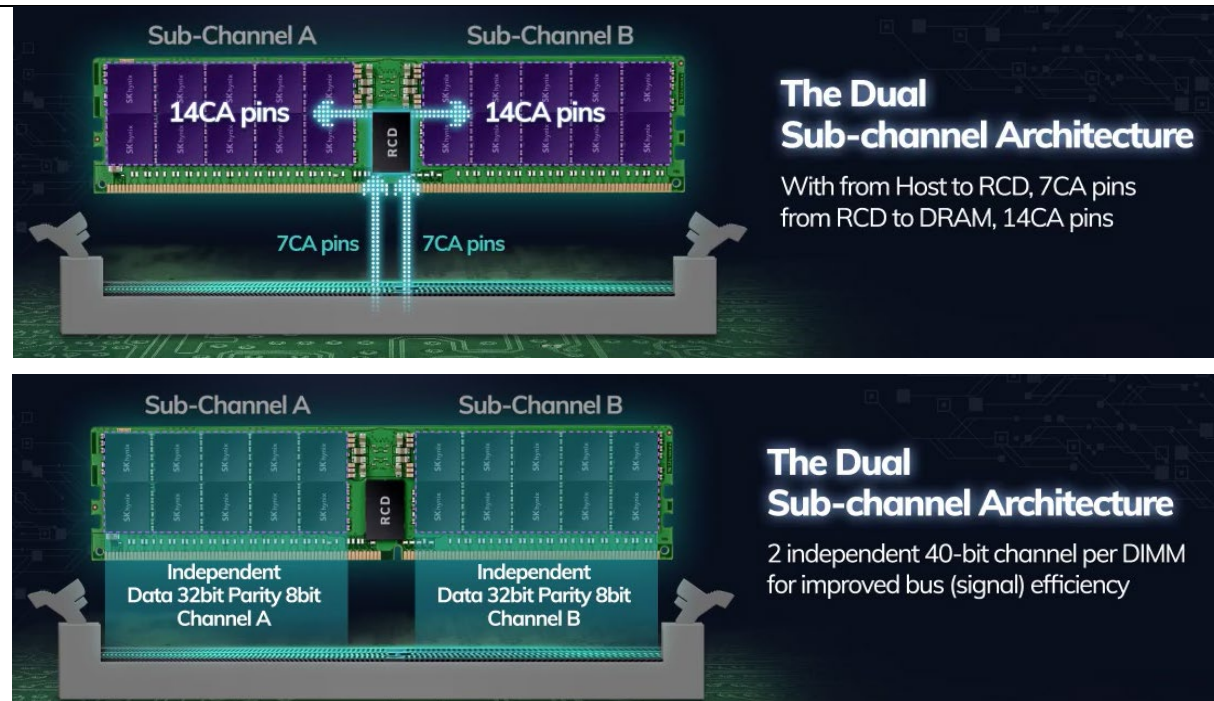
| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK hynix A New Standard) at 5.

SK hynix’s website also shows this element, advertising that “[w]ith a dual sub-channel structure, DDR5 RDIMM increases the number of CA (Command-Address) pins to seven double-data rate CA buses each from host to RCD and 14 single-data-rate buses each on RCD to DRAM.” Ex. CC-41 (SK hynix RDIMM).

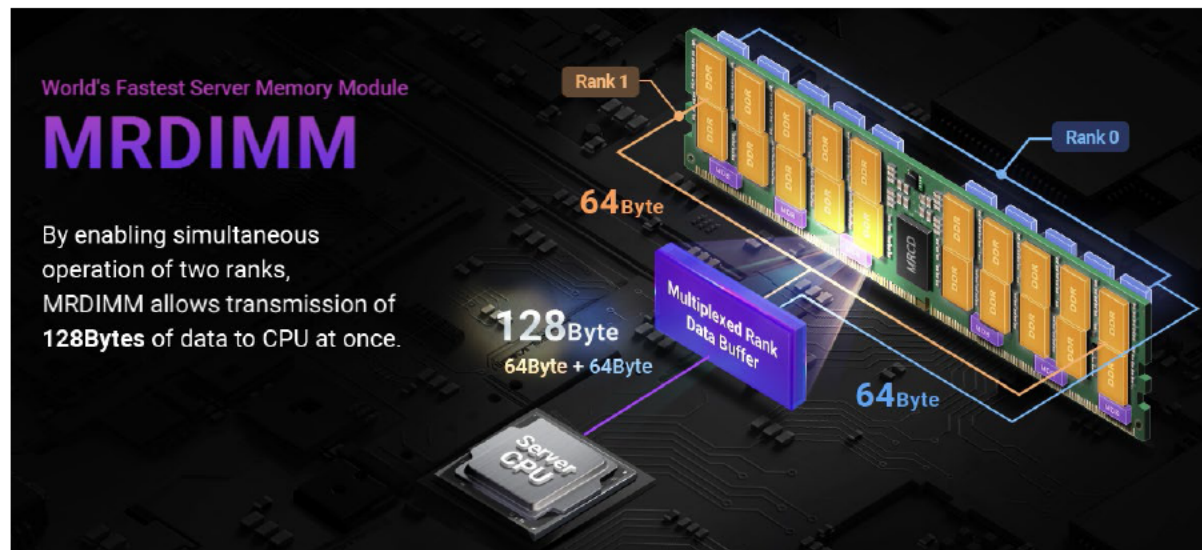
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See CC-41 (SK hynix RDIMM).

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For the MRDIMM, SK hynix also shows the communication bus between a DIMM and server CPU.

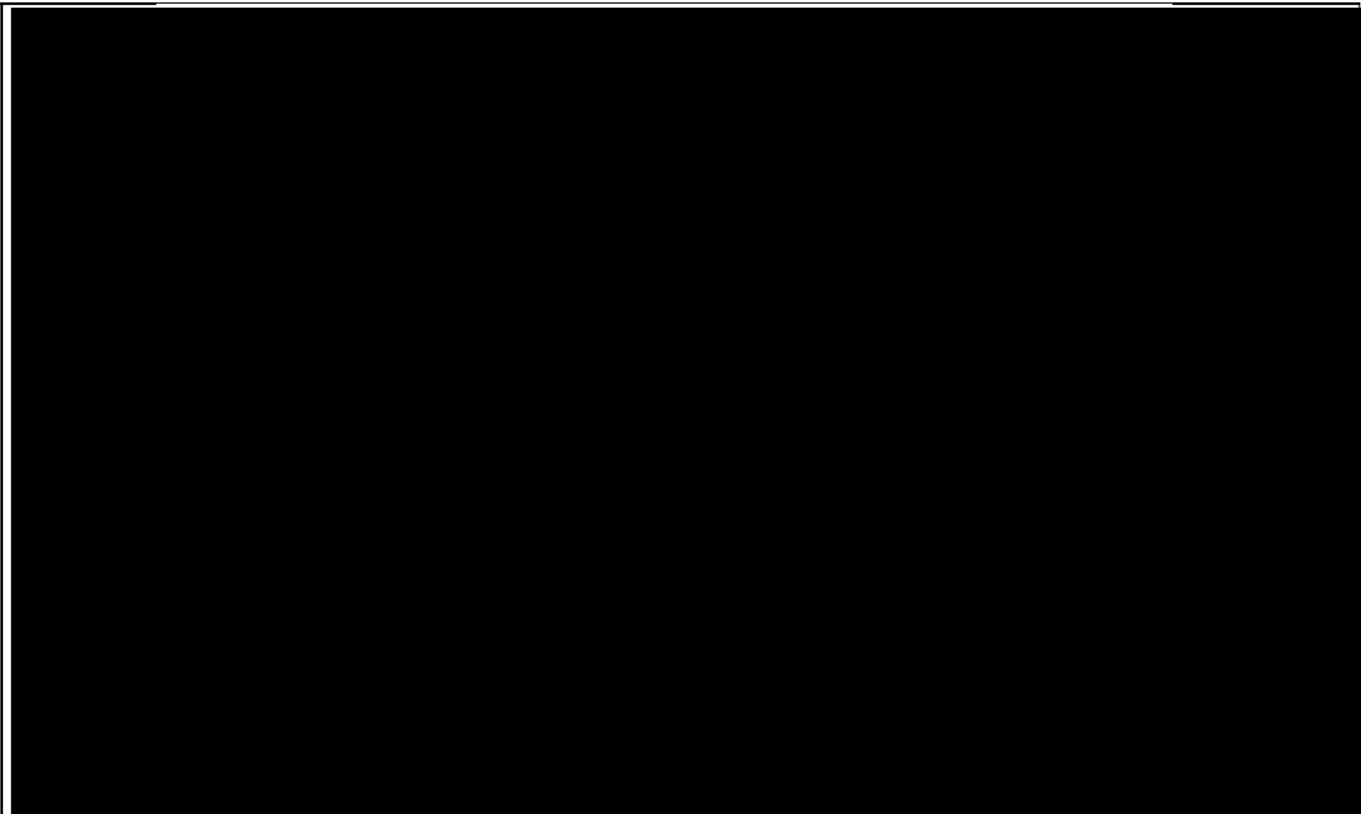


Ex. CC-42 (SK hynix MRDIMM) at 4.

Also,

[REDACTED]

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	 Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 17, 19. [REDACTED]
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7.

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9–10.

As another example,

[REDACTED]

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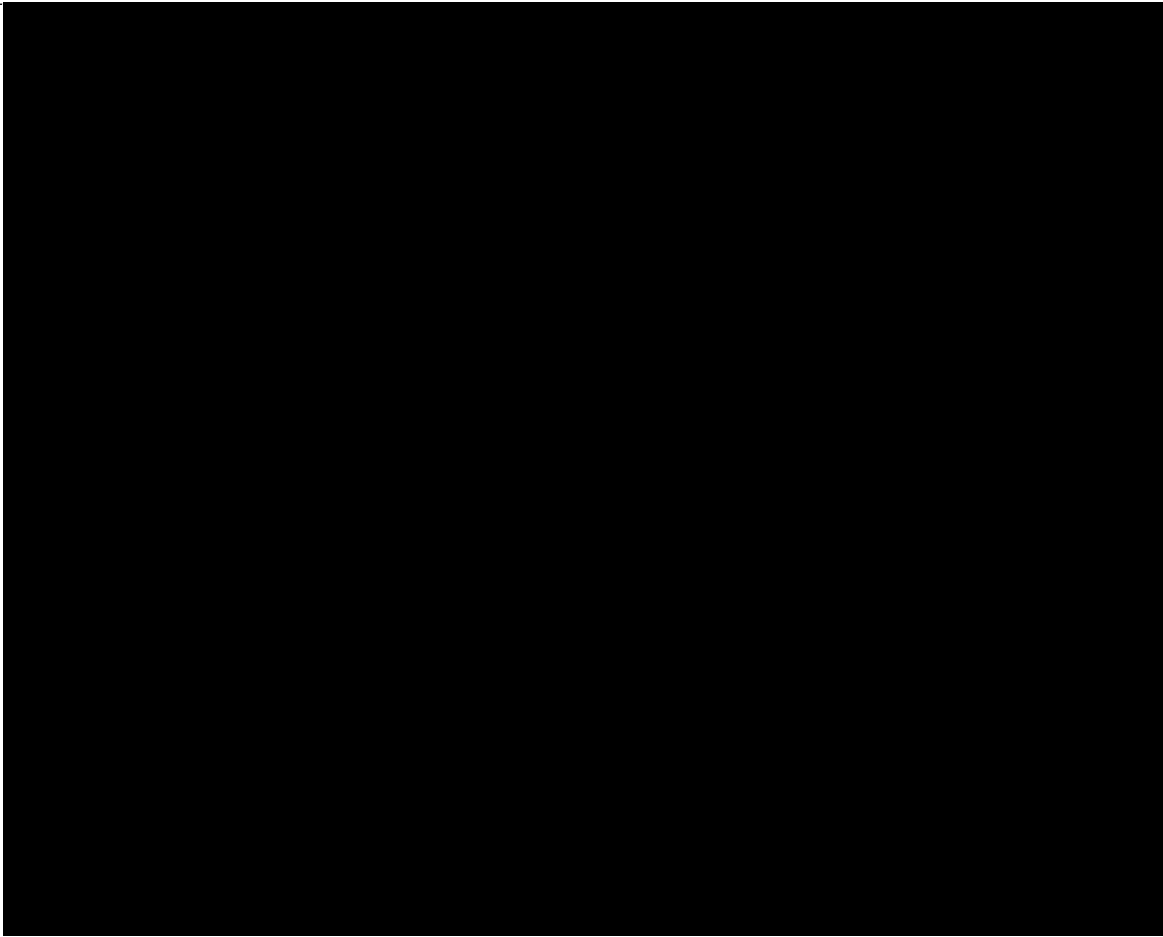
[REDACTED]

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[REDACTED]

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	 Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–6. [REDACTED]
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[REDACTED]

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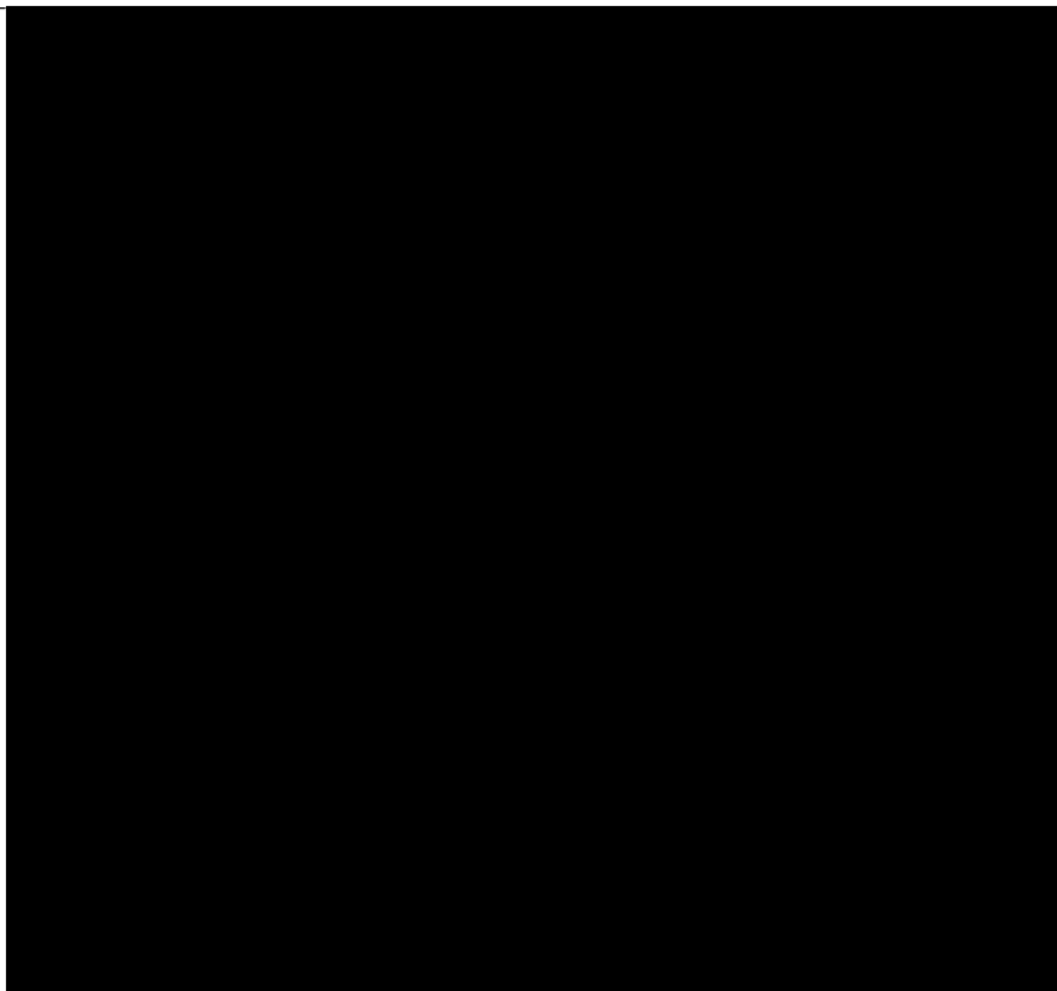
[REDACTED]

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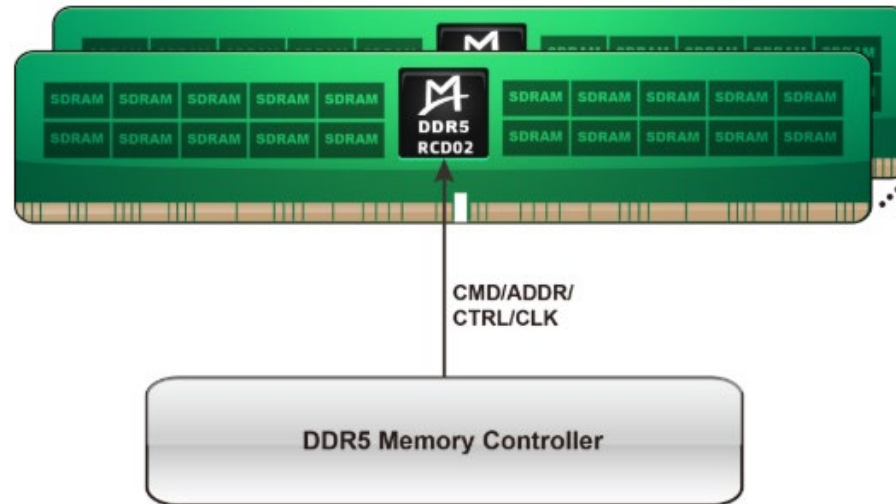


Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10–12.

As another example showing how DIMMs are operable to communicate data with a system memory controller via a memory bus, the DDR5 MRDIMM includes a Montage DDR5 MRCD. *E.g.*, Ex. CC-

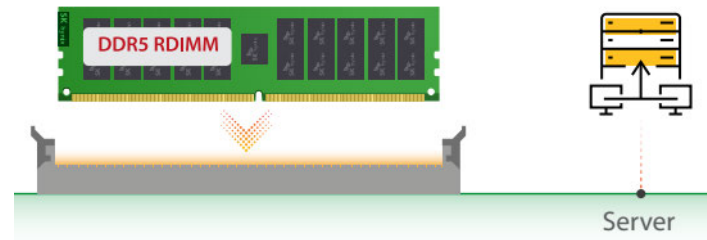
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36 at IMG_5124 (showing part number M88MR5RCD01). Shown below, this part is operable to communicate data with a system memory controller via a memory bus.



Ex. CC-43 (M88DR5RCD02 Flyer) at 2.

Use Cases



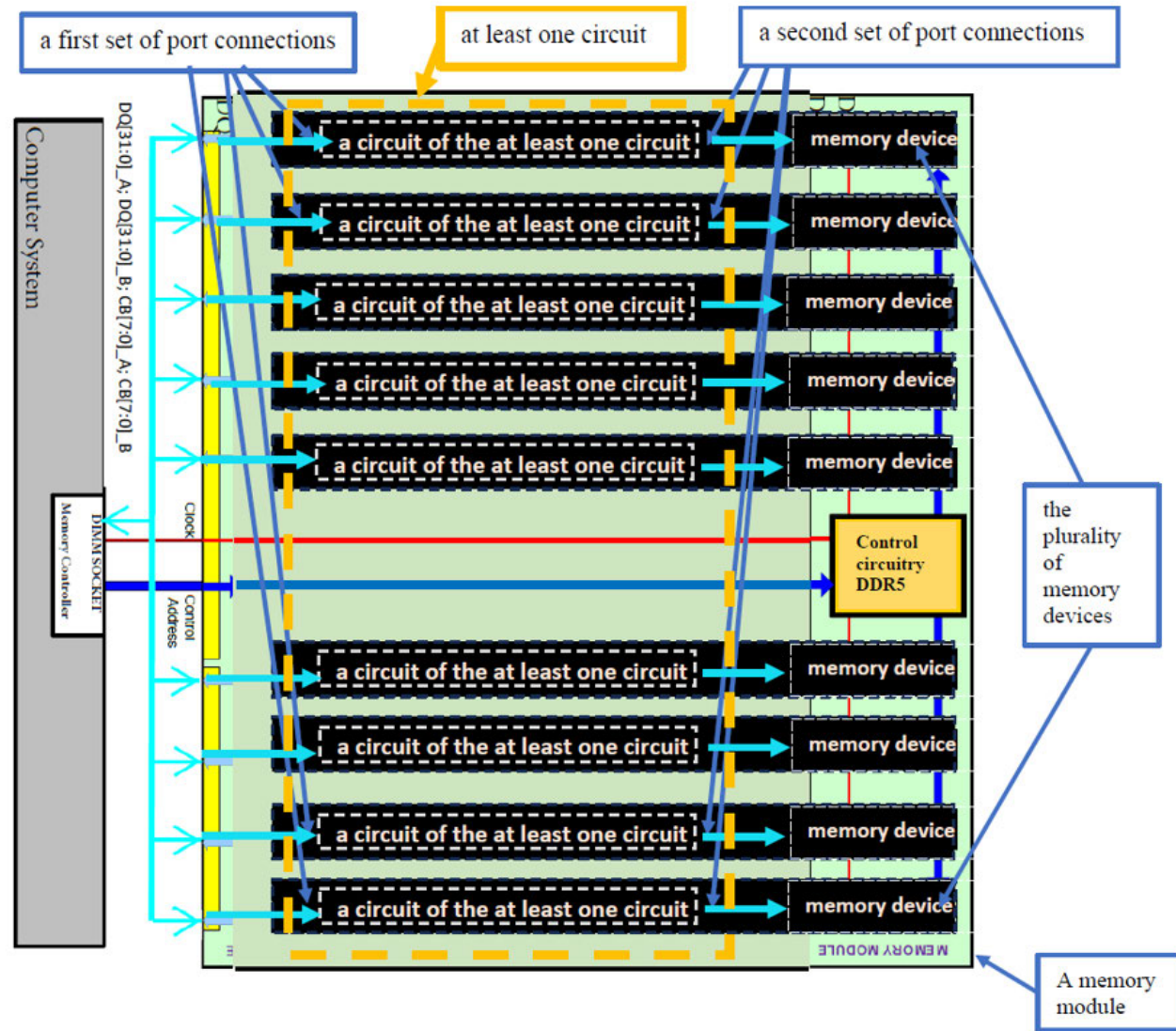
Ex. CC-40 (SK hynix A New Standard) at 6.



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	The data communication between any of the Practicing '731 Products and a system memory controller is [iii] “in response to address and control signals from the system memory controller.” Referring to the representative illustration, address and control signals are transmitted, via at least one of the memory bus lines (e.g., blue line, red line), by the computer system to the memory module.
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As one example,

[REDACTED]

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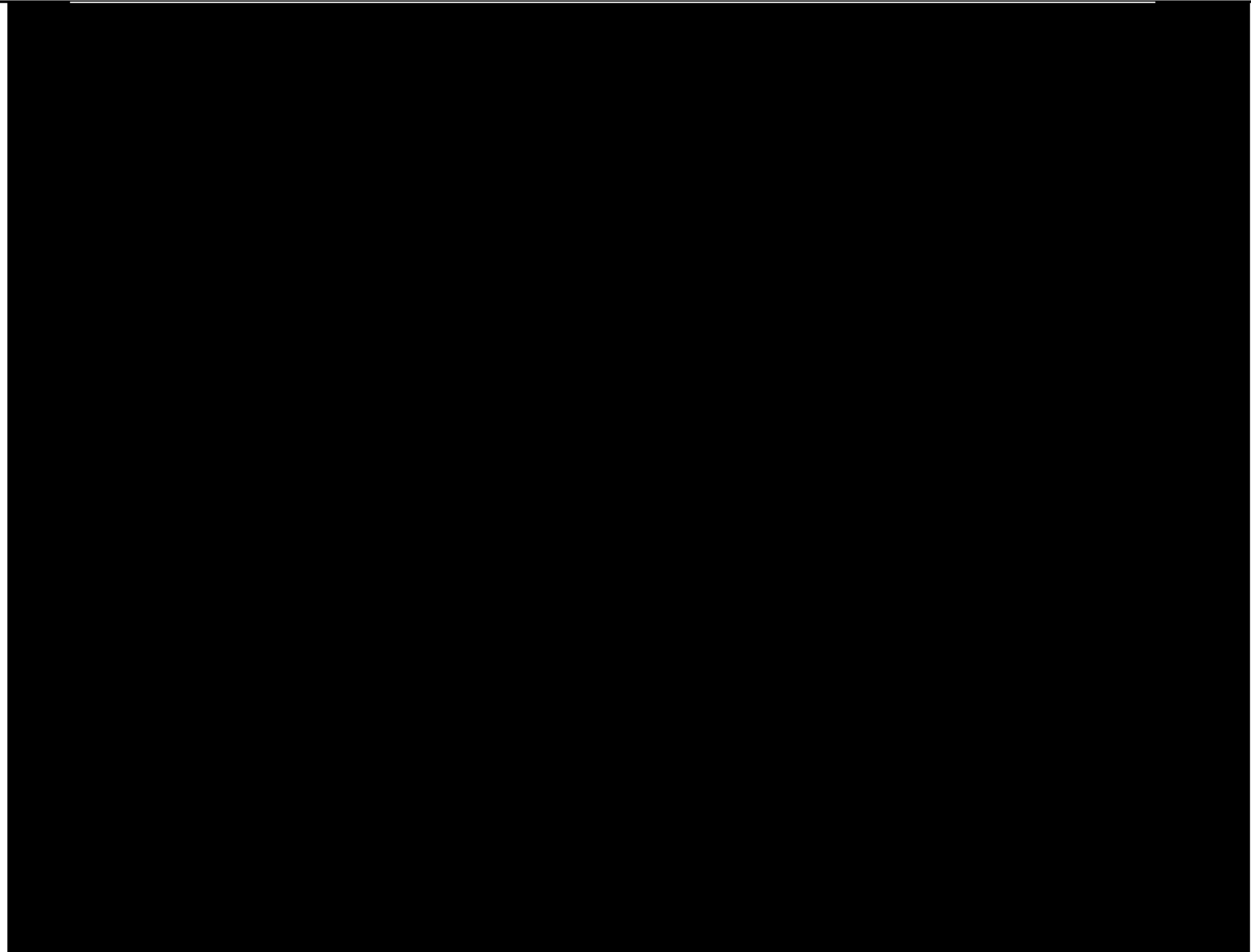
	[REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7. [REDACTED]
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9–10.

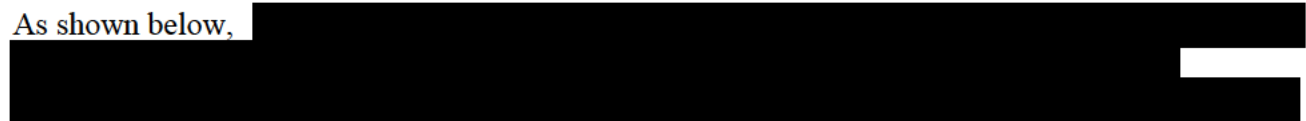
As another example,

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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–5.

As shown below,



[REDACTED]

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[REDACTED]

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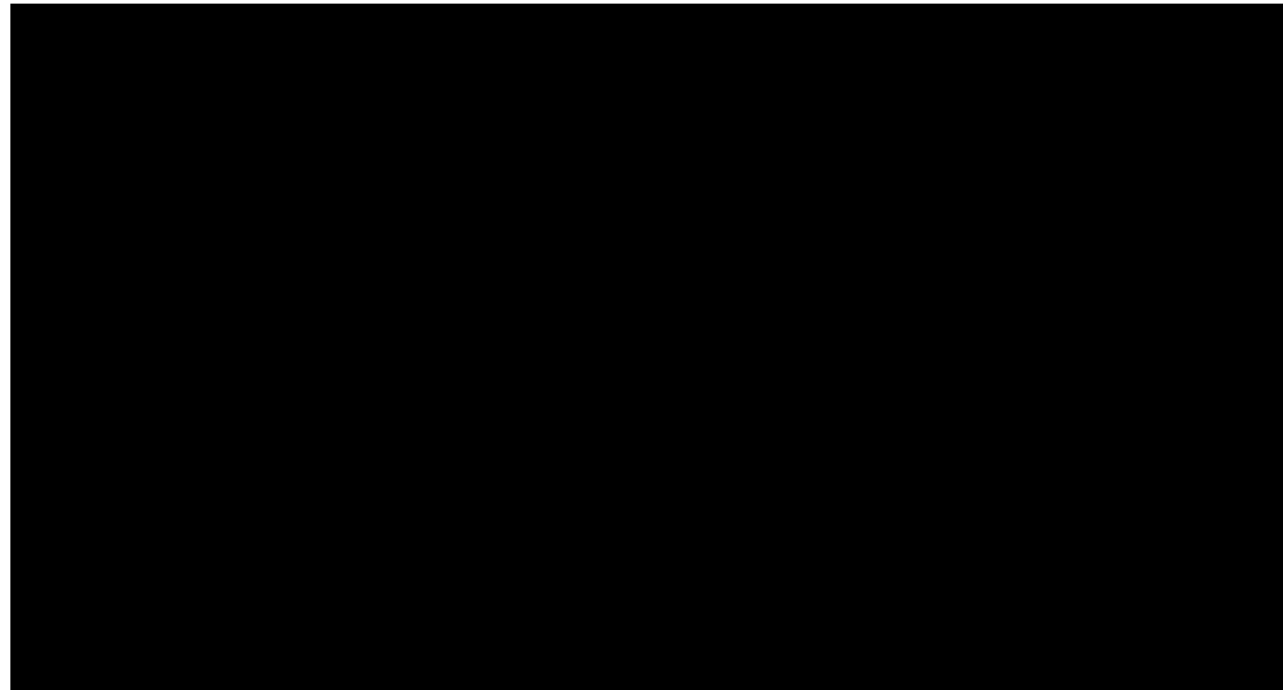
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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10–12.

As further evidence

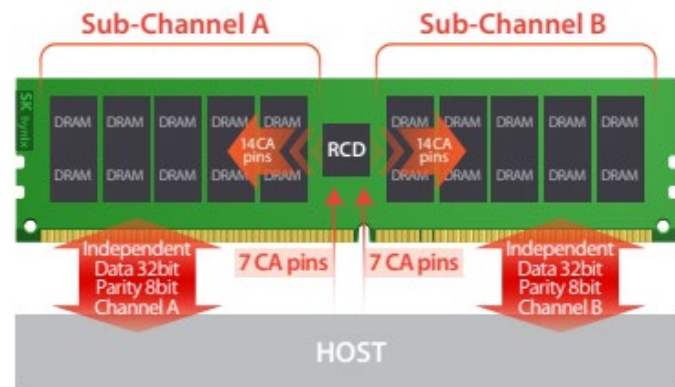
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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 45.

As yet another example, the Practicing '731 Products receive address and control signals and propagate these signals to the various DRAMs to facilitate data read/write between the host and DRAMs.

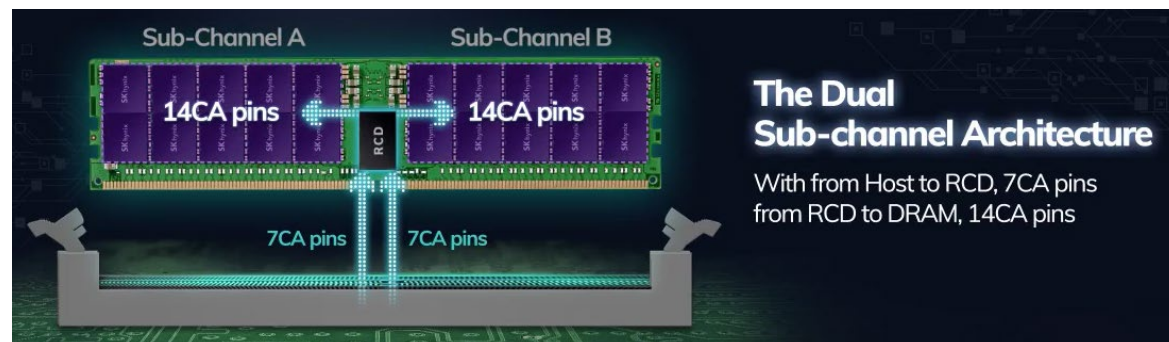
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New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

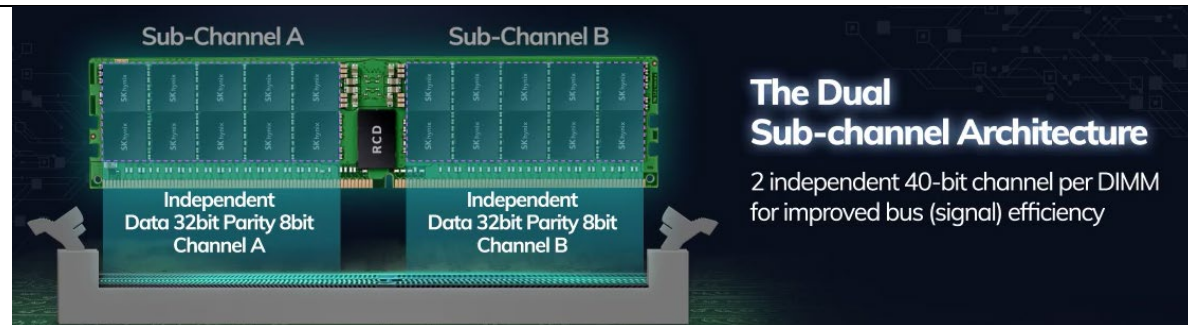
Ex. CC-40 (SK hynix A New Standard) at 5.



The Dual Sub-channel Architecture

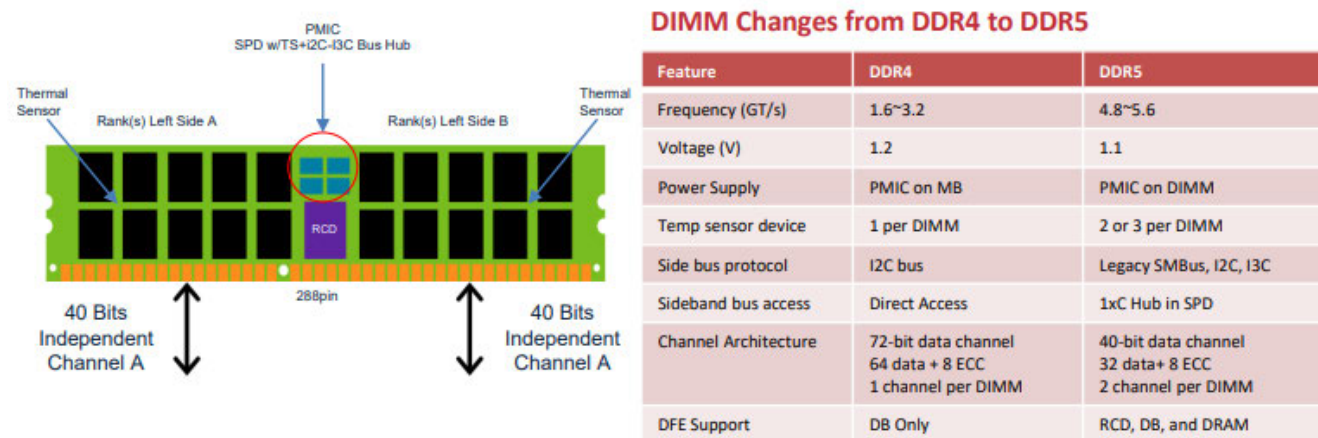
With from Host to RCD, 7CA pins from RCD to DRAM, 14CA pins

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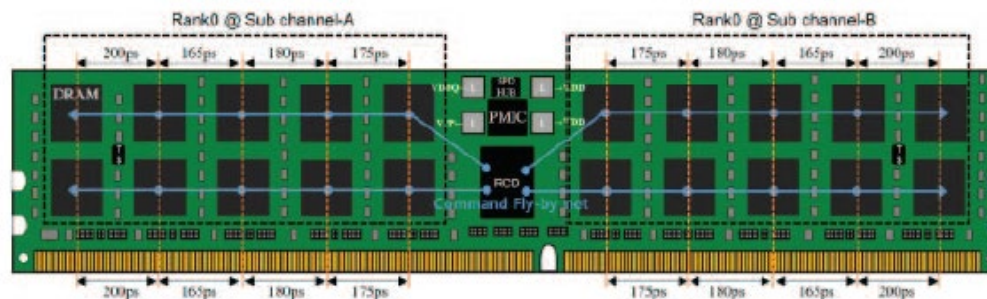
Ex. CC-41 (SK hynix RDIMM).

Furthermore, as a change from DDR4 to DDR5, Netlist Product Brief of DDR5 DIMM Products highlights that a PMIC is on the memory module. Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1. The PMIC is shown below and would be involved in such response signaling.



Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

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Ex. CC-45 (Na *et al.*, “A Novel Simulation Methodology Reflecting System Power Scenario using a Markov-Chain-based Stochastic Random Power Model,” 2022 IEEE 72nd Electronic Components and Technology Conference, 929–936) at 931.

As also shown, those in the industry understood that a DDR5 SDRAM on a memory module receives data from the memory controller (write operation) or outputs data to the memory controller in response to the control and address signals it receives, which are transmitted either directly or through control circuitry to the address/control signal input circuitry of the SDRAM.

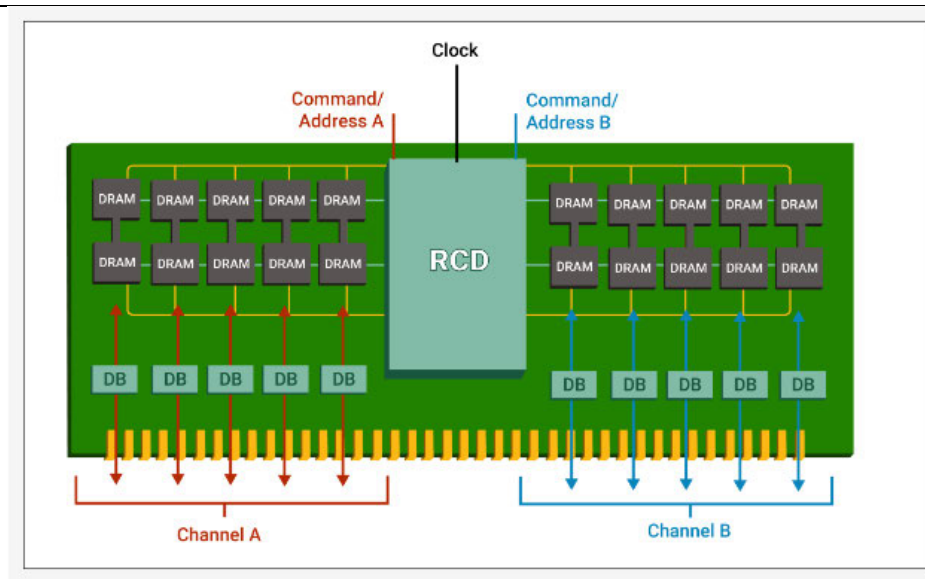
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Function	Abbrevia- tion	CS_n	CA Pins													NOTES	
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	11,17,20,21, 23,24
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/ R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	11,15,18,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	11,15,18,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/ DOPID	8,11,13,20, 22,23,24
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/ DOPID	8,13,20,22, 23,24,25,26
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,12,15,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,12,15,19, 20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	WR Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,15,19,20, 21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL*=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/ DOPID	8,15,19,20, 21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	V	V	CID3	

Ex. CC-46 (DDR5 Command Truth Table).

Ex. CC-46 (DDR5 Command Truth Table).

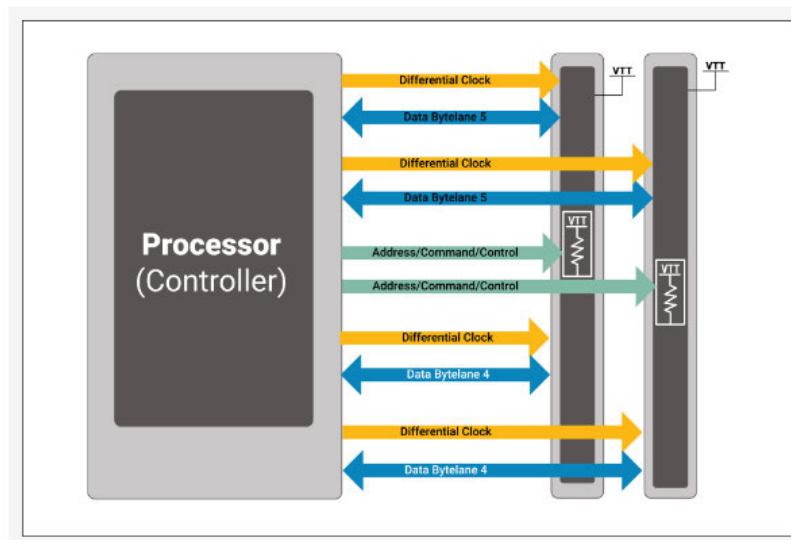
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Ex. CC-47 (DDR Memory and the Challenges in PCB Design).

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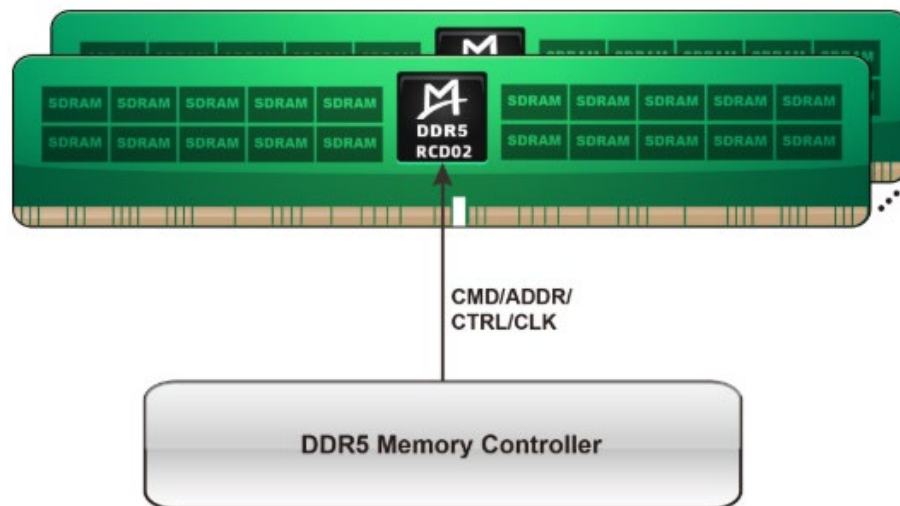
Bus topologies—On-board two-UDIMM



Ex. CC-47 (DDR Memory and the Challenges in PCB Design).

Ex. CC-48 (Montage M88DR5RCD01) (M88D5RCD01 is “a two-channel RCD designed for driving addresses and controlling nets between the host controller and the DDR5 SDRAMs”).

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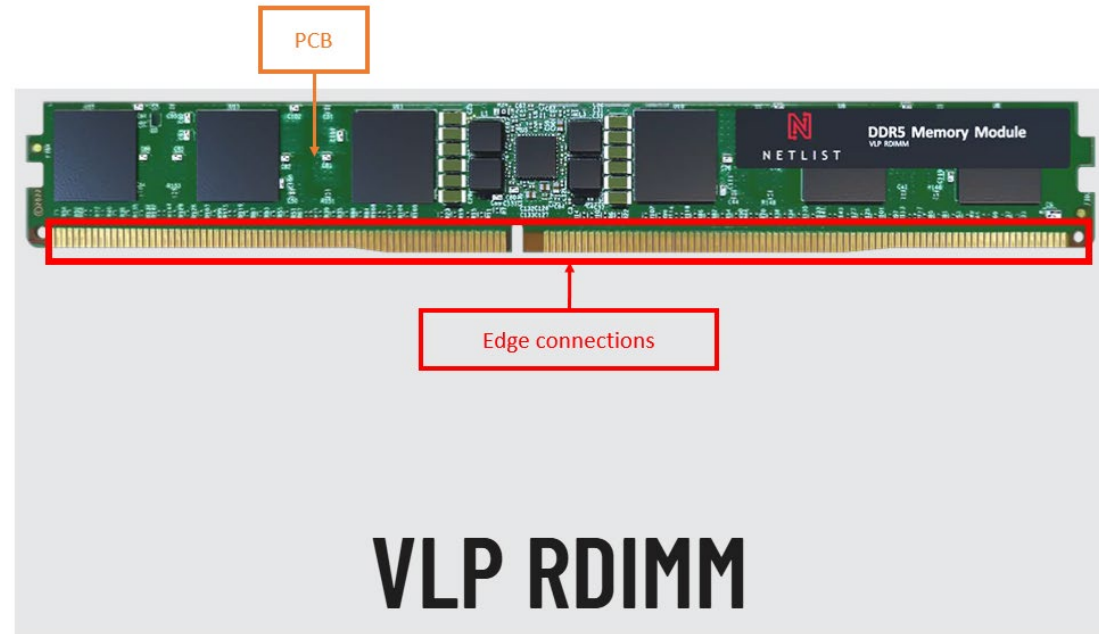
Ex. CC-43 (M88DR5RCD02 Flyer) at 2.

This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.

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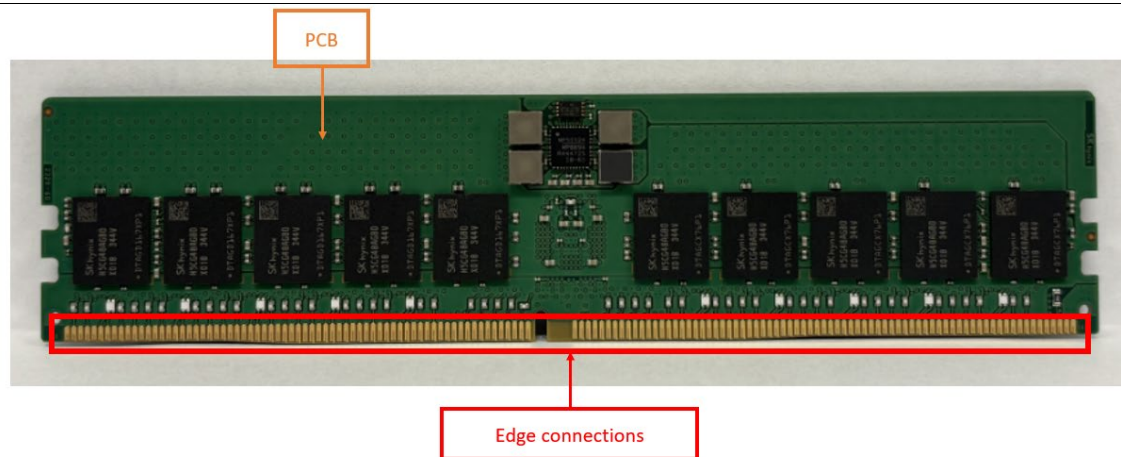
[1.b] [i] a printed circuit board comprising [ii] at least one connector [iii] configured to be operatively coupled to the memory bus, [iv] the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

The Practicing '731 Products each comprise [i] “a printed circuit board” (the green board on which the module components are mounted) comprising [ii] “at least one connector” [iv] “the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board” (e.g., as edge pins or gold fingers placed along the bottom edge).

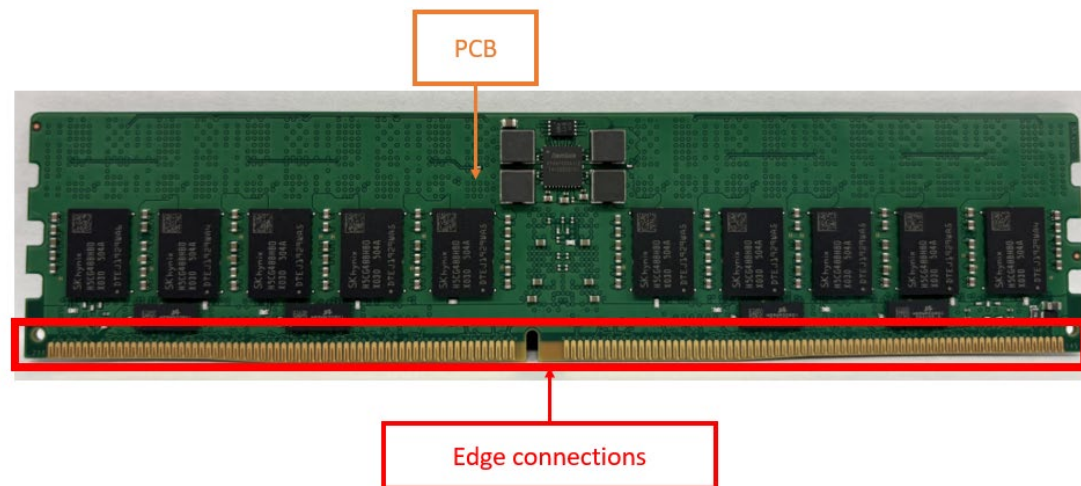


Ex. CC-33 (Netlist DDR5 DIMMs), annotated.

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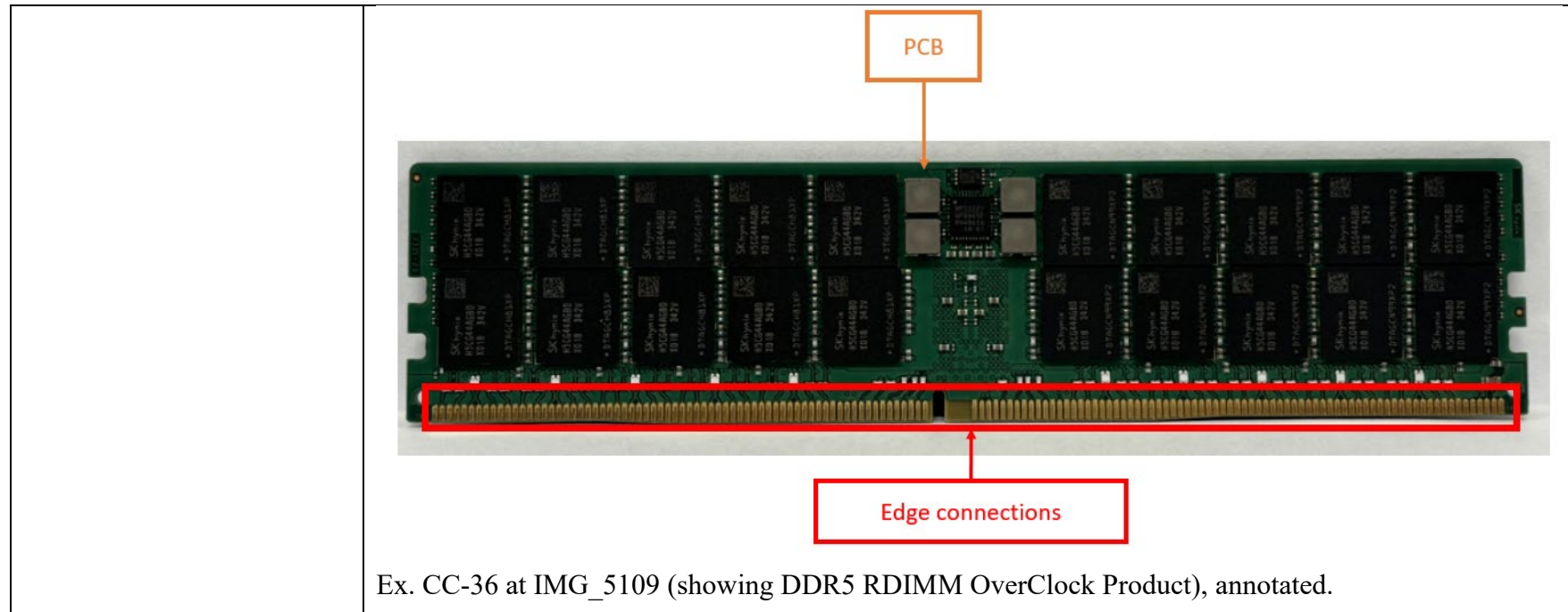


Ex. CC-36 at IMG_5117 (showing DDR5 RDIMM Product), annotated.

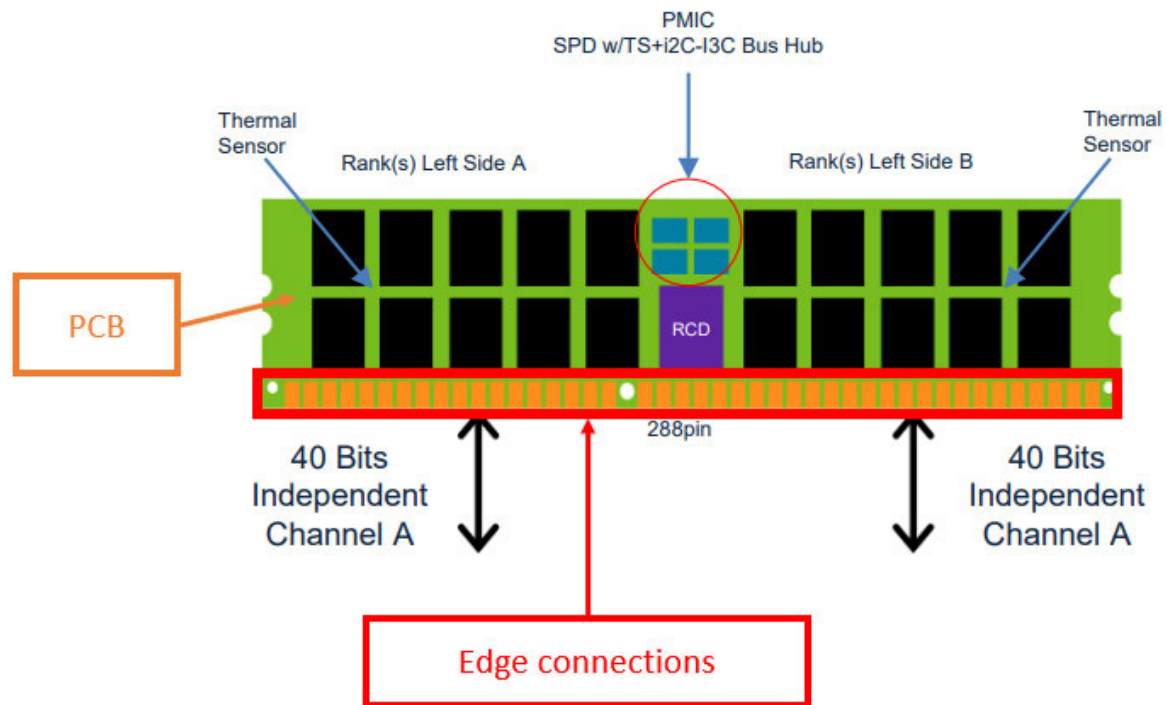


Ex. CC-36 at IMG_5126 (showing DDR5 MRDIMM Product), annotated.

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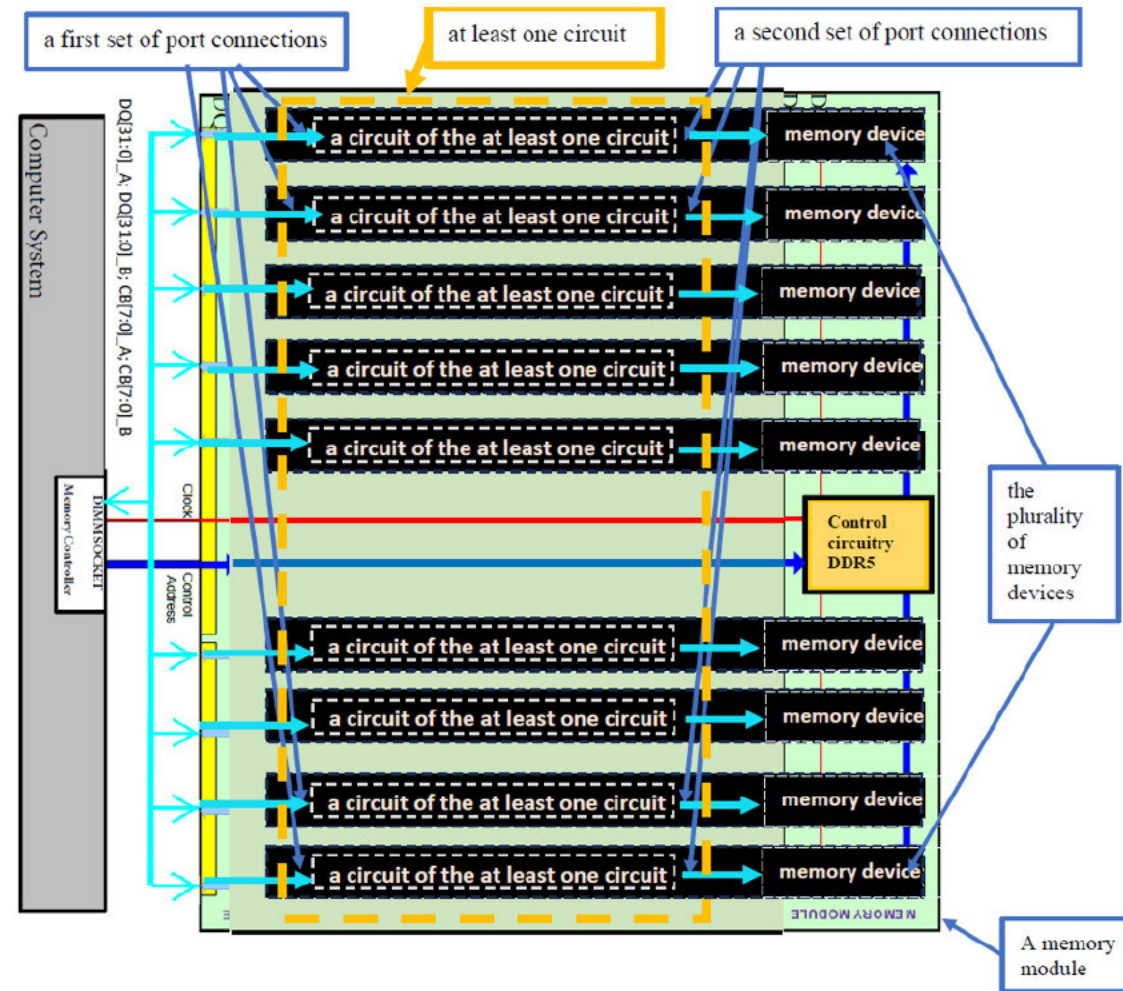


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Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1, annotated.

As an example, [REDACTED]
[REDACTED]



[REDACTED]

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[REDACTED]

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[REDACTED]

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	As another example, [REDACTED] [REDACTED] [REDACTED]
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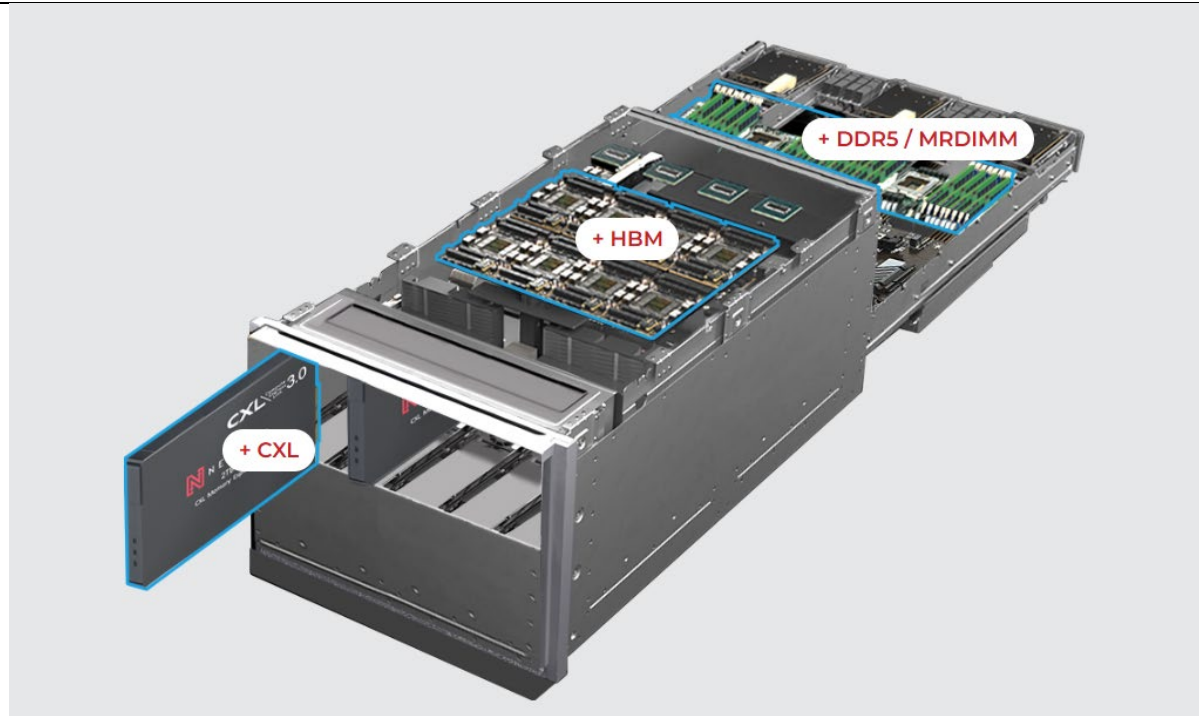
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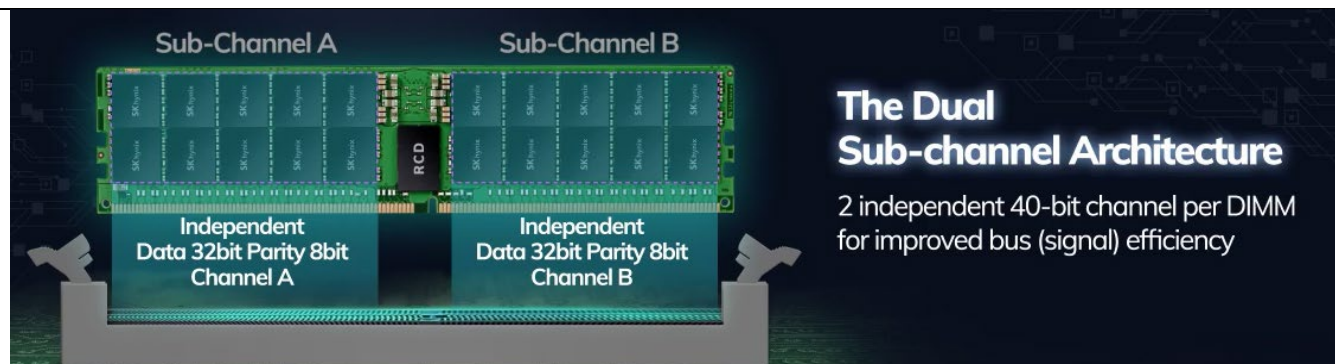
	A large solid black rectangular area used to redact a diagram or image. Ex. CC-39C (Netlist’s DDR5 Overclock RDIMM Datasheet Version 1.0) at 7–9. In the Practicing ’731 Products, the at least one connector is also [iii] “configured to be operatively coupled to the memory bus.” For example, as shown below, the edge connector is configured to be operatively coupled to the memory bus when inserting the module into a memory slot, as illustrated below.
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U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

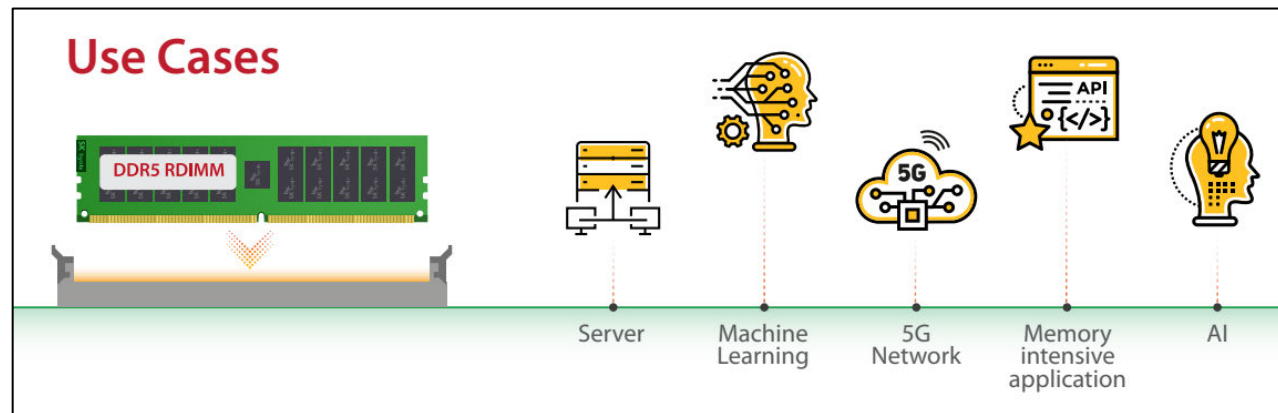


Ex. CC-49 (Netlist AI Server).

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart



Ex. CC-41 (SK hynix RDIMM).



Ex. CC-40 (SK hynix A New Standard) at 6.

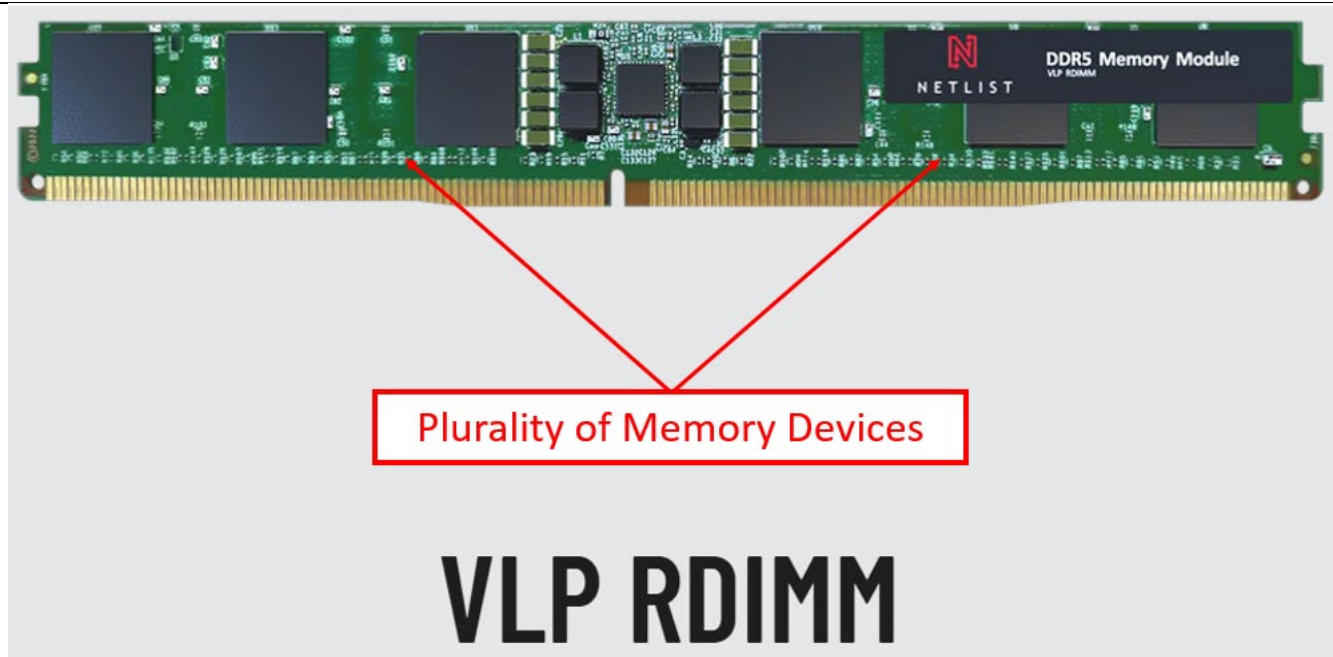
[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

	Ex. CC-50 (TrendForce News MRDIMM-MCRDIMMId). This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.
[1.c] [i] a plurality of memory devices on the printed circuit board, [ii] the	The Practicing '731 Products each have [i] “a plurality of memory devices” (e.g., memory cores of the DDR5 SDRAM chips) “on the printed circuit board.”

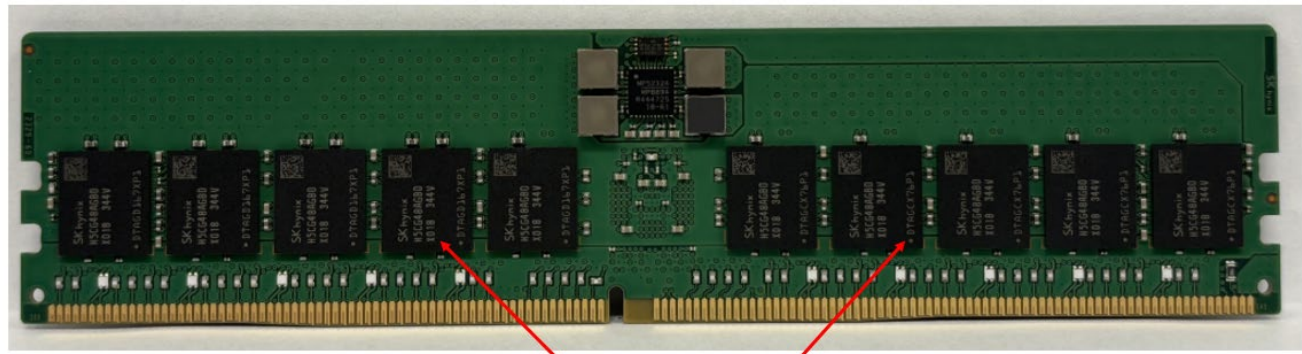
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

plurality of memory devices being arranged in multiple ranks, [iii] each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus;



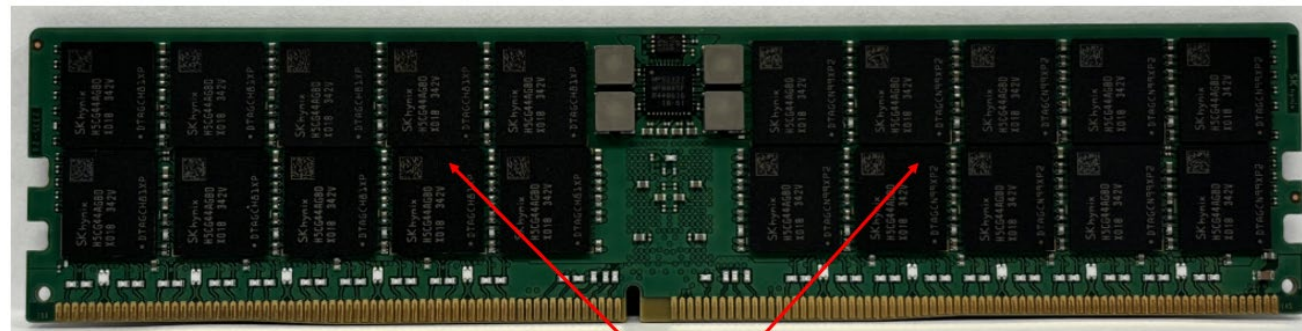
Ex. CC-33 (Netlist DDR5 DIMMs), annotated.

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Plurality of Memory Devices

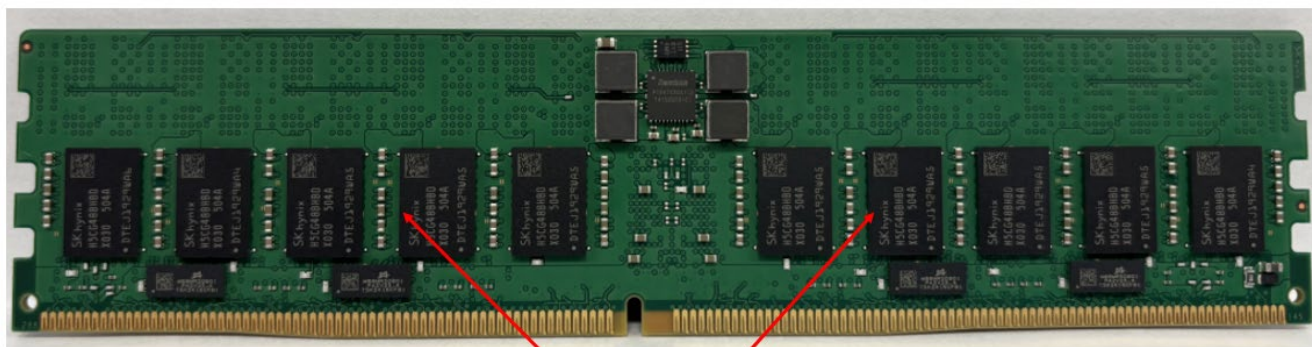
Ex. CC-36 at IMG_5117 (showing DDR5 RDIMM Product), annotated.



Plurality of Memory Devices

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Ex. CC-36 at IMG_5109 (showing DDR5 RDIMM OverClock Product), annotated.



Plurality of Memory Devices

Ex. CC-36 at IMG_5126 (showing DDR5 MRDIMM Product), annotated.

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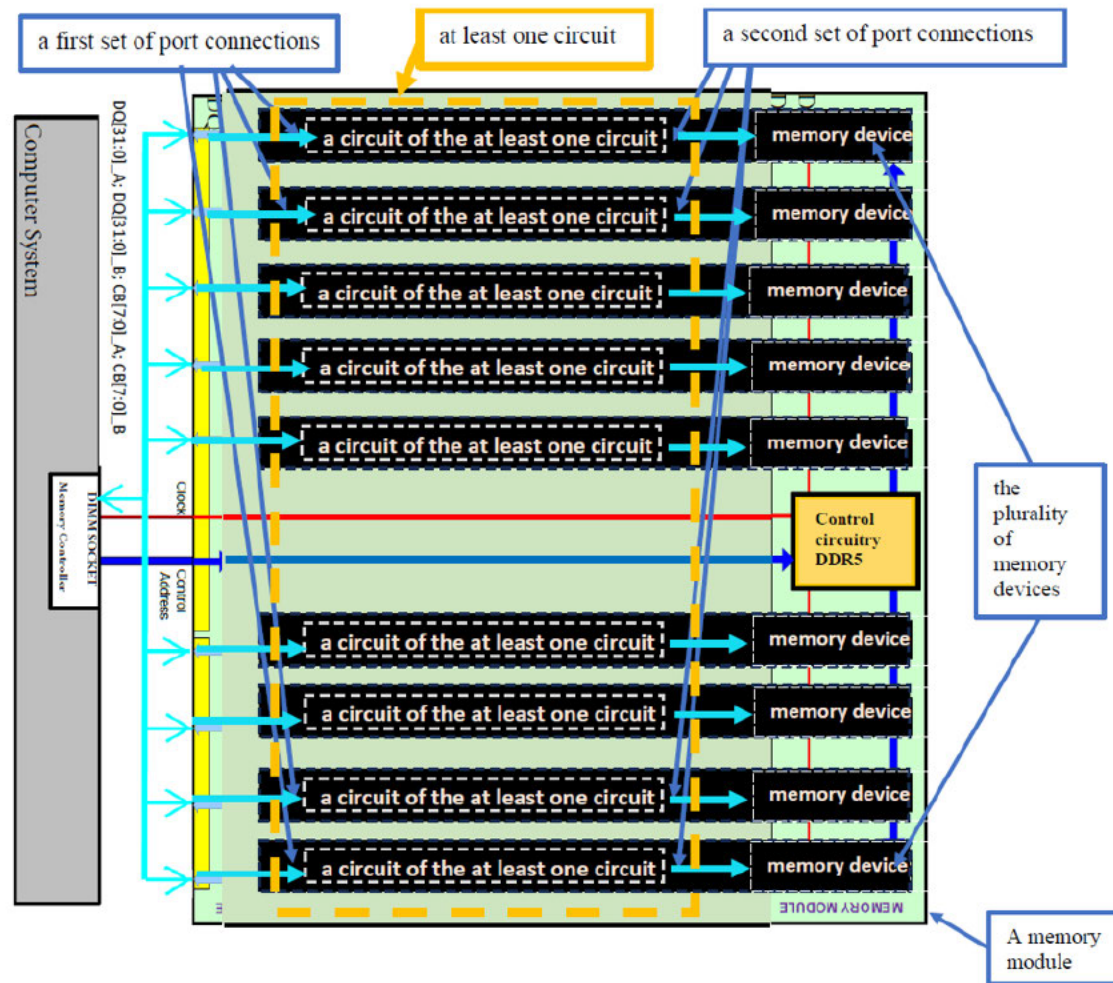
NETLIST DDR5 RDIMM Ordering Information

DIMM Type	Form Factor	Height	Density	Config	Number of DRAM	Speed (MT/s)	Hynix PN	Netlist PN	Description
Server RDIMM	LP	31.25	16GB	1Rx8	10	4800	HMCG78AEBRA	NLR4G0054G0748200HA	16GB 1Rx8 DDR5-4800-40-39-39 RDIMM SK Hynix A-Die SDP
						5600	HMCG78AGBRA	NLR4G0054G0756200HA	16GB 1Rx8 DDR5-5600-46-45-45 RDIMM SK Hynix A-Die SDP
			32GB	1Rx4	20	4800	HMCG84AEBRA	NLR5G0054G0348200HA	32GB 1Rx4 DDR5-4800-40-39-39 RDIMM SK Hynix A-Die SDP
						5600	HMCG84AGBRA	NLR5G0054G0356200HA	32GB 1Rx4 DDR5-5600-46-45-45 RDIMM SK Hynix A-Die SDP
			32GB	2Rx8	20	4800	HMCG88AEBRA	NLR5G0054G0748200HA	32GB 2Rx8 DDR5-4800-40-39-39 RDIMM SK Hynix A-Die SDP
						5600	HMCG88AGBRA	NLR5G0054G0756200HA	32GB 2Rx8 DDR5-5600-46-45-45 RDIMM SK Hynix A-Die SDP
			48GB	1Rx4	20	4800	HMCGY4MEBRB	NLRF00054G0348200HA	48GB 1Rx4 DDR5-4800-40-39-39 RDIMM SK Hynix M-Die SDP
						5600	HMCGY4MGBRB	NLRF00054G0356200HA	48GB 1Rx4 DDR5-5600-46-45-45 RDIMM SK Hynix M-Die SDP
			48GB	2Rx8	20	4800	HMCGY8MEBRB	NLRF00054G0748200HA	48GB 2Rx8 DDR5-4800-40-39-39 RDIMM SK Hynix M-Die SDP
						5600	HMCGY8MGBRB	NLRF00054G0756200HA	48GB 2Rx8 DDR5-5600-46-45-45 RDIMM SK Hynix M-Die SDP
			64GB	2Rx4	20	4800	HMCG94AEBRA	NLR6G0054G0348200HA	64GB 2Rx4 DDR5-4800-40-39-39 RDIMM SK Hynix A-Die SDP
						5600	HMCG94AGBRA	NLR6G0054G0356200HA	64GB 2Rx4 DDR5-5600-46-45-45 RDIMM SK Hynix A-Die SDP
			96GB	2Rx4	20	4800	HMCGM4MEBRB	NLRCG0054G0356200HA	96GB 2Rx4 DDR5-4800-40-39-39 RDIMM SK Hynix M-Die SDP
						5600	HMCGM4MGBRB	NLRCG0054G0356200HA	96GB 2Rx4 DDR5-5600-46-45-45 RDIMM SK Hynix M-Die SDP
			128GB	4Rx4	40	4800	HMCT04MEBRA129N	NLR7G0054G3848209HM	128GB 4Rx4 DDR5-4800-46-39-39 RDIMM SK Hynix M-Die 3DS-2H
						5600	HMCT04AGBRA129N**	NLR7G0054G3848209HA	128GB 4Rx4 DDR5-5600-52-45-45 RDIMM SK Hynix A-Die 3DS-2H
	VLP	18.75	16GB	1Rx8	10	4800	N*/A	NVR4G0054G0748201HA	16GB 1Rx8 DDR5-4800-40-39-39 RDIMM SK Hynix A-Die SDP
						5600	N*/A	NVR4G0054G0756202HA	16GB 1Rx8 DDR5-5600-46-45-45 RDIMM SK Hynix A-Die SDP
			32GB	2Rx8	20	4800	N*/A	NVR5G0054G0748201HA	32GB 2Rx8 DDR5-4800-40-39-39 RDIMM SK Hynix A-Die SDP
						5600	N*/A	NVR5G0054G0756202HA	32GB 2Rx8 DDR5-5600-46-45-45 RDIMM SK Hynix A-Die SDP
			48GB	2Rx8	20	4800	N*/A	NVRFG0054G0748201HA	48GB 2Rx8 DDR5-4800-46-45-45 RDIMM SK Hynix M-Die SDP
						5600	N*/A	NVRFG0054G0756202HA	48GB 2Rx8 DDR5-5600-46-45-45 RDIMM SK Hynix M-Die SDP

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 2, annotated (showing multiple DRAMs).

Shown below, a plurality of memory devices are on the green printed circuit board.

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As an example,

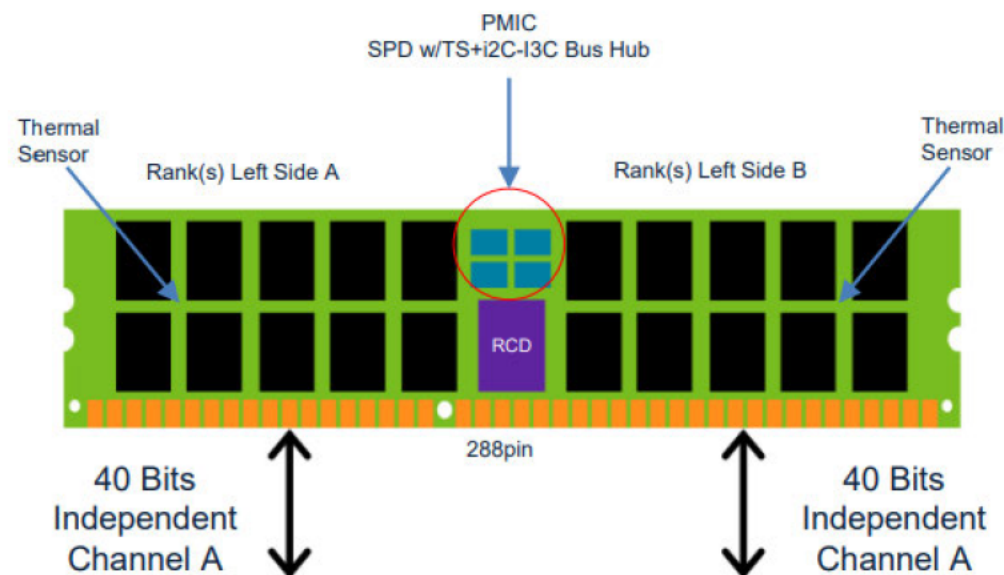
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 1.

Additionally,

Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1.

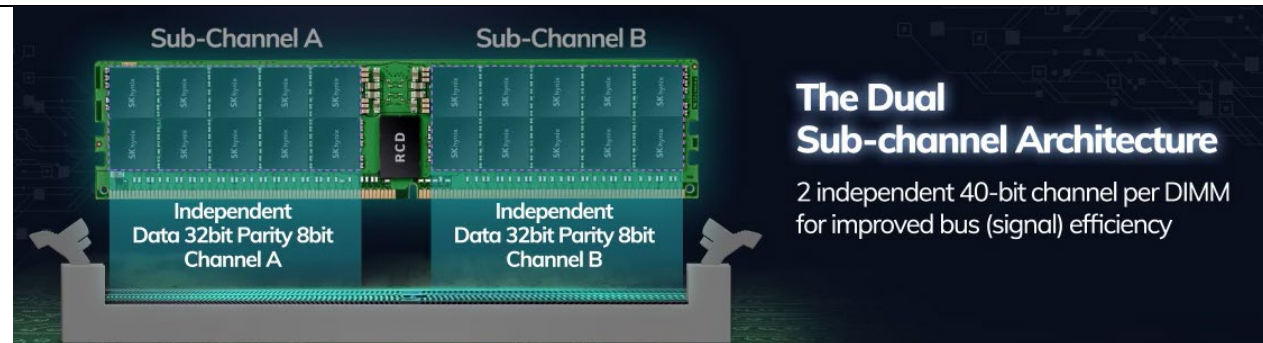
In the Practicing '731 Products, [ii] “the plurality of memory devices” (e.g., memory cores of the SDRAM chips) are “arranged in multiple ranks.”



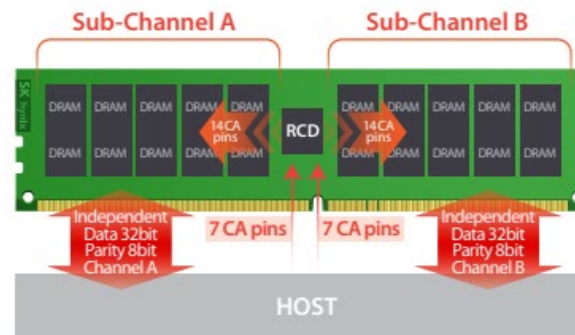
Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

For RDIMMs, SK hynix shows two independent channels, channel A and channel B, each channel corresponding to a plurality of DDR5 DRAMs.

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart



Ex. CC-41 (SK hynix RDIMM).



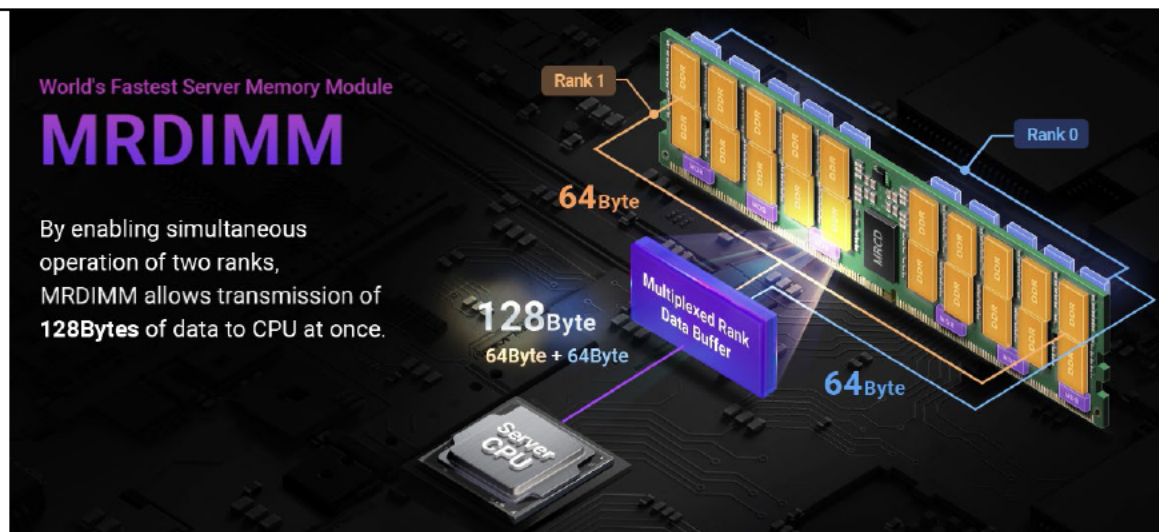
| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK hynix A New Standard) at 5.

For MRDIMMs, SK hynix's website states: "SK Hynix has developed MRDIMM (*Multiplexed Rank Dual In-line Memory Modules*)" and "facilitating simultaneous operation of *two ranks*." Ex. CC-42 (SK hynix MRDIMM) at 2–3. And, SK hynix shows two ranks, Rank 0 and Rank 1, each comprised of a plurality of DRAMs on DDR5 MRDIMMs.

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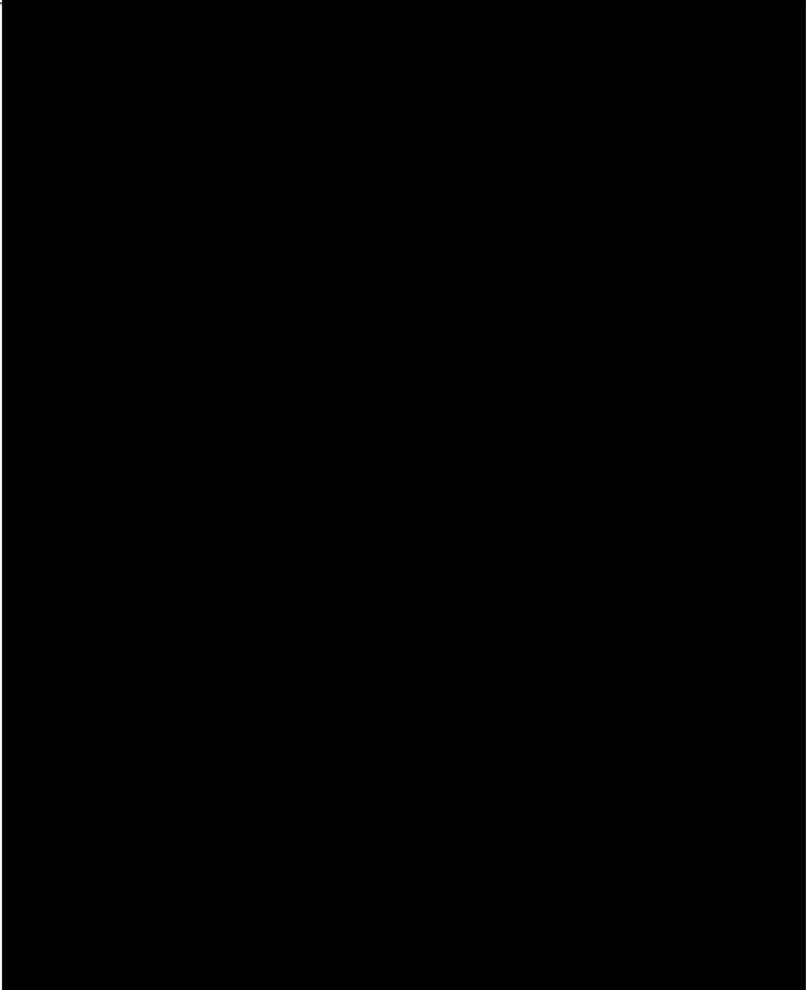


Ex. CC-42 (SK hynix MRDIMM) at 4.

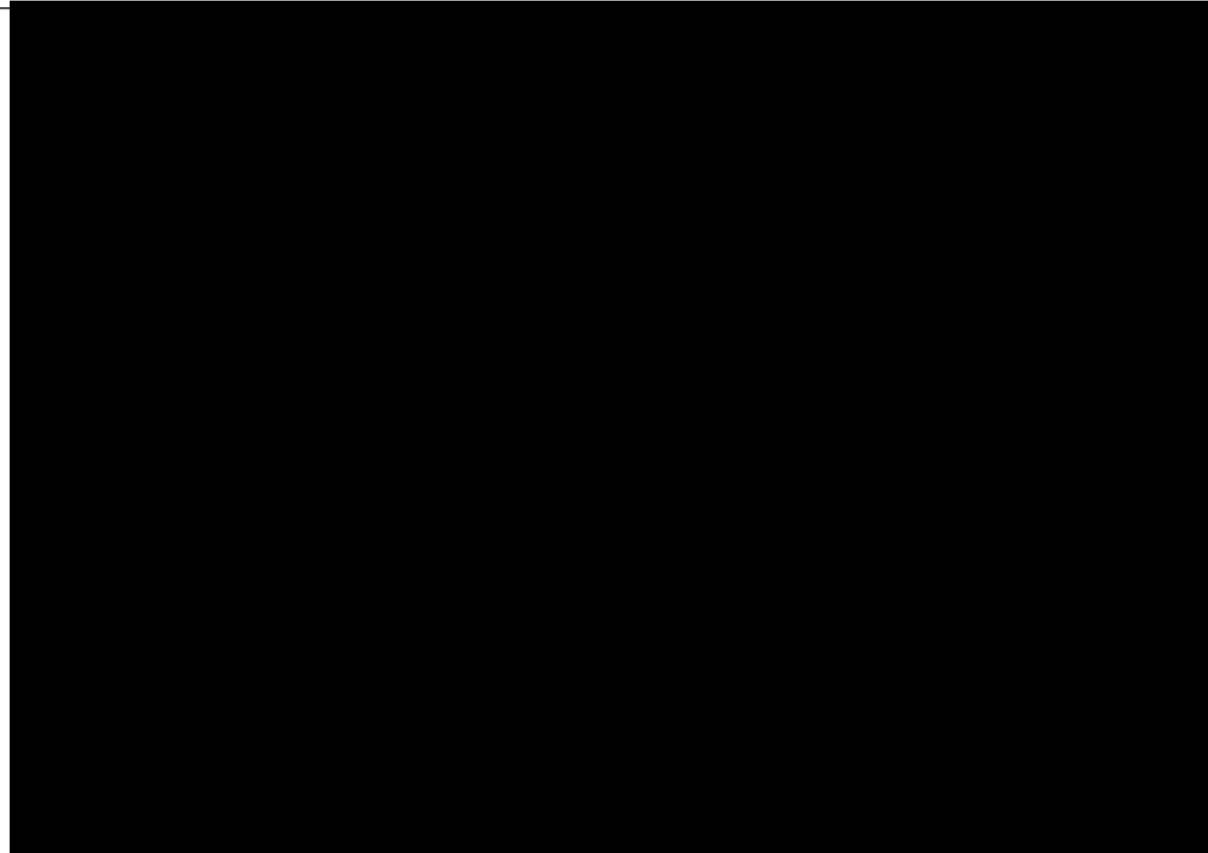
As another example,



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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9, annotated.		

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Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 10, annotated.

As another example,



[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

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[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

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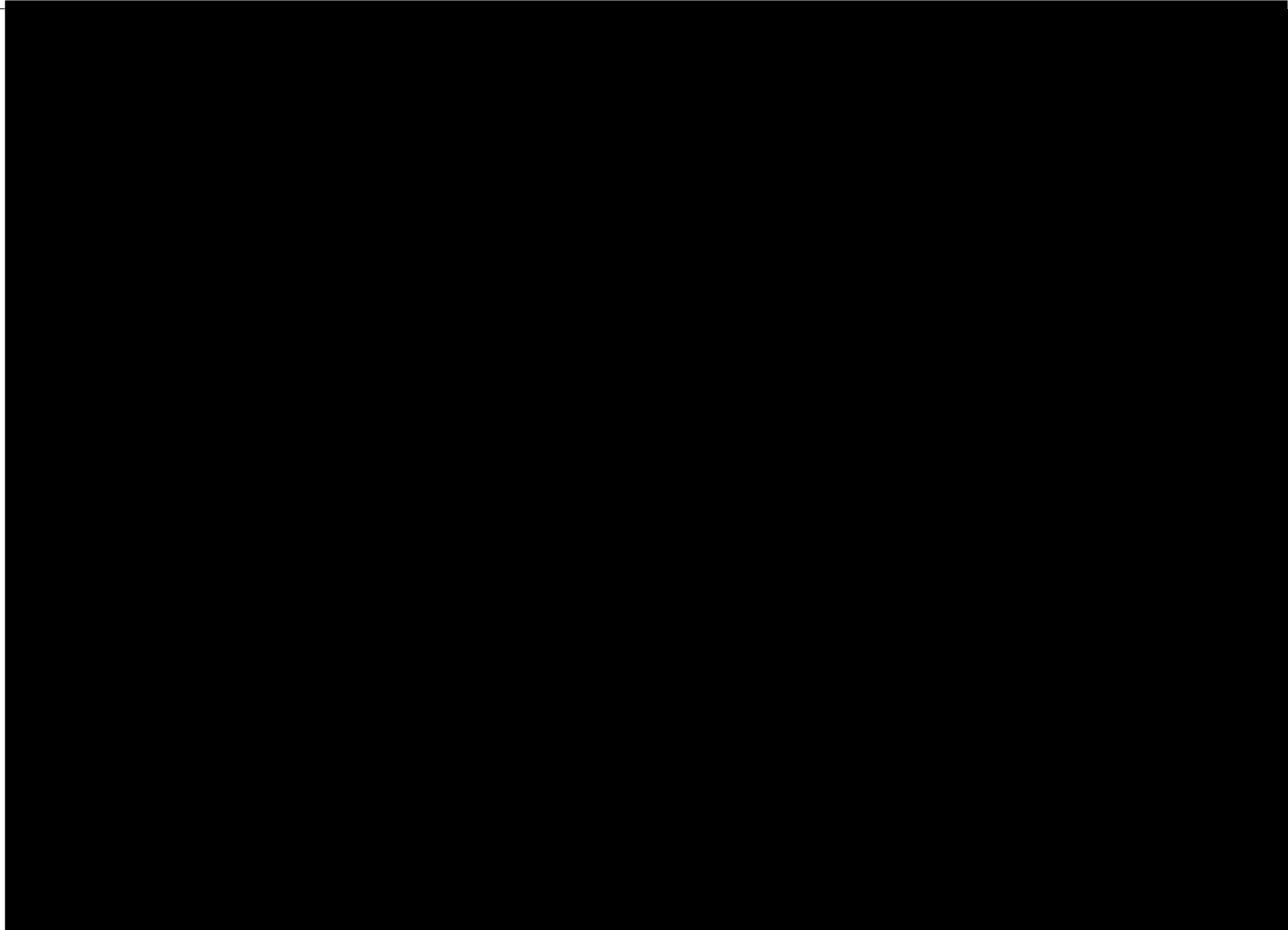
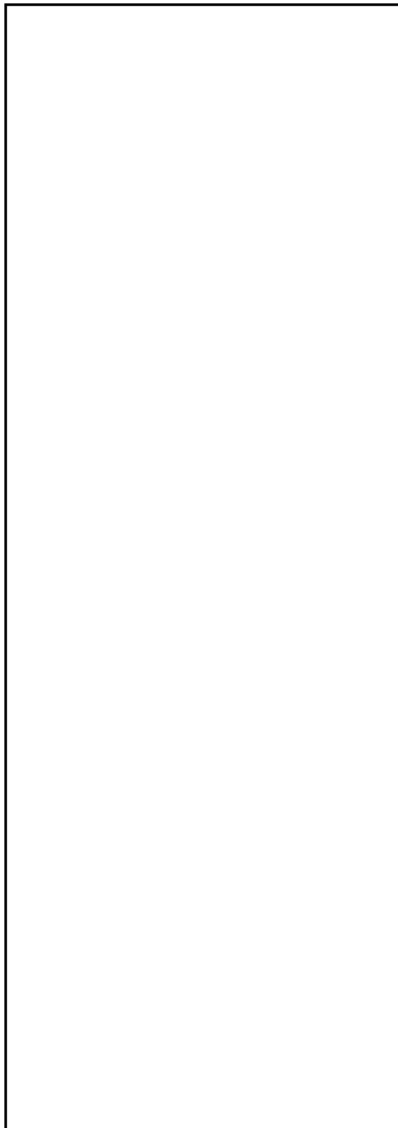
[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

	[REDACTED] Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10–12. In the Practicing '731 Products, [REDACTED]
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[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7.

[REDACTED]

[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

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U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

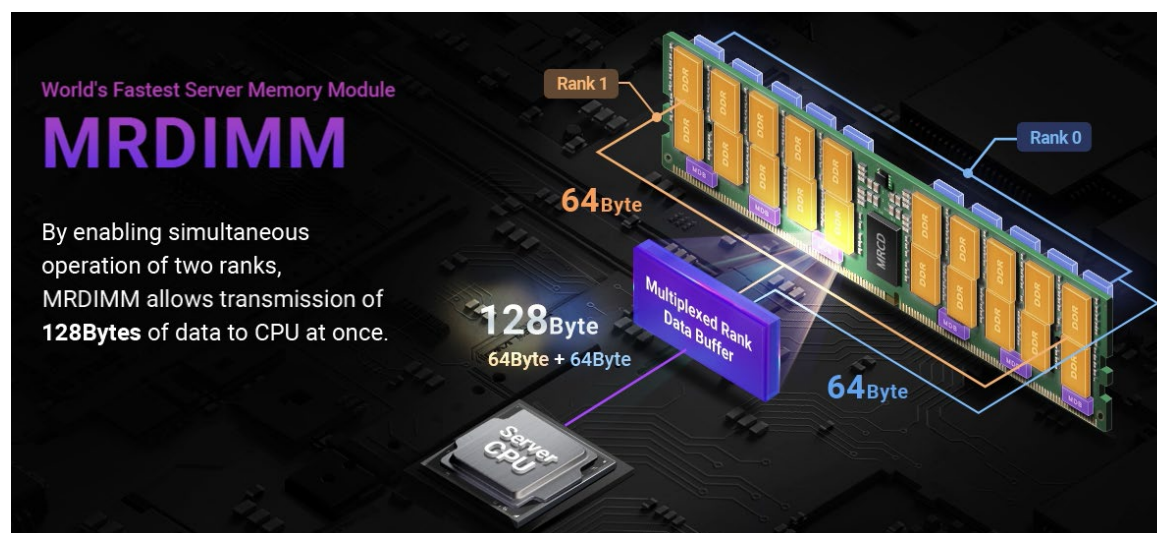
	A large solid black rectangular area used to redact the main content of the table cell. Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4–5. Indeed, the industry understands that DDR5 modules use multiple ranks, each comprising an independent set of memory devices. <i>See, e.g.</i>, Ex. CC-47 (DDR Memory and the Challenges in PCB
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[REDACTED]

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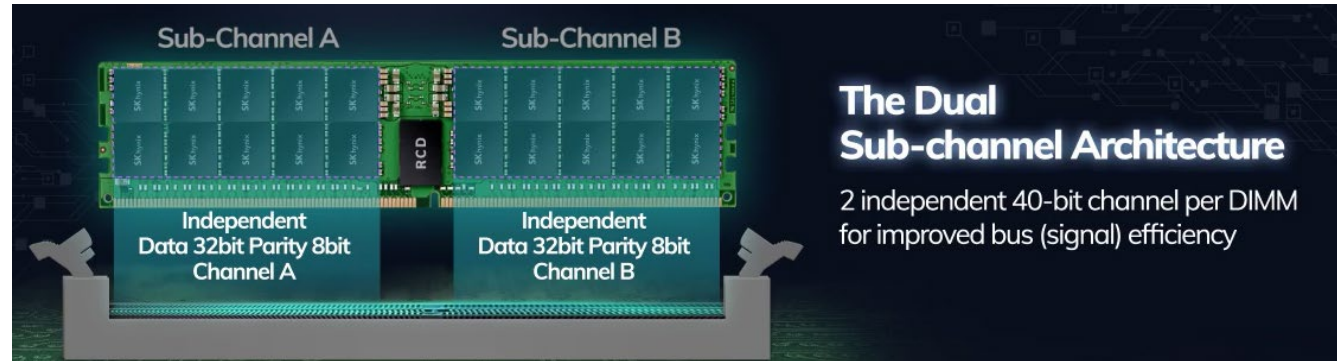
Design) (“In DDR5, each DIMM will have two 40-bit channels (32 data bits and 8 ECC bits). While the data width is the same (64-bit total) having two smaller independent channels improves DDR memory access efficiency. The benefit of this is higher MT/s is amplified by greater efficiency.”); Ex. CC-12 (M88MR5RCD01 Flyer) at 1 (“Two sub channels, each divided into two pseudo-channels to increase the total bandwidth of the host system.”); Ex. CC-43 (M88DR5RCD02 Flyer) (“Two-channel”); Ex. CC-43 (M88DR5RCD02 Flyer) (“the RCD02 contains two separate channels”); Ex. CC-48 (Montage M88DR5RCD01) (“It is a two-channel RCD”); Ex. CC-51 (Gen 2 RCD Renesas RG5R256) (The RCD “contains two separate channels with some common logic such as clocking, but otherwise operate independently of each other”); Ex. CC-52 (Gen 1 RCD Renesas 5RCD0148) (same).

In the Practicing ’731 Products, [iii] “each rank compris[es] an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus.” Shown immediately above, each rank comprises an independent set of memory devices (e.g., memory cores of the DDR5 SDRAM chips). Also shown above for (1.a)[i], the system memory controller is operable to access the DDR5 SDRAM chips. For example, the system memory controller can access each rank to access a full bit-width of the memory bus.

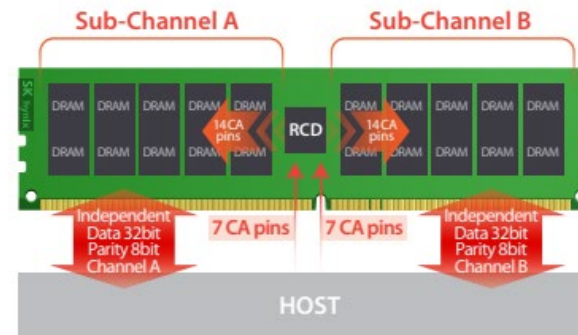


U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

Ex. CC-42 (SK hynix MRDIMM) at 4.



Ex. CC-41 (SK hynix RDIMM).



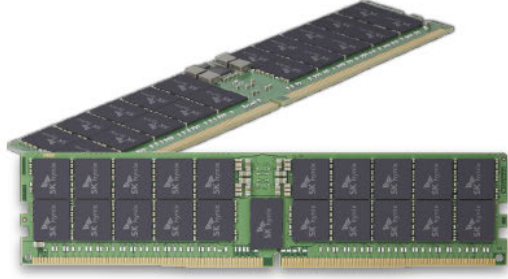
| New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK hynix A New Standard) at 5.

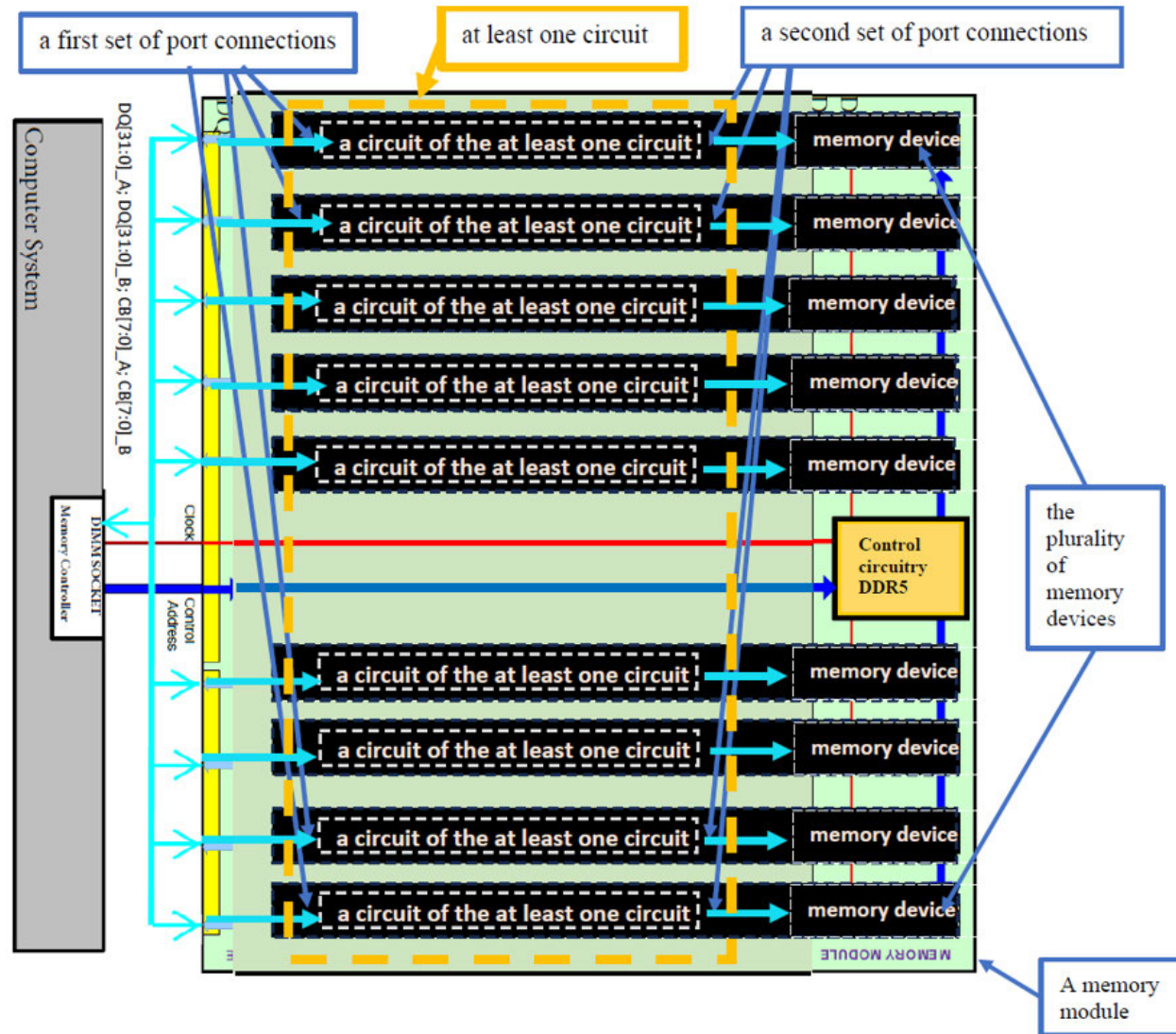
[REDACTED]

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	 DDR5 RDIMM  <table border="1" data-bbox="1119 277 1833 662"> <tr> <td>Module Density</td><td>16GB, 24GB, 32GB, 48GB, 64GB, 96GB, 128GB, 256GB</td></tr> <tr> <td>Configuration</td><td>2Gx80(16GB), 3Gx80(24GB), 4Gx80(32GB), 6Gx80(48GB), 8Gx80(64GB), 12Gx80(96GB), 16Gx80(128GB), 32Gx80(256GB), 4Gx72(32GB), 6Gx72(48GB), 8Gx72(64GB), 12Gx72(96GB)</td></tr> <tr> <td>Package Material</td><td>Lead&Halogen Free</td></tr> <tr> <td>Package Type</td><td>FCBGA</td></tr> <tr> <td>Data Rate</td><td>4,800-6,400Mbps and higher</td></tr> <tr> <td>Data Width</td><td>X80(ECC X36), X72(ECC X8)</td></tr> </table> Ex. CC-40 (SK hynix A New Standard) at 6, annotated. This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.	Module Density	16GB, 24GB, 32GB, 48GB, 64GB, 96GB, 128GB, 256GB	Configuration	2Gx80(16GB), 3Gx80(24GB), 4Gx80(32GB), 6Gx80(48GB), 8Gx80(64GB), 12Gx80(96GB), 16Gx80(128GB), 32Gx80(256GB), 4Gx72(32GB), 6Gx72(48GB), 8Gx72(64GB), 12Gx72(96GB)	Package Material	Lead&Halogen Free	Package Type	FCBGA	Data Rate	4,800-6,400Mbps and higher	Data Width	X80(ECC X36), X72(ECC X8)
Module Density	16GB, 24GB, 32GB, 48GB, 64GB, 96GB, 128GB, 256GB												
Configuration	2Gx80(16GB), 3Gx80(24GB), 4Gx80(32GB), 6Gx80(48GB), 8Gx80(64GB), 12Gx80(96GB), 16Gx80(128GB), 32Gx80(256GB), 4Gx72(32GB), 6Gx72(48GB), 8Gx72(64GB), 12Gx72(96GB)												
Package Material	Lead&Halogen Free												
Package Type	FCBGA												
Data Rate	4,800-6,400Mbps and higher												
Data Width	X80(ECC X36), X72(ECC X8)												
[1.d] [i] at least one circuit coupled between the at least one connector and the plurality of memory devices, [ii] the at least one circuit comprising a first set of port connections coupled to respective ones of the plurality of edge connections and a second set of port connections coupled to the plurality of	The Practicing '731 Products have [i] “at least one circuit coupled between the at least one connector and the plurality of memory devices.” For example, the Practicing '731 Products have a collection of DFE and ODT/RTT circuits coupled between the edge connectors and the plurality of memory devices (e.g., the memory cores of the DDR5 SDRAM chips). The representative illustration depicts the at least one circuit between the at least one connector and the plurality of memory devices, below.												

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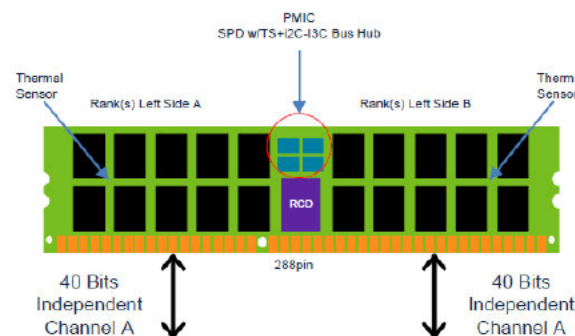
memory devices,



[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

supported via registering clock drivers, data buffers, and DRAMs.



DIMM Changes from DDR4 to DDR5

Feature	DDR4	DDR5
Frequency (GT/s)	1.6~3.2	4.8~5.6
Voltage (V)	1.2	1.1
Power Supply	PMIC on MB	PMIC on DIMM
Temp sensor device	1 per DIMM	2 or 3 per DIMM
Side bus protocol	I2C bus	Legacy SMBus, I2C, I3C
Sideband bus access	Direct Access	1xC Hub in SPD
Channel Architecture	72-bit data channel 64 data + 8 ECC 1 channel per DIMM	40-bit data channel 32 data + 8 ECC 2 channel per DIMM
DFE Support	DB Only	RCD, DB, and DRAM

Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1, annotated; *see also id.* at 2 (DFE listed as a feature).

For another example, [REDACTED]

[REDACTED]

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

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[REDACTED]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 17, 19, 22, 24. As another example, x. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 1.
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[REDACTED]

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DDR5 Ultra Low Latency RDIMM Features

Features	Description	Features	Description
Data Rate (Mbps)	5600~6800	Sideband bus access	IxC Hub in SPD
Voltage (VDDQ)	1.1 (Native)	Channel Architecture	40-bit data channel, 32 data+ 8 ECC 2 channel per DIMM
Clock Driver	@6400Mbps	DFE Support	RCD and DRAM
Power Supply	PMIC on DIMM	PCB Mechanical Outline	JEDEC (Height 31.25mm)
Temp sensor device	2 or 3 per DIMM	Environmental	0°C – 95°C, RoHS Compliant and Halogen-Free
Side bus protocol	Legacy SMBus, I2C, I3C		

Ex. CC-56 (Netlist Ultra Low Latency DDR5 RDIMM Product Brief) at 1.

DDR5 RDIMM Features

Features	Description
Frequency (GT/s)	4.8~6.4
Voltage (VDDQ)	1.1 (Native) 1.4V (XMP)
Power Supply	PMIC on DIMM
Temp sensor device	2 or 3 per DIMM
Side bus protocol	Legacy SMBus, I2C, I3C
Sideband bus access	I°C Hub in SPD
Channel Architecture	40-bit data channel, 32 data+ 8 ECC 2 channel per DIMM
DFE Support	RCD and DRAM
PCB Mechanical Outline	JEDEC (Height 31.25mm)
Environmental	0°C – 95°C, RoHS Compliant and Halogen-Free

[REDACTED]

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Ex. CC-57 (Netlist Lightning DDR5 High Performance Module Product Brief) at 1 annotated.

In addition, each SDRAM includes on-die termination (“ODT”) circuitry, since “ODT is a crucial component for high-speed DDR5 memory systems since it aids in enhancing signal quality, decreasing signal ringing, and eventually allowing for higher data transfer speeds with less signal deterioration.”

Ex. CC-10 (Intellect Partners DD5's Secret Weapon: ODT) (DDR5's Secret Weapon: On-Die Termination). Furthermore, ODT is not only crucial but also a mandatory requirement by the JEDEC standard. Ex. CC-05 (JESD79-5C.01_v1.31) at 313 (stating that “On-Die Termination (ODT) is a feature of the DDR5 SDRAM”):

5.1 On-Die Termination for DQ, DQS, DM, and TDQS

On-Die Termination (ODT) is a feature of the DDR5 SDRAM that allows the DRAM to change termination resistance for each DQ, DQS, DM and TDQS pin. Unlike previous DDR technologies, DDR5 no longer has a physical ODT pin and all ODT based control is now command & mode register based. For x4 and x8 configuration devices, ODT is applied to each DQ, DQS_t, DQS_c and DM_n, as well as TDQS_t, TDQS_c for the x8 configuration. For x16 configuration devices, ODT is applied to each DQU, DQL, DQSU_t, DQSU_c, DQSL_t, DQSL_c, DMU_n and DML_n. ODT is enabled via non-target Read/Mode Register Read commands, Write and non-target Write commands, or is set to the default “PARK” value through settings in MR33-MR35.

The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to independently change termination resistance for any or all DRAM devices. In addition to the control capability of the DQ ODT, the DQS ODT will now be independently programmed via MR33:OP[5:3] and held static. All ODT control will be targeted for the DQs. This addition allows for adjusting the delay common in an unmatched architecture. DQS RTT offset control mode is enabled via MR40:OP[2:0].

The ODT feature is turned off and not supported in Self-Refresh mode, but does have an optional mode when in Power Down. A simple functional representation of the DRAM ODT feature is shown in **Figure 184**.

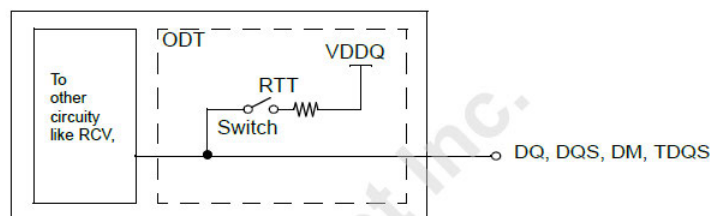


Figure 184 — Functional Representation of ODT

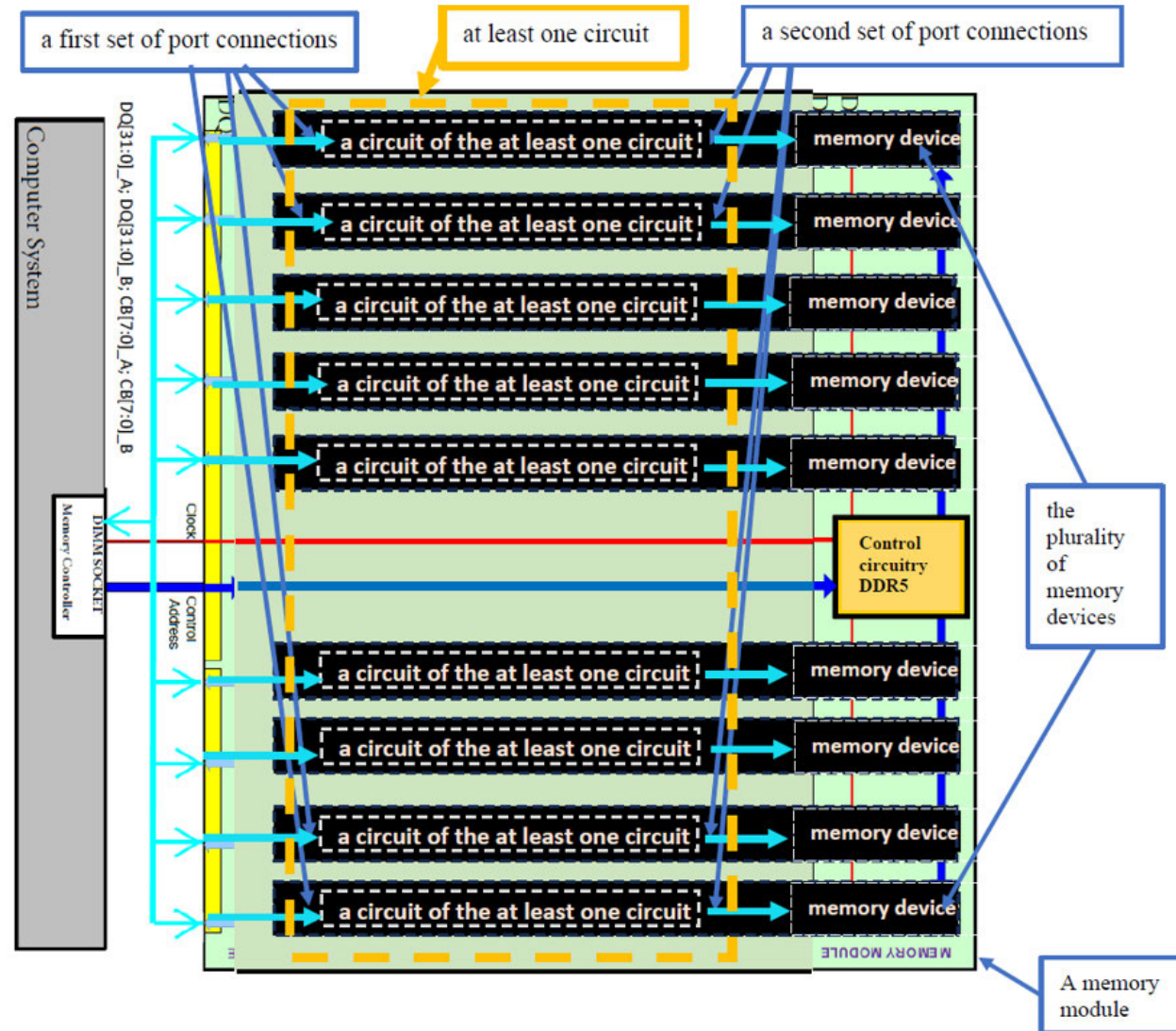
The switch is enabled by the internal ODT control logic, which uses command decode, Mode Register Settings and other control information, see below. The value of RTT is determined by the settings of Mode Register bits.

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	Ex. CC-55 (JEDEC JESD79-5B_v1.20) at 314. Each SDRAM of the Practicing '731 Products includes ODT circuitry based at least on the products being JEDEC compliant. <i>See</i> Ex. CC-33 (Netlist DDR5 DIMMs); Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 2; Ex. CC-37 (Netlist Lightning DDR5 DIMM) (Netlist DDR5 Lightning RDIMMs being JEDEC compliant and “utilizing JEDEC-standard PCB, DRAM, RCD, PMIC, and clock components”); Ex. CC-56 (Netlist Ultra Low Latency DDR5 RDIMM Product Brief) at 1–2 (advertising Netlist’s DDR5 RDIMM Ultra Low Latency Products as JEDEC RDIMM); Ex. CC-57 (Netlist Lightning DDR5 High Performance Module Product Brief) at 1–2 (Netlist’s DDR5 RDIMM OverClock Products having default JEDEC-compliant configurations and XMP profiles built upon the JEDEC standard). To the extent the whole or part of the DFE and/or ODT/RTT circuitry resides outside the SDRAM chips (e.g., in multiplexing data buffers (MDBs) of the MRDIMM Products),¹ that location would still be coupled between the at least one connector (e.g., the edge connectors on the printed circuit board) and the plurality of memory devices (e.g., the memory cores of the DDR5 SDRAM chips). Specifically, if the DFE and/or ODT/RTT circuitry is in MDBs or some other circuit external to the SDRAM chips, that circuit would be coupled between the PCB’s edge connectors and the SDRAM chips. In this configuration, the memory cores of the DDR5 SDRAM chips <i>or</i> the SDRAM chip packages as a whole would be the claimed “memory device.” In the Practicing '731 Products, [ii] “the at least one circuit” (e.g., the DFE and ODT/RTT circuits) “compris[es] a first set of port connections coupled to respective ones of the plurality of edge connections and a second set of port connections coupled to the plurality of memory devices.” As described above and shown below, the circuitry comprises a first set of port connections (e.g., data ports on the host side) coupled to the edge connections of the printed circuit board and a second set of port connections (e.g., data ports on the DRAM side) coupled to the DDR5 SDRAM memory cores.
--	--

¹ Montage, a supplier of MDBs, reported: “We have introduced multiple innovative circuits and algorithms to address the signal integrity challenges of DDR5’s high-speed parallel bus. On the circuit level, we developed high-speed, low-noise transceivers, supplemented by multi-tap decision feedback equalization (DFE) to compensate for far-end crosstalk and inter-symbol interference.” Ex. CC-71 (Montage HK Proof) at 180.

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If the DFE and/or ODT/RTT circuitry resides outside the SDRAM chips (e.g., in multiplexing data

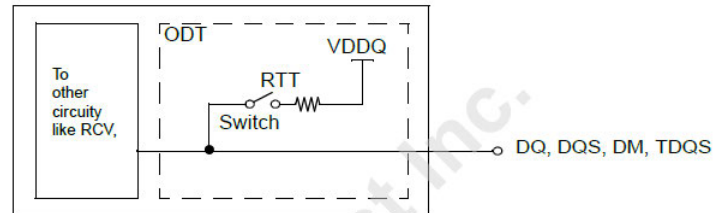
[REDACTED]

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	buffers (MDBs) of the MRDIMM Products), that configuration would also include a first set of port connections (e.g., data ports on the host side) coupled to the edge connections and a second set of port connections (e.g., data ports on the DRAM side) coupled to the DDR5 SDRAM chips. This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.
[1.e] each circuit of the at least one circuit further comprising a set of correction circuits, each correction circuit of the set of correction circuits being configured to make corrections in one or more signals transmitted between a respective port connection of the first set of port connections and a corresponding port connection in the second set of port connections,	In the Practicing '731 Products, each circuit of the at least one circuit (e.g., the DFE and ODT/RTT circuitry) further comprises “a set of correction circuits, each correction circuit of the set of correction circuits being configured to make corrections in one or more signals transmitted between a respective port connection of the first set of port connections and a corresponding port connection in the second set of port connections.” As explained for (1.d), each circuit of the [REDACTED] [REDACTED] [REDACTED] [REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 19.

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Additionally, the ODT/RTT circuitry improves signal integrity: “The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to independently change termination resistance for any or all DRAM devices.” Ex. CC-55 (JEDEC JESD79-5B_v1.20) at 314. For example, it corrects timing: “RTT_WR: The rank that is being written to provide termination and *adjusts timing* based on ODT Control Mode Register.” Ex. CC-55 (JEDEC JESD79-5B_v1.20) at 315.

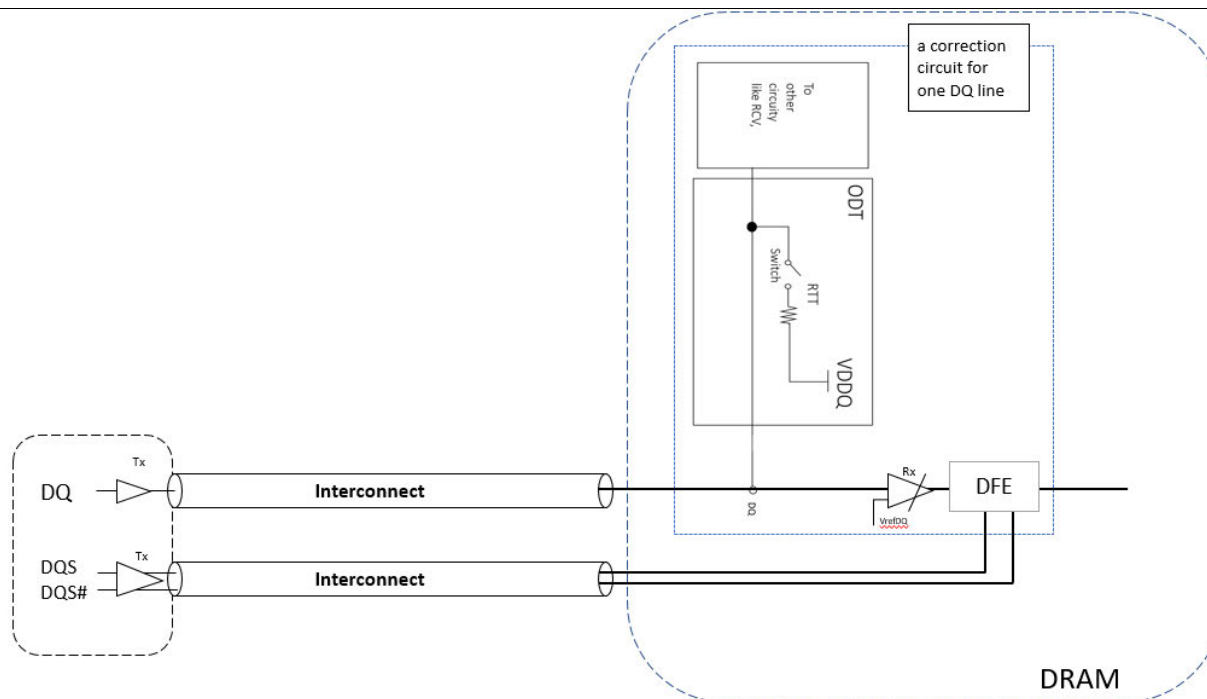


Ex. CC-55 (JEDEC JESD79-5B_v1.20) at 314.

Shown below, the DFE and ODT/RTT circuitry are collectively a “correction circuit.”

[REDACTED]

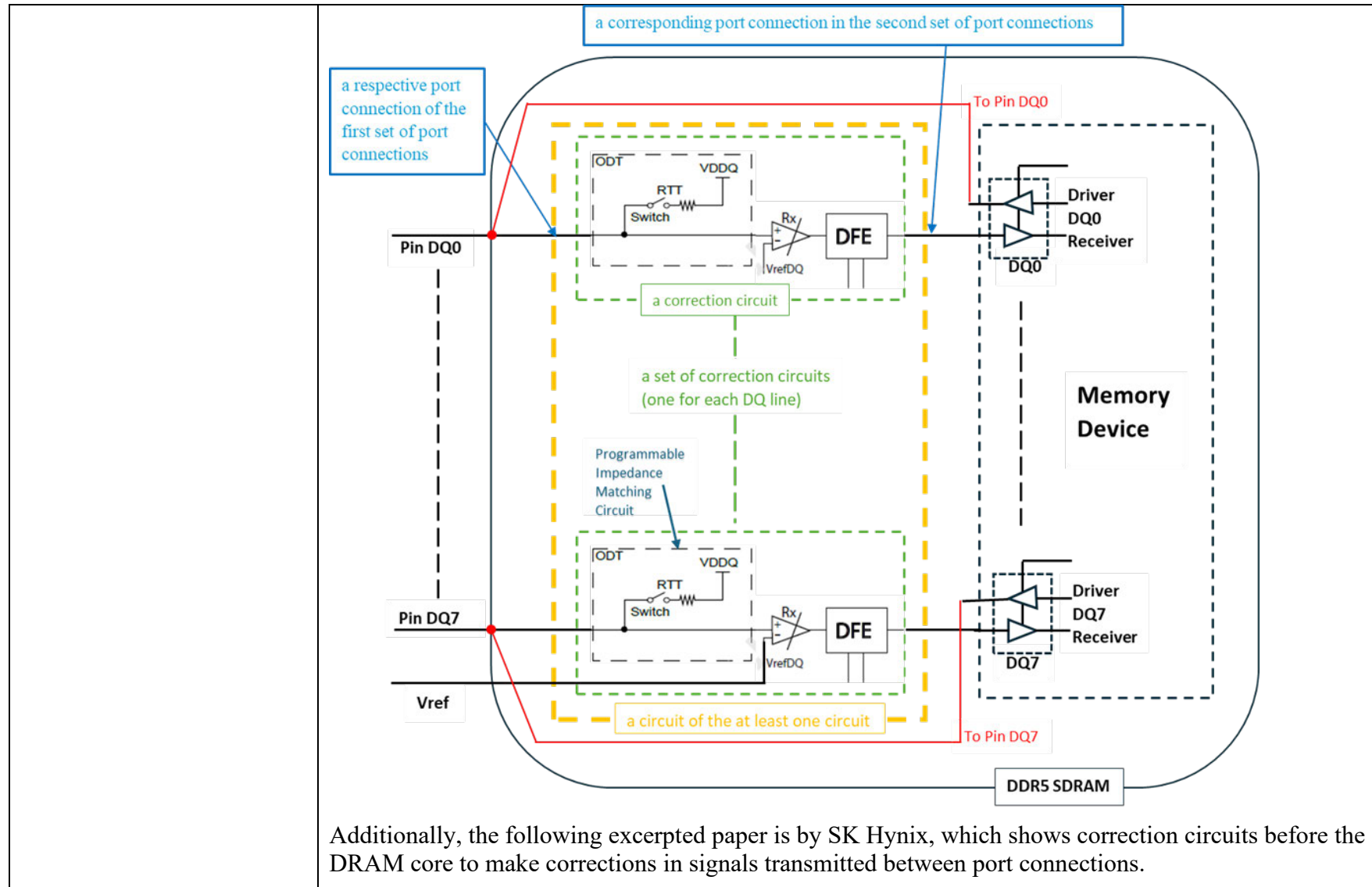
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See also Ex. CC-56 (Netlist Ultra Low Latency DDR5 RDIMM Product Brief) at 2 (x4 and x8 configurations); Ex. CC-57 (Netlist Lightning DDR5 High Performance Module Product Brief) at 1-2 (x4 and x8 configurations); Ex. CC-40 (SK hynix A New Standard) at 3 (x4 and x8 configurations); Ex. CC-41 (SK hynix RDIMM) (x4 and x8 configurations).

Additionally, as explained for (1.d) and shown below, the DFE and ODT/RTT circuits are located between a respective port connection of the first set of port connections and a corresponding port connection in the second set of port connections. They would therefore “make corrections in one or more signals transmitted between these connections.”

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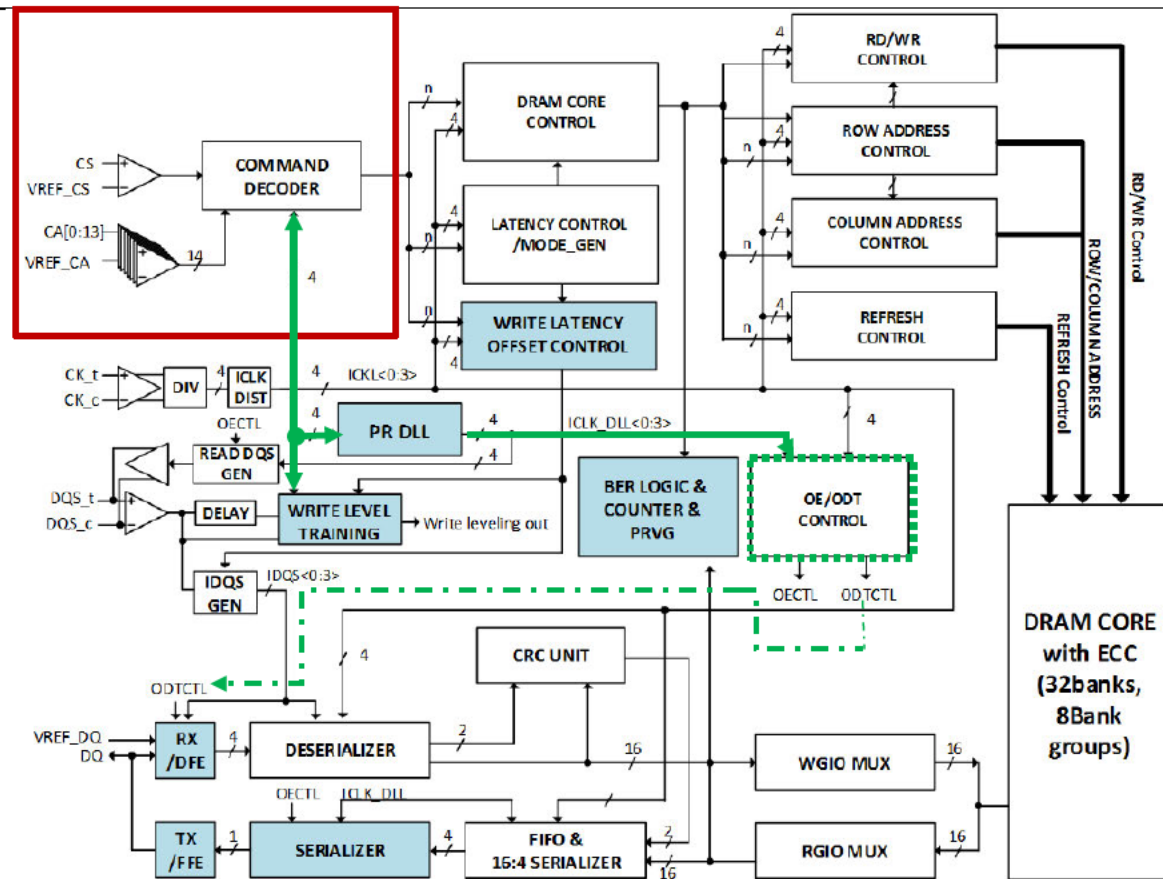
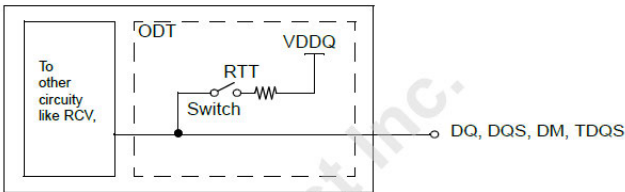


Fig. 2. DDR5 functional block diagram.

Ex. CC-58 (D. Kim *et al.*, "A 1.1-V 10-nm Class 6.4-Gb/s/Pin 16-Gb DDR5 SDRAM With a Phase Rotator-ILO DLL, High-Speed SerDes, and DFE/FFE Equalization Scheme for Rx/Tx," IEEE Journal of Solid-State Circuits, vol. 55, no. 1, pp. 167–177 (Jan. 2020)) at 168, annotated.

This claim element is literally present, as described herein, but is also satisfied under the doctrine of

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	equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.
[1.f] the each correction circuit of the set of correction circuits including at least one programmable impedance matching circuit; and,	In the Practicing '731 Products, each correction circuit of the set of correction circuits includes “at least one programmable impedance matching circuit.” As explained above for (1.d) and (1.e), each correction circuit of the set of correction circuits includes an ODT/RTT circuit. An ODT/RTT circuit is a “programmable impedance matching circuit.” 5.1 On-Die Termination for DQ, DQS, DM, and TDQS On-Die Termination (ODT) is a feature of the DDR5 SDRAM that allows the DRAM to change termination resistance for each DQ, DQS, DM and TDQS pin. Unlike previous DDR technologies, DDR5 no longer has a physical ODT pin and all ODT based control is now command & mode register based. For x4 and x8 configuration devices, ODT is applied to each DQ, DQS_t, DQS_c and DM_n, as well as TDQS_t, TDQS_c for the x8 configuration. For x16 configuration devices, ODT is applied to each DQU, DQL, DQSU_t, DQSU_c, DQSL_t, DQSL_c, DMU_n and DML_n. ODT is enabled via non-target Read/Mode Register Read commands, Write and non-target Write commands, or is set to the default “PARK” value through settings in MR33-MR35. The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to independently change termination resistance for any or all DRAM devices. In addition to the control capability of the DQ ODT, the DQS ODT will now be independently programmed via MR33:OP[5:3] and held static. All ODT control will be targeted for the DQs. This addition allows for adjusting the delay common in an unmatched architecture. DQS RTT offset control mode is enabled via MR40:OP[2:0]. The ODT feature is turned off and not supported in Self-Refresh mode, but does have an optional mode when in Power Down. A simple functional representation of the DRAM ODT feature is shown in Figure 184.  Figure 184 — Functional Representation of ODT The switch is enabled by the internal ODT control logic, which uses command decode, Mode Register Settings and other control information, see below. The value of RTT is determined by the settings of Mode Register bits. Ex. CC-55 (JEDEC JESD79-5B_v1.20) at 314; <i>see also id.</i> at 315.

[REDACTED]
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart**Power Efficiency**

The **Dynamic On-Die Termination (DODT)** feature of DDR5 memory enables dynamic management of the termination impedance.

Dynamic On-Die Termination (DODT)

It is a more sophisticated ODT technology that was introduced with DDR5 memory. When using DODT, the termination impedance can be changed dynamically, in contrast to PODT. According to the settings of the memory controller and the precise data transfer requirements at any given time, the termination resistor can be changed or turned on or off. With the aid of this dynamic management, signal integrity can be improved for a range of data rates and load situations.

Signal Reflection Reduction

Due to the nature of high-speed digital transmissions, signal reflections and impedance mismatches occur when data signals are carried across the memory bus. These reflections may deteriorate the quality of the delivered data and distort the signal. To lessen signal reflections and minimize data errors, ODT offers termination resistors that are directly attached to the memory chips and match the characteristic impedance of the transmission lines.

Ex. CC-10 (Intellect Partners DD5's Secret Weapon: ODT).

[REDACTED]

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	Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 36–37.
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	A large black rectangular area redacting the majority of the content in the first column of the table. Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 22. The following excerpts also show how the impedance matching circuit is programmable.
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Table 177 — Termination State Table

Command	Mode Register Configuration Settings				Results		Note
	RTT_PARK	RTT_WR	RTT_NOM_WR	RTT_NOM_RD	Target DRAM Term	Non-Target DRAM Term	
ANY	Disabled				HI-Z (ODT OFF)		3
Any Non-Term CMD	Enabled	Don't Care			RTT_PARK	RTT_PARK	4
WR	Disabled			Don't Care	HI-Z (ODT OFF)	HI-Z (ODT OFF)	
	Disabled		Enabled		HI-Z (ODT OFF)	RTT_NOM_WR	
	Disabled	Enabled	Disabled		RTT_WR	HI-Z (ODT OFF)	
	Don't Care	Enabled			RTT_WR	RTT_NOM_WR	2
	Enabled		Disabled		RTT_WR	HI-Z (ODT OFF)	2,5
	Enabled	Disabled	Enabled		RTT_PARK	RTT_NOM_WR	
	Enabled	Disabled			RTT_PARK	HI-Z (ODT OFF)	5
RD/MRR	Enabled	Don't Care		Disabled	HI-Z (ODT OFF)	HI-Z (ODT OFF)	1,5
	Don't Care			Enabled	HI-Z (ODT OFF)	RTT_NOM_RD	1

NOTE 1 When read command is executed, DRAM termination state of target rank will be HI-Z for defined period independent of MR setting of RTT_PARK/RTT_NOM_RD/RTT_NOM_WR.

NOTE 2 If RTT_WR is enabled, RTT_WR will be activated by Write command for defined period time independent of MR setting of RTT_PARK / RTT_NOM.

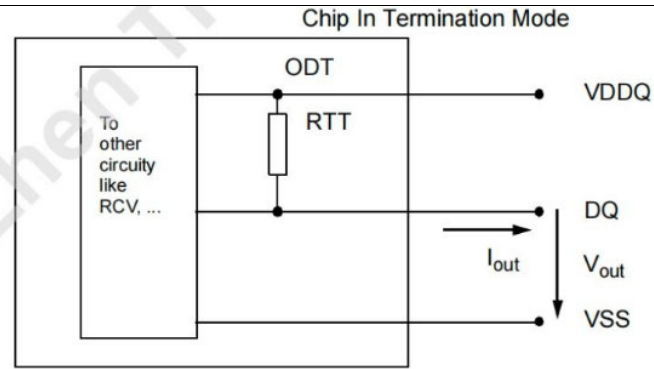
NOTE 3 If all RTT configs are disabled, ODT receiver power will be turned off to save power.

NOTE 4 If RTT_PARK is enabled, DRAM RTT_PARK termination will be enabled while WR/RD/MRR are not being executed.

NOTE 5 When a Non-Target ODT command is executed and the RTT_NOM_WR or RTT_NOM_RD is disabled, the non-target rank will be HI-Z for a defined period, independent of the MR setting of RTT_PARK.

Ex. CC-59 (JEDEC D5 Chapter 5).

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Figure 171 — On Die Termination 知乎 @sakui

Ex. CC-59 (JEDEC D5 Chapter 5).

- RTT_WR: The rank that is being written to provide termination and adjusts timing based on ODT Control Mode Register settings.
- RTT_NOM_RD: DRAM turns ON RTT_NOM_RD if it sees CS asserted during the second pulse of the READ command (except when ODT is disabled by MR35:OP[5:3]).
- RTT_NOM_WR: DRAM turns ON RTT_NOM_WR if it sees a CS asserted during the second pulse of the WRITE command (except when ODT is disabled by MR35:OP[2:0]).
- RTT_PARK: Default parked value set via MR34:OP[2:0] and is to be enabled when a READ or WRITE is not active.
- DQS_RTT_PARK: Default parked value set for DQS via MR33:OP[5:3] and is to be enabled when a READ is not active.
- Data Termination Disable: DRAM driving data upon receiving READ command disables the termination after RL-1 and stays off for a duration of BL/2.
- Strobe Termination Disable: DRAM driving strobe upon receiving READ command disables the termination after RL-1-tRPRE and stays off for a duration of BL/2+tRPST.

Ex. CC-60 (CSDN Blog, JESD79-5).

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Those RTT values have priority as follows:

- 1 Data Termination Disable & Strobe Termination Disable
- 2 RTT_WR
- 3 RTT_NOM_RD
- 4 RTT_NOM_WR
- 5 RTT_PARK

This means that if there is a WRITE command, then the DRAM turns on RTT_WR, not RTT_NOM_WR or RTT_NOM_RD, and also if there is a READ command, then the DRAM disables data termination and goes into Driving mode. If during the second pulse of a READ or WRITE command, a CS enable is sent, then Non-Target ODT is enabled and the appropriate RTT_NOM_RD or RTT_NOM_WR is enabled for the non-target rank. This provides additional and potentially different termination options for the other ranks on the channel.

Ex. CC-60 (CSDN Blog, JESD79-5).

Application: Controller can control each RTT condition with WR/RD command and **ODT** pin

- RTT_WR: The rank that is being written to provide termination regardless of **ODT** pin status (either HIGH or LOW)
- RTT_NOM: DRAM turns ON RTT_NOM if it sees **ODT** asserted (except **ODT** is disabled by MR1).
- RTT_PARK: Default parked value set via MR5 to be enabled and **ODT** pin is driven LOW.
- Data Termination Disable: DRAM driving data upon receiving READ command disables the termination after RL-X and stays off for a duration of BL/2 + X clock cycles.
X is 2 for 1tCK and 3 for 2tCK preamble mode.

- The Termination State Table is shown in **Table 72**.

Those RTT values have priority as following.

1. Data Termination Disable
2. RTT_WR
3. RTT_NOM
4. RTT_PARK

which means if there is WRITE command along with **ODT** pin HIGH, then DRAM turns on RTT_WR not RTT_NOM, and also if there is READ command, then DRAM disables data termination regardless of **ODT** pin and goes into Driving mode.

Ex. CC-60 (CSDN Blog, JESD79-5).

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Table 384 — Termination State Table

Command	Mode Register Configuration Settings				Results		Note
	RTT_PA	RTT_W	RTT_NOM_	RTT_NOM_	Target DRAM Term	Non-Target DRAM	
ANY	Disabled				HI-Z (ODT OFF)		3
Any Non-	Enabled	Don't Care			RTT_PARK	RTT_PARK	4
WR	Disabled			Don't Care	HI-Z (ODT OFF)	HI-Z (ODT OFF)	
	Disabled		Enabled		HI-Z (ODT OFF)	RTT_NOM_WR	
	Disabled	Enabled	Disabled		RTT_WR	HI-Z (ODT OFF)	
	Don't	Enabled			RTT_WR	RTT_NOM_WR	2
	Enabled		Disabled		RTT_WR	HI-Z (ODT OFF)	2,5
	Enabled	Dis-	Enabled		RTT_PARK	RTT_NOM_WR	
	Enabled	Disabled			RTT_PARK	HI-Z (ODT OFF)	5
RD/MRR	Enabled	Don't Care		Disabled	HI-Z (ODT OFF)	HI-Z (ODT OFF)	1,5
	Don't Care			Enabled	HI-Z (ODT OFF)	RTT_NOM_RD	1
NOTE 1 When read command is executed, DRAM termination state of target rank will be Hi-Z for defined period independent of MR setting of RTT_PARK/RTT_NOM_RD/RTT_NOM_WR.							
NOTE 2 If RTT_WR is enabled, RTT_WR will be activated by Write command for defined period time independent of MR setting of RTT_PARK /RTT_NOM.							
NOTE 3 If all RTT configs are disabled, ODT receiver power will be turned off to save power.							
NOTE 4 If RTT_PARK is enabled, DRAM RTT_PARK termination will be enabled while WR/RD/MRR are not being executed.							
NOTE 5 When a Non-Target ODT command is executed and the RTT_NOM_WR or RTT_NOM_RD is disabled, the DRAM termination state of the non-target rank will be Hi-Z for a defined period, independent of the MR setting of RTT_PARK.							

Ex. CC-60 (CSDN Blog, JESD79-5).

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On-Die Termination effective resistance R_{TT} is defined by MR bits.

ODT is applied to the DQ, DM, DQS_T/DQS_C and TDQS_T/TDQS_C (x8 devices only) pins.

A functional representation of the on-die termination is shown in Figure 153.

$$R_{TT} = \frac{V_{DDQ} - V_{out}}{|I_{out}|}$$

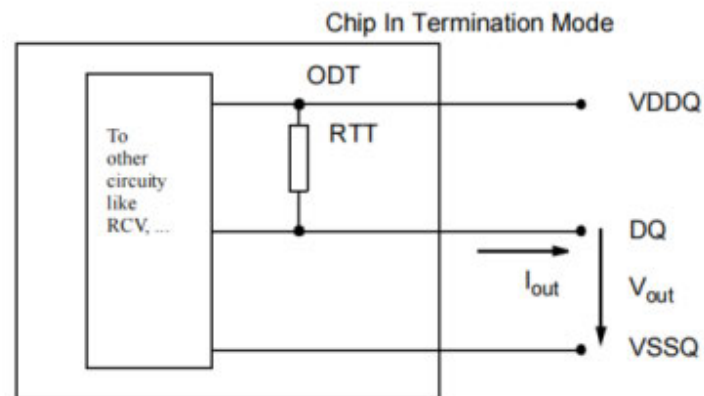


Figure 153 — On Die Termination

On die termination effective R_{tt} values supported are 240, 120, 80, 60, 48, 40, 34 ohms.

Ex. CC-60 (CSDN Blog, JESD79-5).

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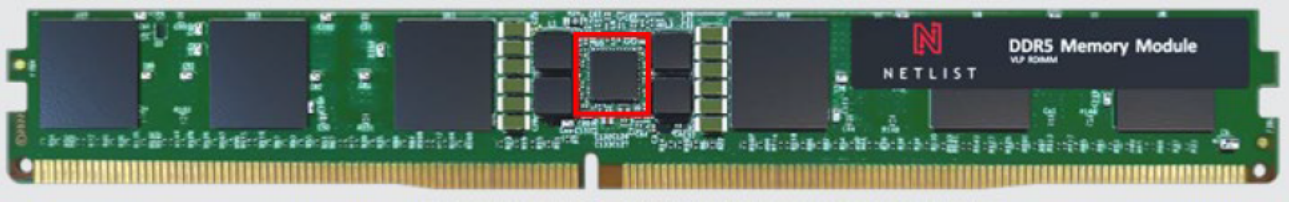
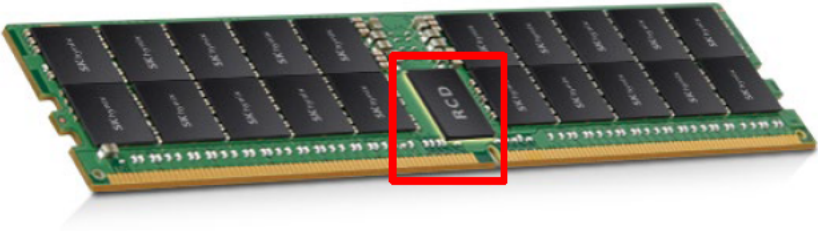
Table 385 — ODT Electrical Characteristics RZQ=240Ω +/-1% Entire Temperature Operation Range; after Proper ZQ Calibration

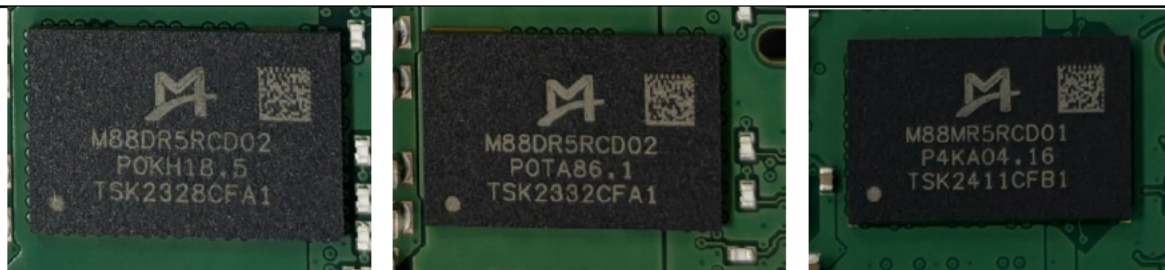
RTT	Vout	Min	Nom	Max	Unit	NOTE
240Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ	1,2,3
120Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/2	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/2	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/2	1,2,3
80Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/3	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/3	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/3	1,2,3
60Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/4	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/4	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/4	1,2,3
48Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/5	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/5	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/5	1,2,3
40Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/6	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/6	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/6	1,2,3
34Ω	VOLdc= 0.5* VDDQ	0.9	1	1.25	RZQ/7	1,2,3
	VOMdc= 0.8* VDDQ	0.9	1	1.1	RZQ/7	1,2,3
	VOHdc= 0.95* VDDQ	0.8	1	1.1	RZQ/7	1,2,3
DQ-DQ Mismatch within byte	VOMdc = 0.8* VDDQ	0	-	8	%	1,2,4,5,6
NOTE 1 The tolerance limits are specified after calibration with stable voltage and temperature. For the behavior of the tolerance limits if temperature or voltage changes after calibration, see Chapter 6.						
NOTE 2 Pull-up ODT resistors are recommended to be calibrated at 0.8*VDDQ. Other calibration schemes may be used to achieve the linearity spec shown in the table, e.g., calibration at 0.5*VDDQ and 0.95*VDDQ.						
NOTE 3 The tolerance limits are specified under the condition that VDDQ=VDD and VSSQ=VSS						
NOTE 4 DQ to DQ mismatch within byte variation for a given component including DQS_T and DQS_C (characterized)						

Ex. CC-60 (CSDN Blog, JESD79-5).

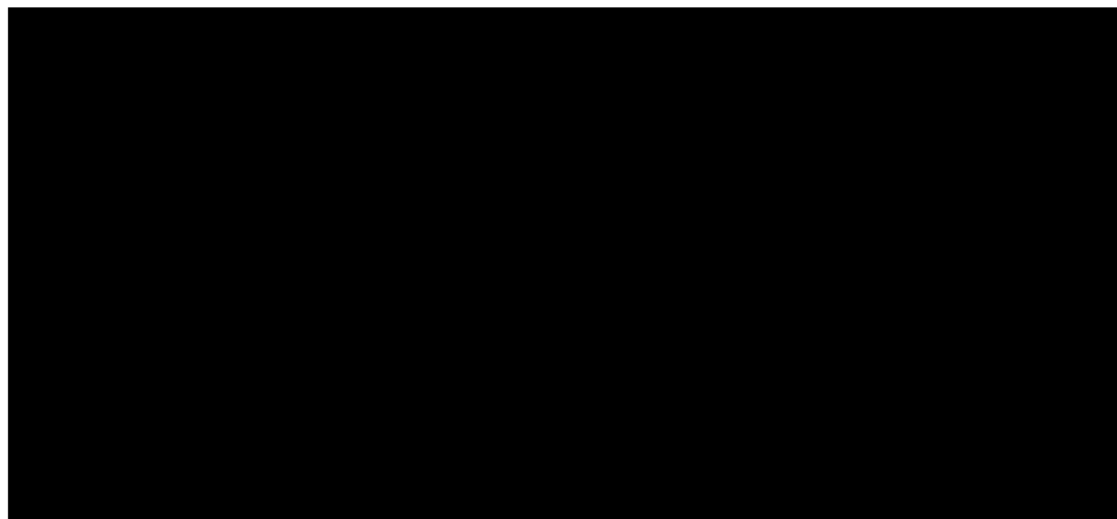
This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

	same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.
[1.g] control circuitry configured to receive the address and control signals from the system memory controller and to control the plurality of memory devices in response to the address and control signals,	The Practicing '731 Products include “control circuitry” (e.g., the RCD or MRCD and/or the control/address circuitry on the SDRAM chips).  VLP RDIMM Ex. CC-33 (Netlist DDR5 DIMMs), annotated.  SERVER DDR5 RDIMM Ex. CC-34 (SK hynix Products), annotated.

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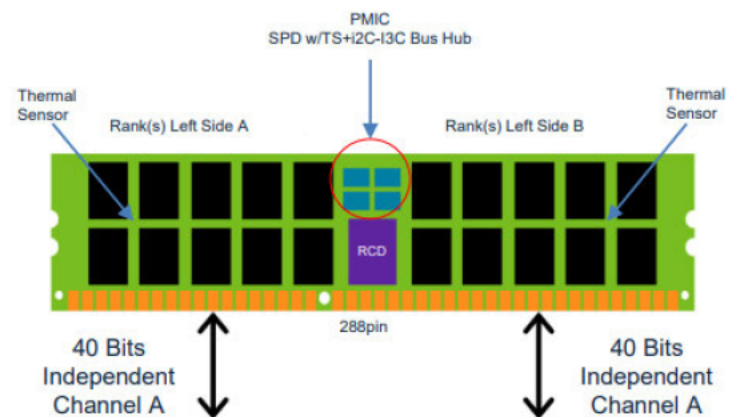
Ex. CC-36 at IMG_5108 (Montage M88DR5RCD02 for DDR5 RDIMM OverClock Product), IMG_5116 (Montage M88DR5RCD02 for DDR5 RDIMM), IMG_5124 (Montage M88MR5RCD01 for DDR5 MRDIMM), excerpts.



Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 45, annotated.

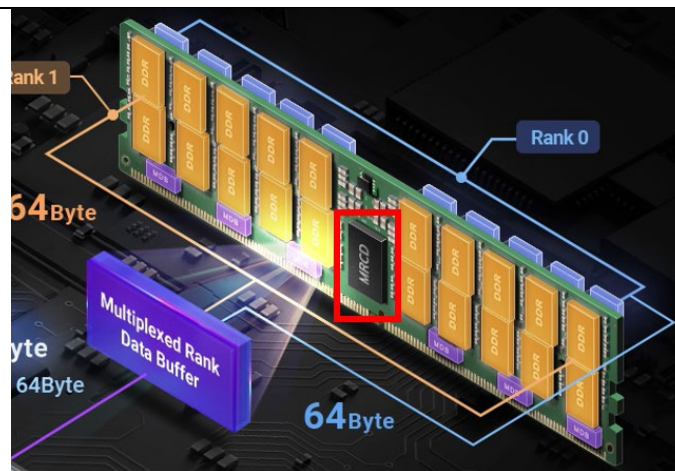
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Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 25.



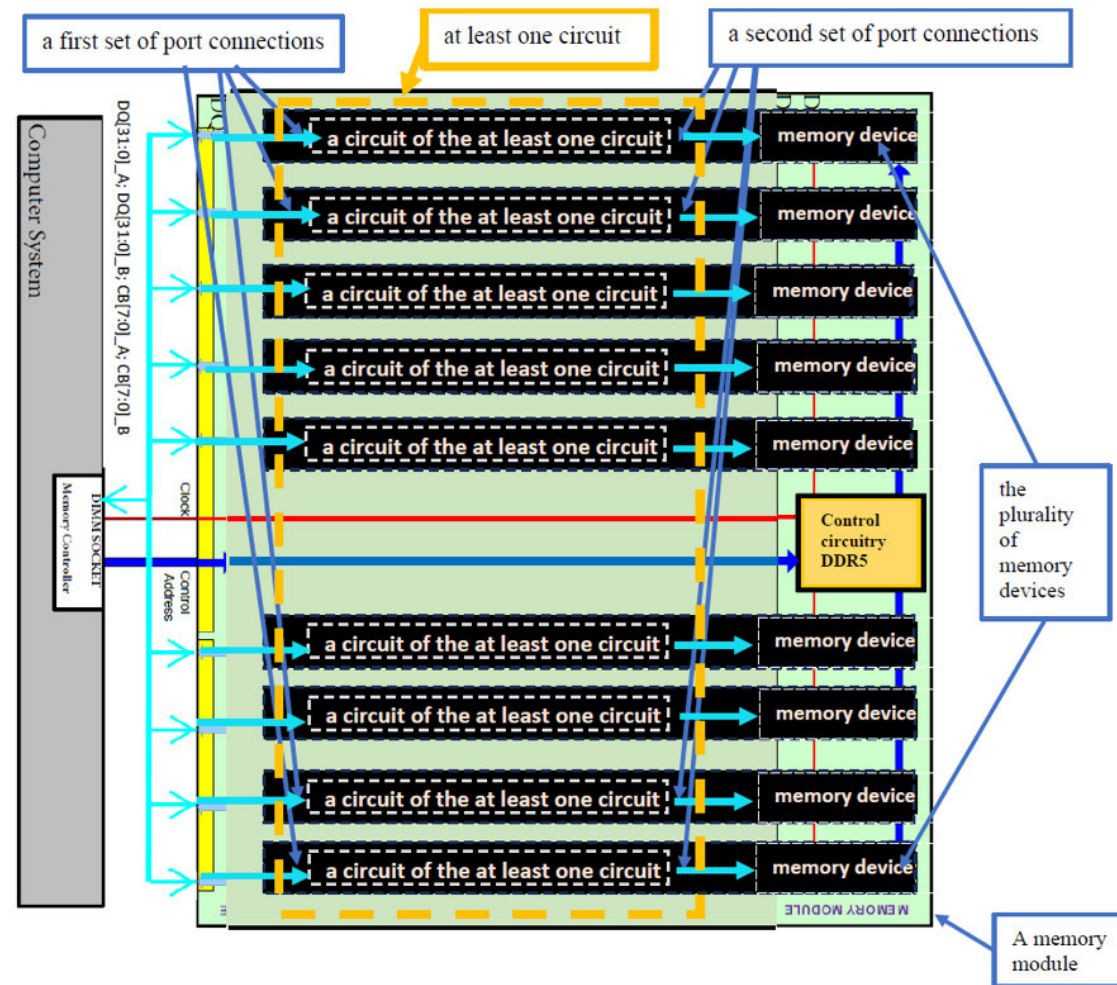
Ex. CC-44 (Netlist DDR5 DIMM Product Brief) at 1.

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Ex. CC-42 (SK hynix MRDIMM) at 4, excerpted and annotated.

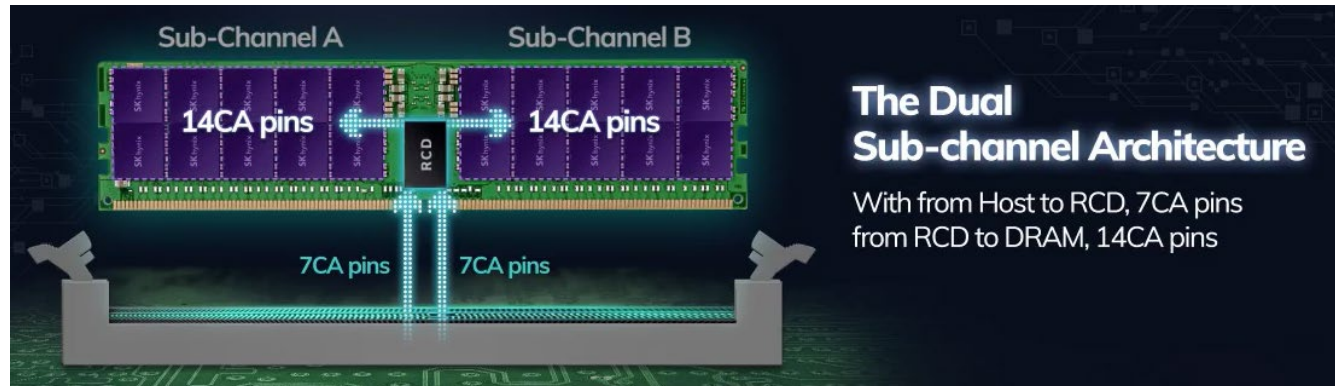
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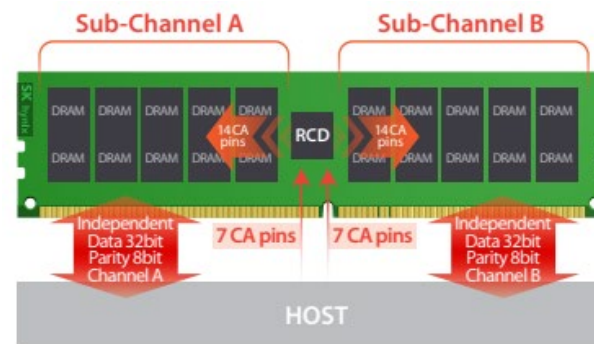
In the Practicing '731 Products, the control circuitry is "configured to receive the address and control signals from the system memory controller and to control the plurality of memory devices in response to the address and control signals."

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For example, the RCD receives address and control signals from the host. The RCD then controls the plurality of memory devices in response to the address and control signals.



Ex. CC-41 (SK hynix RDIMM).



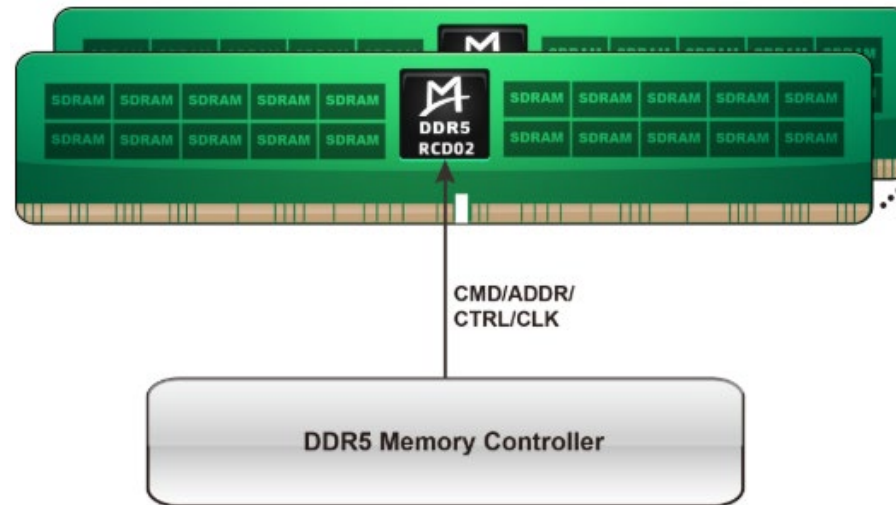
New Sub-channel DRAM Architecture

The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

Ex. CC-40 (SK hynix A New Standard) at 5.

[REDACTED]

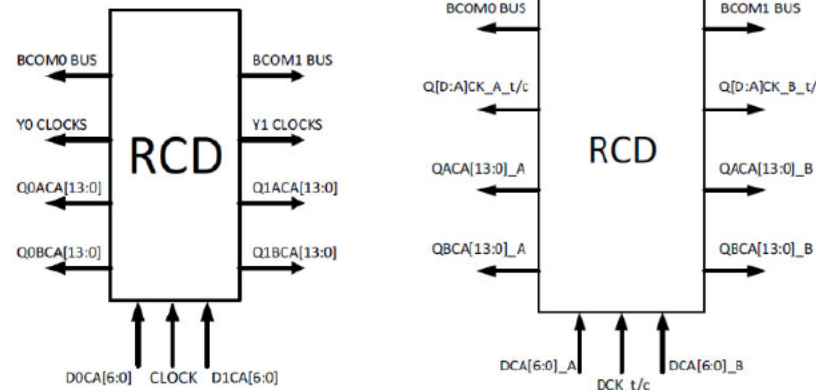
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Ex. CC-43 (M88DR5RCD02 Flyer); *see also* Ex. CC-36 at IMG_5112, 5114, 5122 (Montage supplying RCDs for the Practicing '731 Products).

For example, the 5RCD0148HC2 (RCD) is a registering clock driver used on DDR5 RDIMMs. *“Its primary function is to buffer the Command Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus to control the data buffers for LRDIMMs. [¶] The [5RCD0148HC2] contains two separate channels with some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip-select inputs, produces two copies of 14-bit single data rate CA bus outputs and two copies of the chip-select outputs. The RCD has a common clock input and PLL but produces separate clock outputs to the DRAM channels.”* Ex. CC-51 (Gen 2 RCD Renesas RG5R256).

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Ex. CC-52 (Gen 1 RCD Renesas 5RCD0148); Ex. CC-51 (Gen 2 RCD Renesas RG5R256); *see also* Ex. CC-53C (0887-99A-1M6060 Prelim 02-08-22) at 3

Ex. CC-54C (0888-97A-1A6B61-X00) at 2

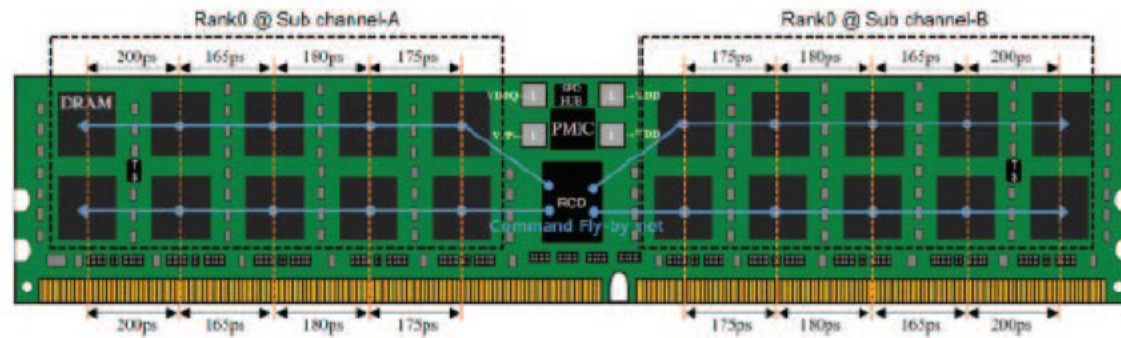
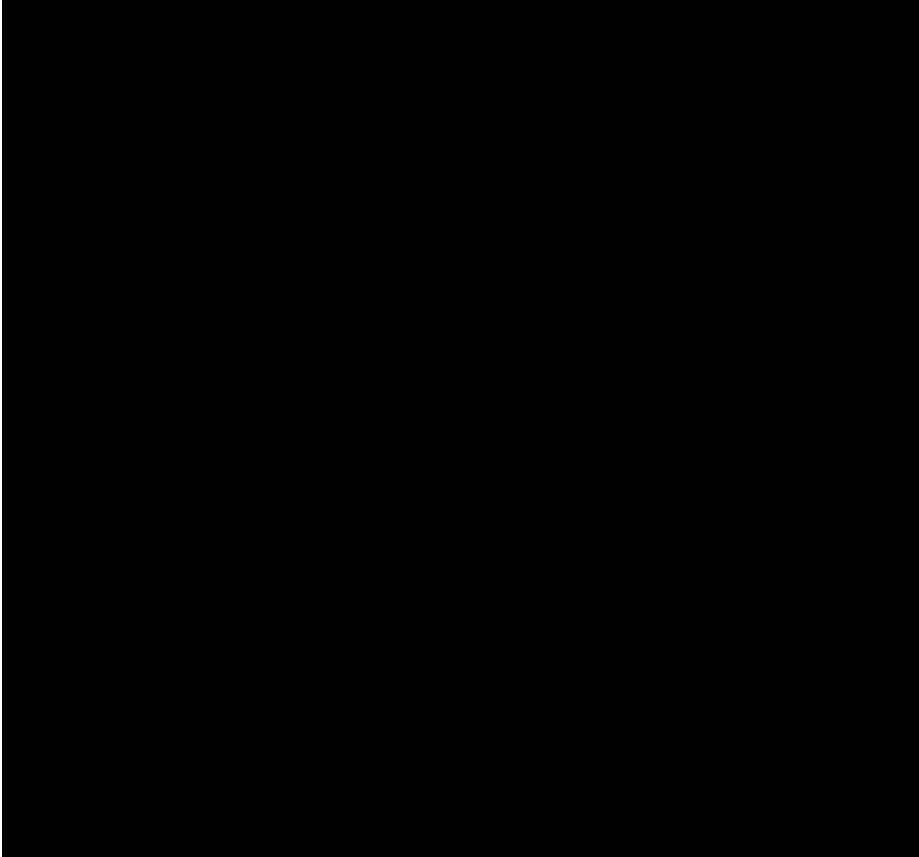


Figure 3. The operation delay due to fly-by topology of command and address nets in DDR5 RDIMM 2Rx4.

Ex. CC-45 (Na *et al.*, "A Novel Simulation Methodology Reflecting System Power Scenario using a Markov-Chain-based Stochastic Random Power Model," 2022 IEEE 72nd Electronic Components and

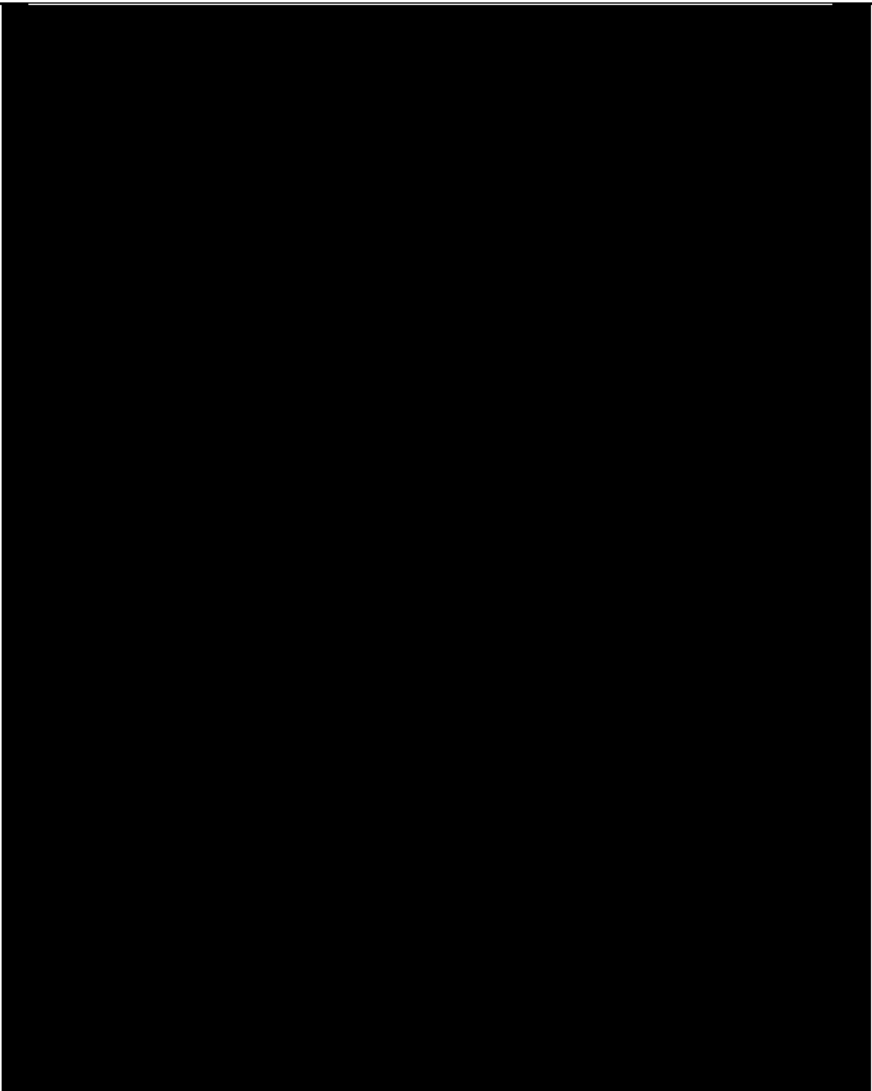


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	Technology Conference, 929–936) at 931.  Ex. CC-38C (DDR5 VLP RDIMM Datasheet Version 1.1) at 9.
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		 A large black rectangular area redacting the majority of the table's content.	
		Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10.	

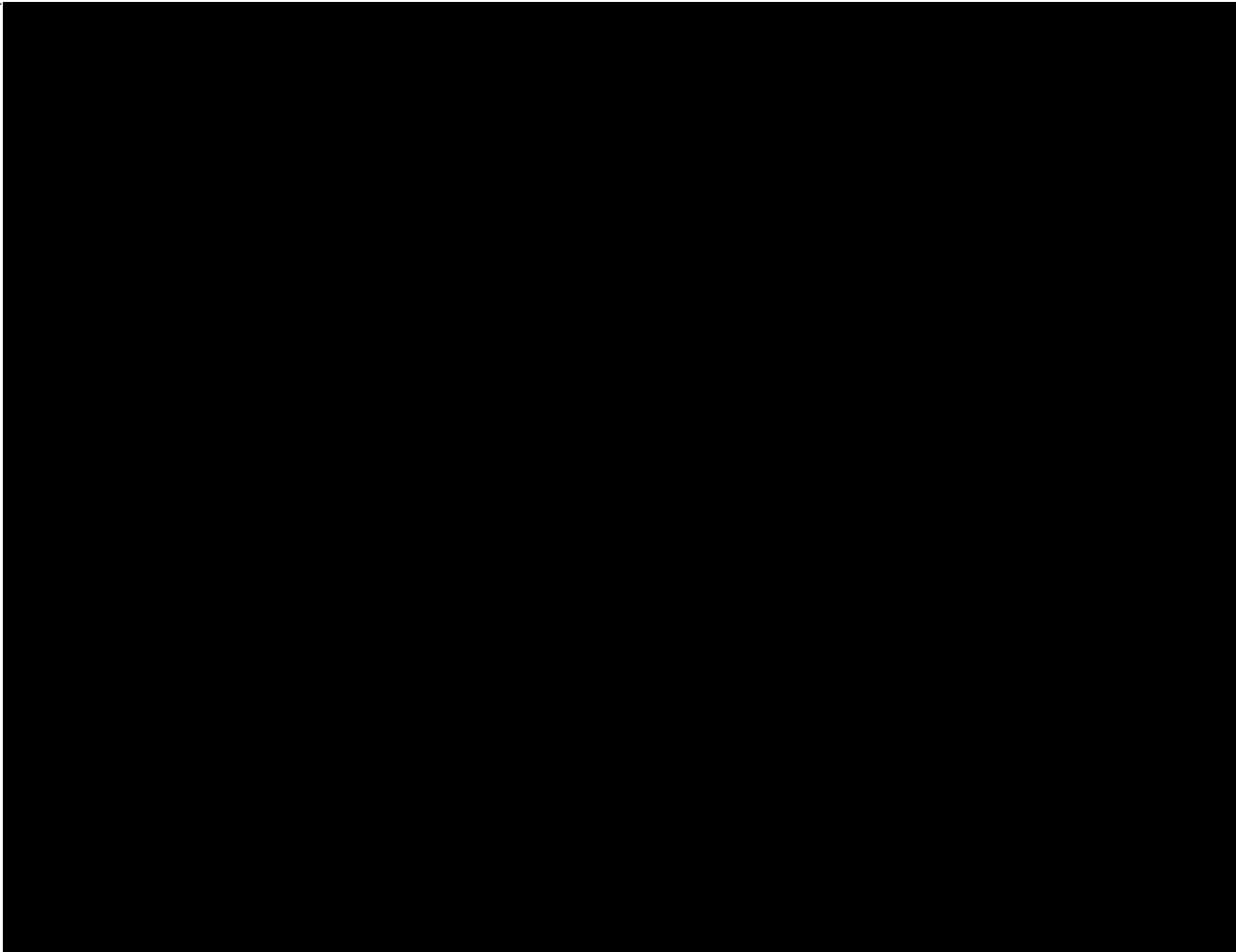
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Function	Abbreviation	CS_n	CA Pins													NOTES	
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	11,17,20,21,23,24
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	11,15,18,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	11,15,18,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/DOPIO	8,11,13,20,22,23,24
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/DOPIO	8,13,20,22,23,24,25,26
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,12,15,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR_Partial=L	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,12,15,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	WR_Partial=L	V	CID3	
Read	RD	L	H	L	H	H	H	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,15,19,20,21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DOPIO	8,15,19,20,21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	V	V	CID3	

Ex. CC-46 (DDR5 Command Truth Table) (showing that commands for Activate, Read, and Write are composed of C/A[0:13] and CS signal).

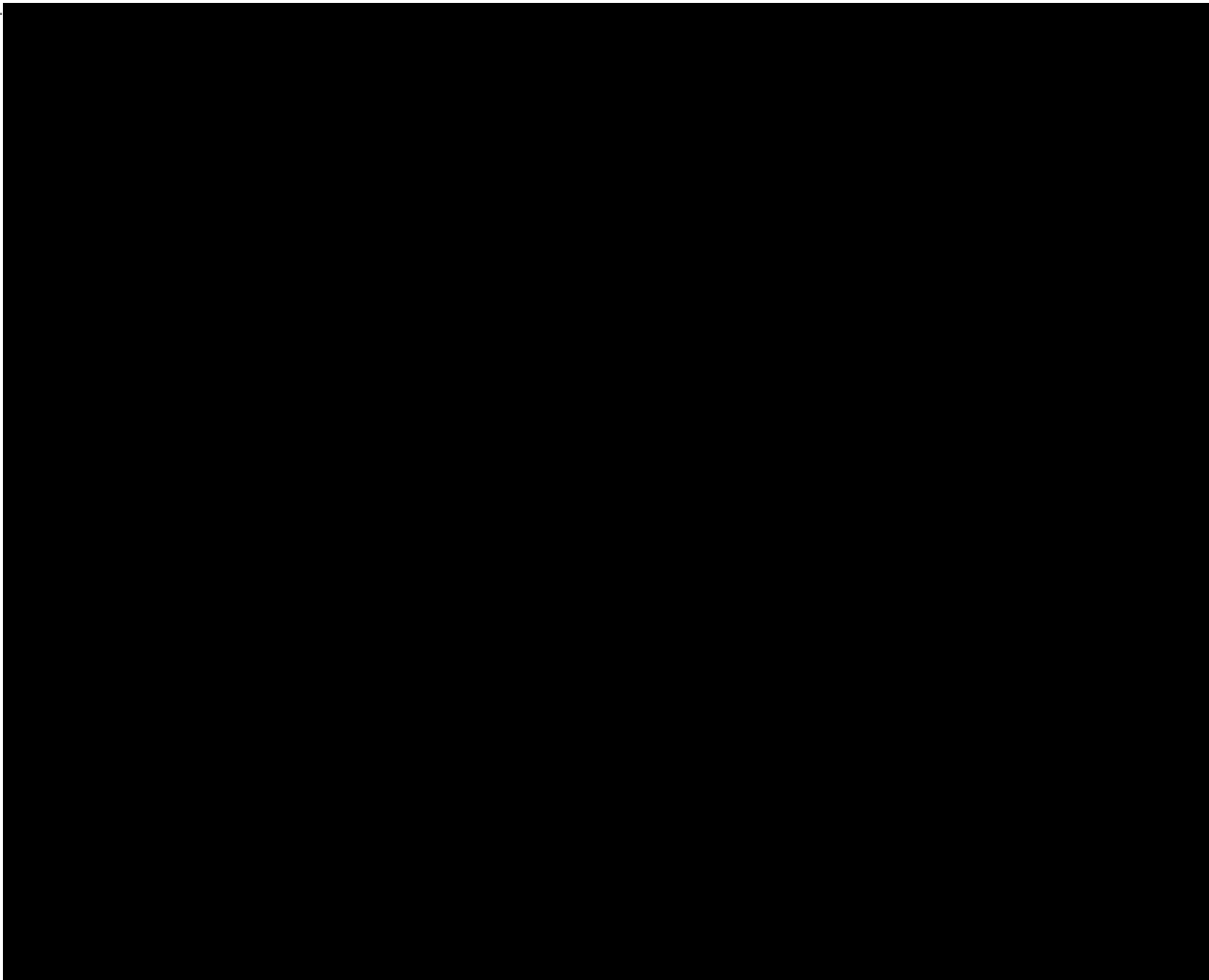


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	 Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 4, 7.
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[REDACTED]

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	 Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 4.
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U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

	A large solid black rectangular area used to redact the main content of the table. Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 5.
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U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

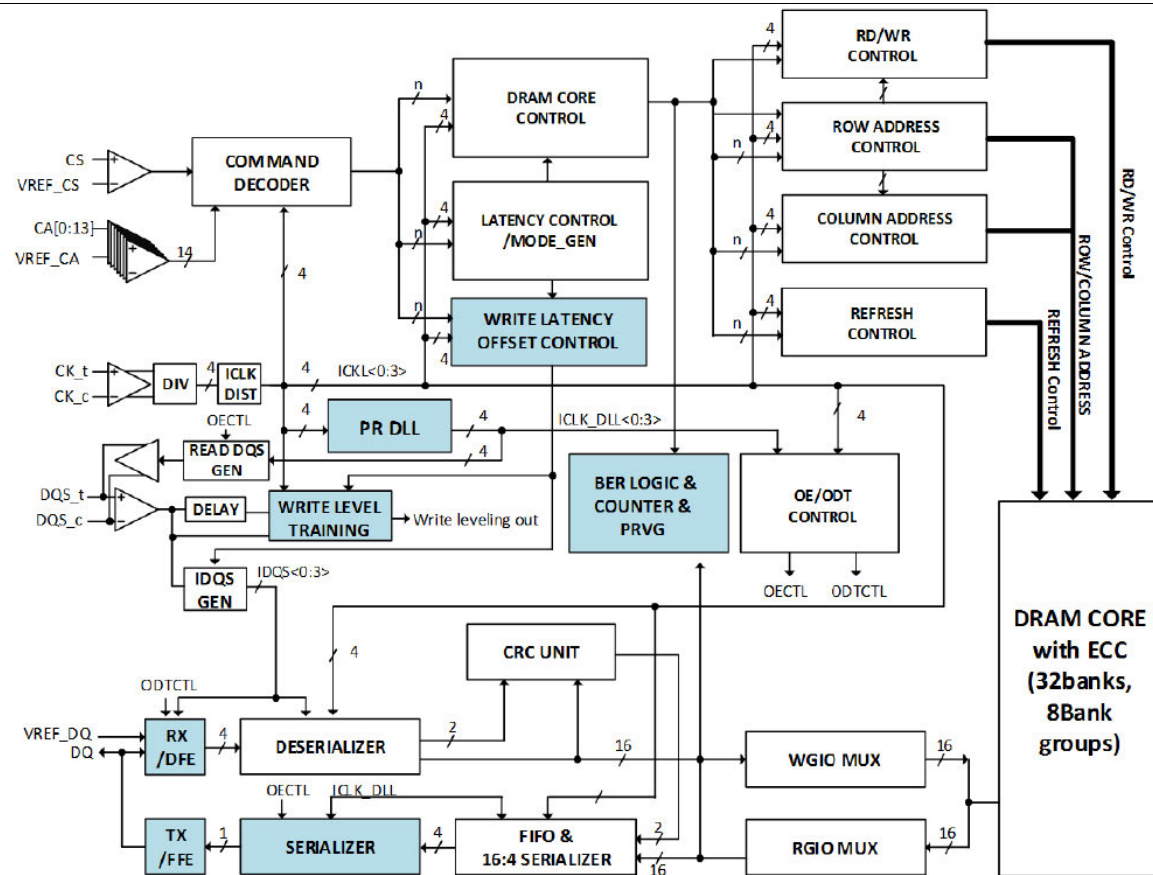


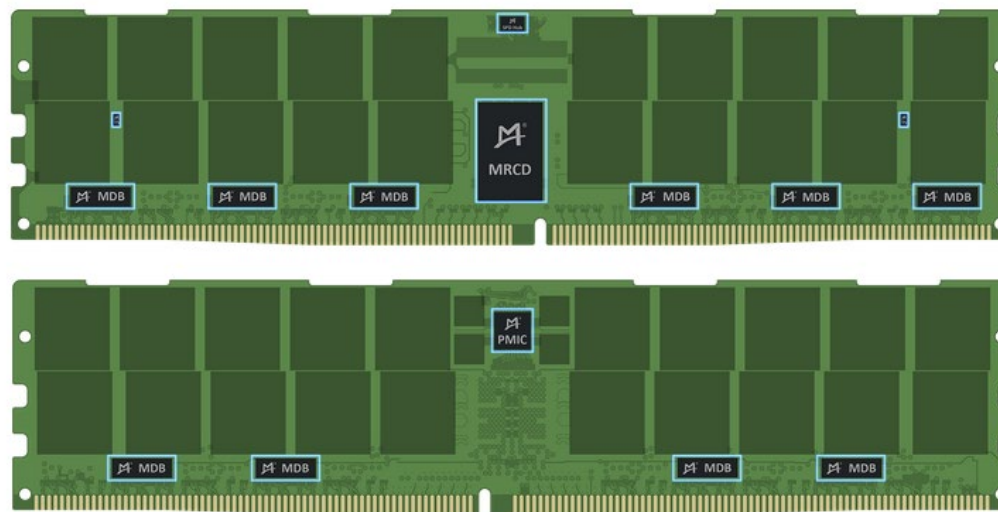
Fig. 2. DDR5 functional block diagram.

Ex. CC-58 (D. Kim *et al.*, "A 1.1-V 10-nm Class 6.4-Gb/s/Pin 16-Gb DDR5 SDRAM With a Phase Rotator-ILO DLL, High-Speed SerDes, and DFE/FFE Equalization Scheme for Rx/Tx," IEEE Journal of Solid-State Circuits, vol. 55, no. 1, pp. 167–177 (Jan. 2020)) at 168.

For another example of control circuitry, the MRDIMM Products have an MRCD, MDBs, and other

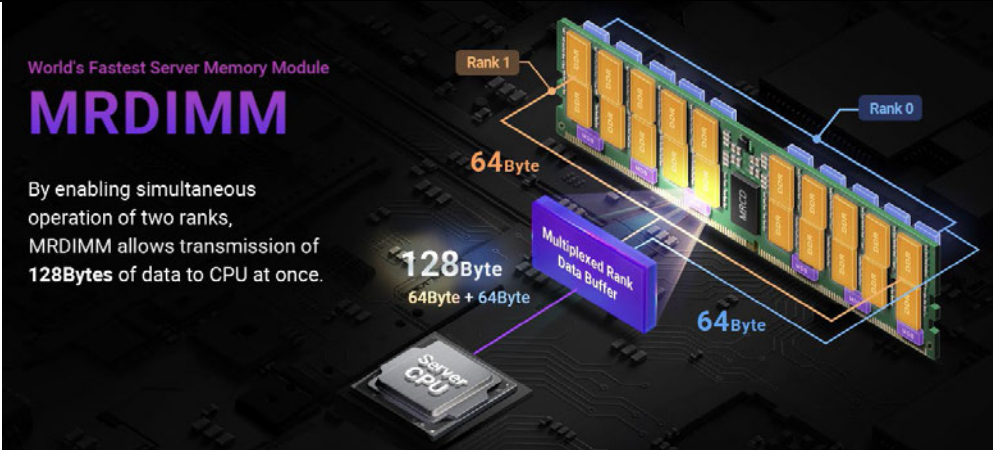
U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

circuitry (shown below), which are configured to receive address and control signals from the system memory controller and used to control the plurality of memory devices in response to the address and control signals. For example, Montage's M88MR5MRCD01 MRCD supports "[t]wo pseudo-channels splitting the incoming CA inputs and generating separate CA outputs," which, as advertised in Montage's flyer for Montage M88MR5DB01 MDB, works in conjunction with Montage's MDB to "concurrently access two DRAM memory ranks through the MDB chips." Ex. CC-12 (M88MR5RCD01 Flyer) at 1; Ex. CC-22 (M88MR5DB01 Flyer) at 1.



Ex. CC-23 (Montage DDR5 Products); *see also* Ex. CC-36 at IMG_5122–28.

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	 World's Fastest Server Memory Module MRDIMM By enabling simultaneous operation of two ranks, MRDIMM allows transmission of 128Bytes of data to CPU at once. Rank 1 Rank 0 64Byte 128Byte 64Byte + 64Byte Multiplexed Rank Data Buffer 64Byte Server CPU Ex. CC-42 (SK hynix MRDIMM) at 4. This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.
[1.h] wherein the control circuitry is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit based on which of the multiple ranks is selected to communicate	In the Practicing '731 Products, “the control circuitry is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit based on which of the multiple ranks is selected to communicate data with the system memory controller in response to the address and control signals.” As explained for (1.f), each correction circuit of the set of correction circuits in the each circuit of the at least one circuit includes a programmable impedance matching circuit—an ODT/RTT circuit. The control circuitry (e.g., RCD or MRCD) is configured to “dynamically control” this ODT/RTT circuit.

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data with the system memory controller in response to the address and control signals.	Power Efficiency The Dynamic On-Die Termination (DODT) feature of DDR5 memory enables dynamic management of the termination impedance. Dynamic On-Die Termination (DODT) It is a more sophisticated ODT technology that was introduced with DDR5 memory. When using DODT, the termination impedance can be changed dynamically, in contrast to PODT. According to the settings of the memory controller and the precise data transfer requirements at any given time, the termination resistor can be changed or turned on or off. With the aid of this dynamic management, signal integrity can be improved for a range of data rates and load situations. Ex. CC-10 (Intellect Partners DD5's Secret Weapon: ODT).
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5.1 On-Die Termination for DQ, DQS, DM, and TDQS

On-Die Termination (ODT) is a feature of the DDR5 SDRAM that allows the DRAM to change termination resistance for each DQ, DQS, DM and TDQS pin. Unlike previous DDR technologies, DDR5 no longer has a physical ODT pin and all ODT based control is now command & mode register based. For x4 and x8 configuration devices, ODT is applied to each DQ, DQS_t, DQS_c and DM_n, as well as TDQS_t, TDQS_c for the x8 configuration. For x16 configuration devices, ODT is applied to each DQU, DQL, DQSU_t, DQSU_c, DQSL_t, DQSL_c, DMU_n and DML_n. ODT is enabled via non-target Read/Mode Register Read commands, Write and non-target Write commands, or is set to the default "PARK" value through settings in MR33-MR35.

The ODT feature is designed to improve signal integrity of the memory channel by allowing the DRAM controller to independently change termination resistance for any or all DRAM devices. In addition to the control capability of the DQ ODT, the DQS ODT will now be independently programmed via MR33:OP[5:3] and held static. All ODT control will be targeted for the DQs. This addition allows for adjusting the delay common in an unmatched architecture. DQS RTT offset control mode is enabled via MR40:OP[2:0].

The ODT feature is turned off and not supported in Self-Refresh mode, but does have an optional mode when in Power Down. A simple functional representation of the DRAM ODT feature is shown in **Figure 184**.

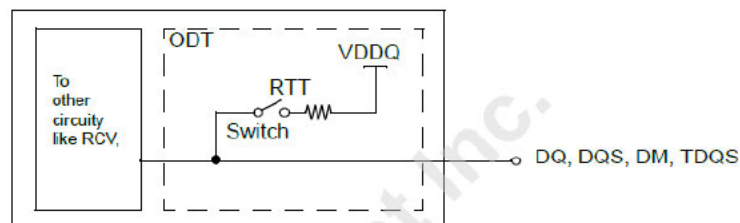


Figure 184 — Functional Representation of ODT

The switch is enabled by the internal ODT control logic, which uses command decode, Mode Register Settings and other control information, see below. The value of RTT is determined by the settings of Mode Register bits.

Ex. CC-55 (JEDEC JESD79-5B_v1.20) at 314; *see also id.* at 315.

As explained for (1.c),

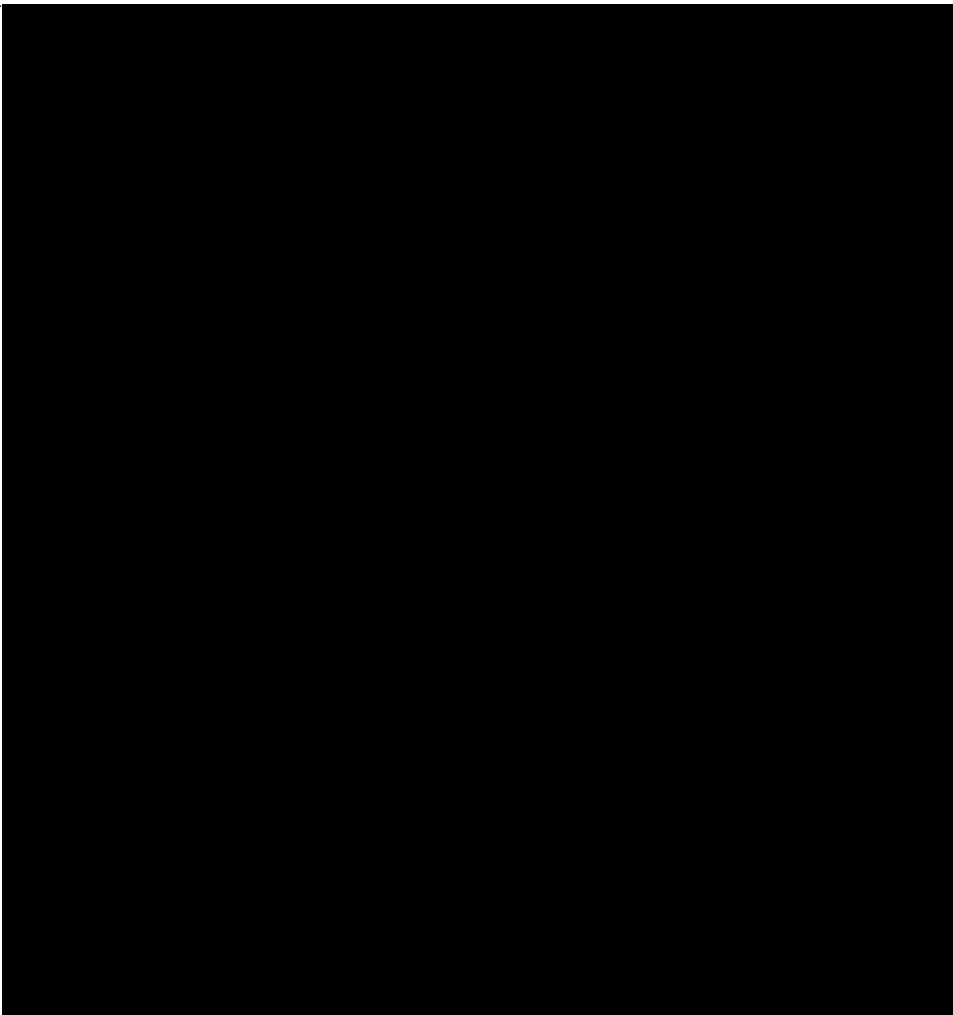
[REDACTED]

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	[REDACTED] [REDACTED] Ex. CC-38C (Netlist DDR5 VLP RDIMM Datasheet Version 1.1) at 9.
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[REDACTED]

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	Ex. CC-39C (Netlist DDR5 Overclock RDIMM Datasheet Version 1.0) at 10.		

U.S. Patent No. 10,025,731 Domestic Industry Claim Chart

3.3 Register Operation**3.3.1 Chip Select Operation**

The component has two chip select inputs per channel, DCS0_n and DCS1_n, and two copies of each chip select output, QCS0_A_n, QCS1_A_n, QCS0_B_n and QCS1_B_n. Commands are sent to a single rank or multiple ranks, as determined by the DCS[1:0]_n inputs. The number of input chip selects matches the number of output chip selects in each of the two sets (A-outputs and B-outputs).

Table 3 — DCS - QxCS Mapping

Input CS	Output CS
DCS0_N_n	QxCS0_N_n
DCS1_N_n	QxCS1_N_n

3.3.2 ODT Operation

DDR5 does not have a separate ODT (On Die Termination) signal going to the DIMM or the DRAMs. The ODT control is encoded into the chip selects. All commands that utilize the data bus require 2UI on the DCA bus in order to transfer the appropriate number of command and addressing bits. This requirement is used to signal ODT control by how the chip selects are asserted for the two UI.

Table 4 — ODT encoding on the chip select

DCS_n on 1st UI	DCS_n on 2nd UI	QxCS[1:0]_n	DRAM DQ Terminations
Active	Inactive	Target of command	Read: none (device drives data) Write: RIT_WR.
Active	Active	Non-Target Termination should be applied.	Read: RIT_NOM_RD Write: RIT_NOM_WR
Inactive	Inactive	DRAM ignores commands and the Parked termination will be maintained	RIT_PARK
Inactive	Active	Illegal	Illegal

The host controller will determine the termination at the time of the command, by activating the chip selects appropriately.

Ex. CC-32 (JESD82-512) at 34, annotated.

Likewise, the C/A and CS circuits on the SDRAMs would decode the incoming C/A and CS signals with proper delay to control the ODT, including the values that it will take:

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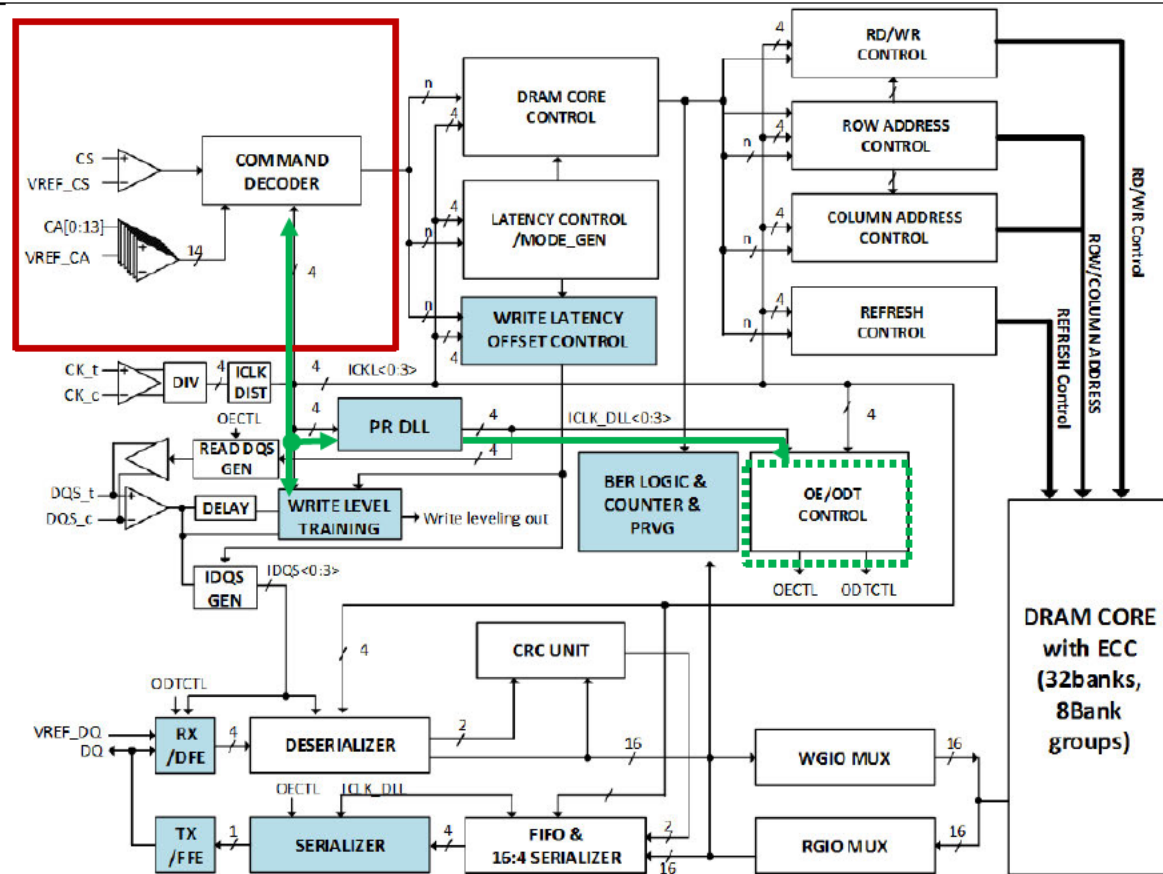


Fig. 2. DDR5 functional block diagram.

Ex. CC-58 (D. Kim *et al.*, "A 1.1-V 10-nm Class 6.4-Gb/s/Pin 16-Gb DDR5 SDRAM With a Phase Rotator-ILO DLL, High-Speed SerDes, and DFE/FFE Equalization Scheme for Rx/Tx," IEEE Journal of Solid-State Circuits, vol. 55, no. 1, pp. 167–177 (Jan. 2020)) at 168.

In the operation below, the first write operation is to Rank0 and the second to Rank1. As seen in the

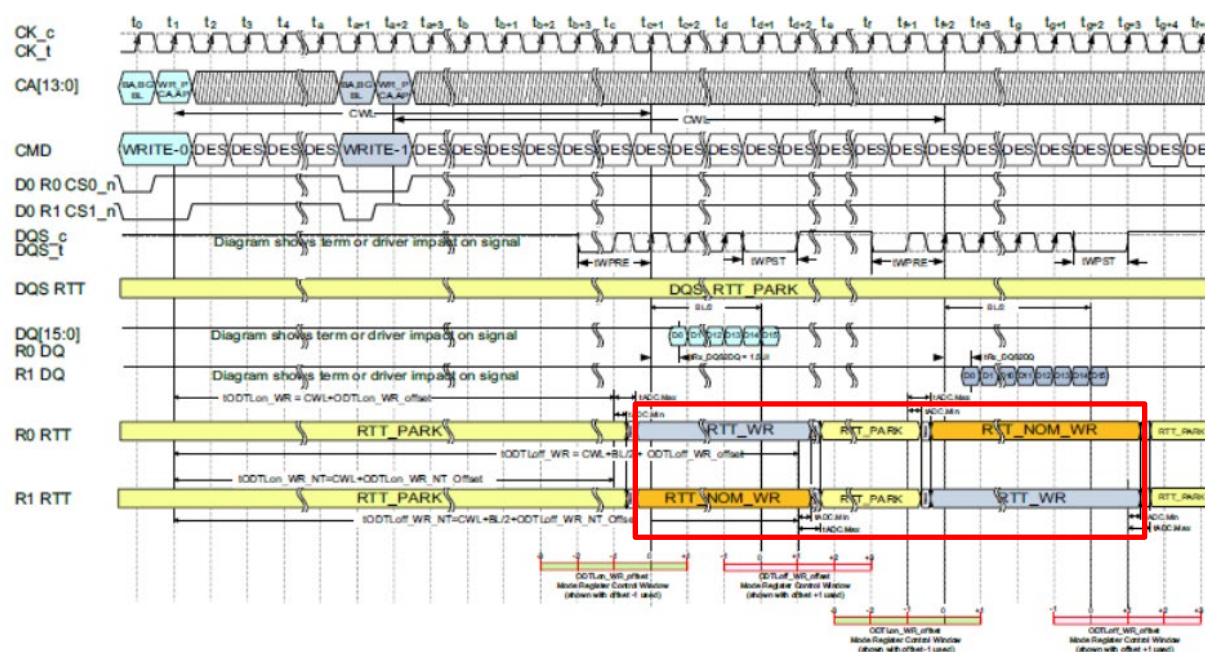
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timing diagram below, the RTT values for Rank0 (R0) and Rank1 (R1) differ depending on which of the two ranks is selected for the write operation. The transition from one RTT value to another is dynamic, as there are no intermediate MRW operations, and it depends on the selected rank.

JEDEC Standard No. 79-5C_v1.30

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5.3.3 ODT Timing Diagrams (cont'd)



NOTES:

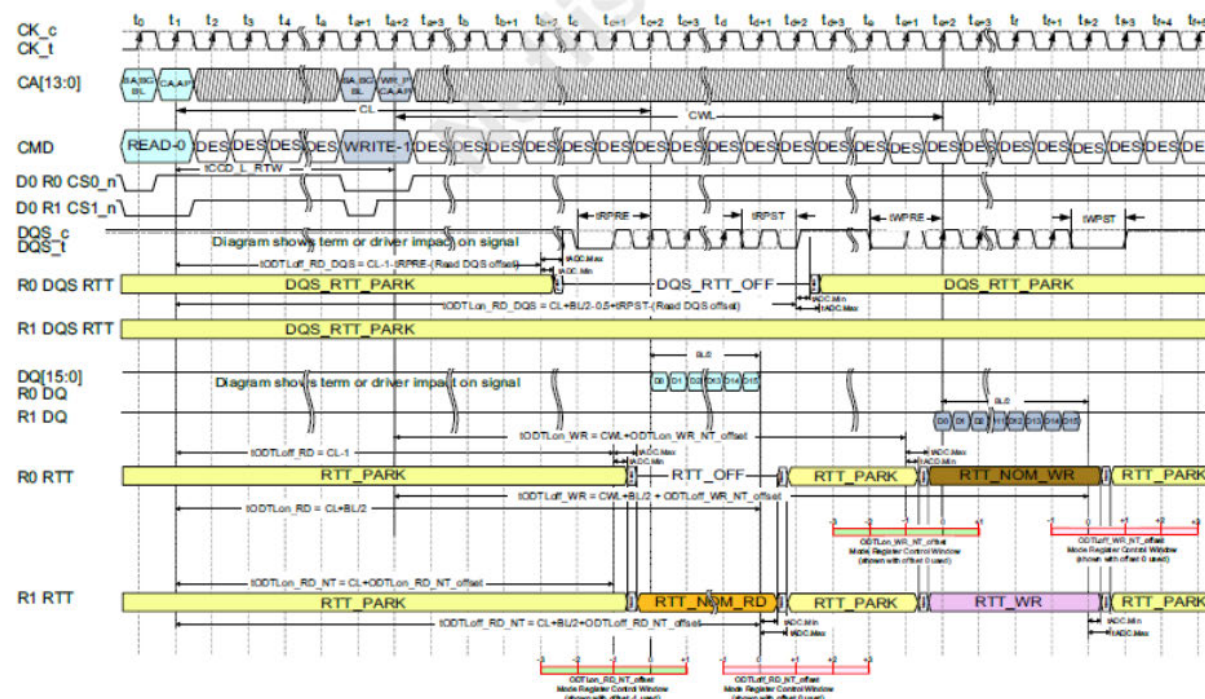
1. ODTLon_WR, ODTLon_WR_NT, ODTLoFF_WR and ODTLoFF_WR_NT are based on Mode Register settings that can push out or pull in the RTT enable and disable time.
2. The entire range of ODTL control is not shown for simplicity.

Figure 197 — Example of Write to Write Turn Around, Different Ranks

Ex. CC-61 (JESD79-5C_v1.30) at 323, annotated.

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As another example, the timing diagram below shows that in the period when read is operated on Rank0, the RTT for Rank0 is off, and after the read operation is completed, the RTT for Rank0 dynamically transitions to RTT_Park, and then to RTT_NOM_WR when a write operation is performed on Rank1. For Rank1, the RTT value transitions to RTT_NOM_RD during read operation for Rank0, transitions to RTT_Park during the inactive period, and transitions to RTT_WR during the Rank1 write operation.



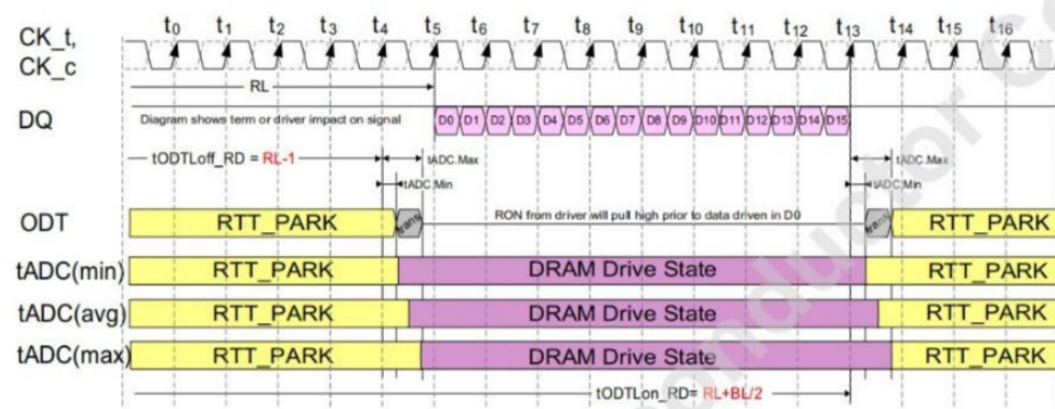
NOTES:

1. ODTLoFF_WR, ODTLoFF_WR_NT, ODTLoFF_RD and ODTLoFF_RD_NT are based on Mode Register settings that can push out or pull in the RTT enable and disable time.
2. ODTLoFF_RD_NT and ODTLoFF_RD_NT are based on Mode Register settings that can push out or pull in the RTT enable and disable time.
3. ODTLoFF_WR_offset and ODTLoFF_RD_offset not shown for simplicity.
4. Example shown with near ideal timings for termination settings, exact offset configurations will vary based on system designs.

Figure 202 — Example of Read to Write Turn Around, Different Ranks

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Ex. CC-61 (JESD79-5C_v1.30) at 325.



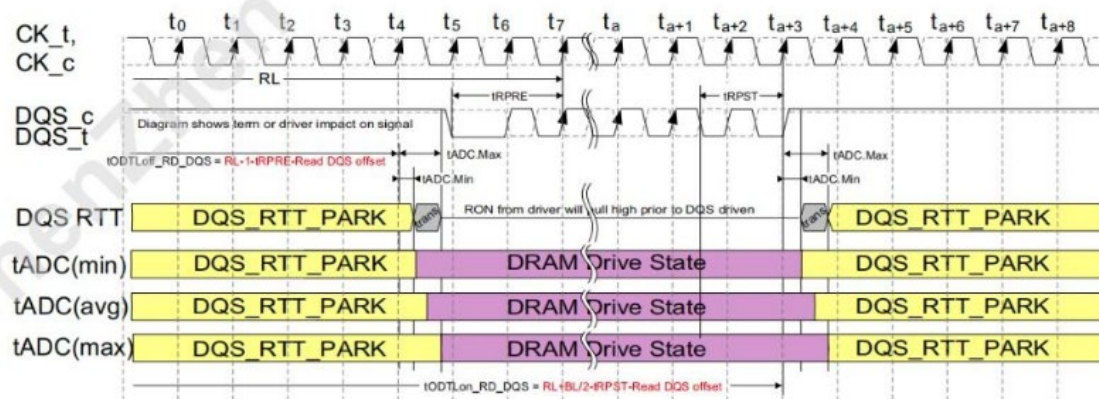
NOTE 1 The diagram shows a transition from RTT_PARK to Read DRAM Drive state. When tADC transitions from RTT to Read Drive state, the DRAM RON from the driver will keep the DQ signal high prior to the data driven in D0. No High-Z time during tADC is allowed in this example.

NOTE 2 In the case of Term to Write, the host will keep the DQ signal HIGH 4UI prior to the data driven in D0.

NOTE 3 The DFE should assume that 4UI prior to D0 the signal is HIGH.

Ex. CC-59 (JEDEC D5 Chapter 5).

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NOTE 1 The diagram shows a transition from DQS_RTT_PARK to Read DRAM Drive state. When tADC transitions from RTT to Read Drive state, the DRAM RON from the driver will keep the DQS signal high prior to the DQS driven at t5. No High Z time during tADC is allowed in this example.

Ex. CC-59 (JEDEC D5 Chapter 5).

See also Claim 2.

This claim element is literally present, as described herein, but is also satisfied under the doctrine of equivalents, and the Practicing '731 Products are manufactured in a manner that is not substantially different from the requirements of this limitation because they achieve substantially or exactly the same function, in substantially or exactly the same way as set forth in the claim element (as described herein), and achieve substantially or exactly the same result.